MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3860 REAGENTS – LIMESTONE EQUIPMENT
DESIGN CALCULATION OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of **3860 REAGENTS – LIMESTONE EQUIPMENT** platform foundation. The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD Pro. RCC column provided below the steel structure. The columns are RCC cast-in-situ. Columns rest on founfation, suitably designed to transfer design load on soil.
Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet.
Only for Lime Slaking Silo foundation base pressure check has been done by STAAD Pro plate mat analysis method.
For strength calculation of Raft & Column Bending Moment & Shear Force values are directly taken from STAAD & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A03 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	Minimum Cove (mm)
Exposed to Soil	Foundation	50
	Column	50

5. LOAD

5.1 Primary Loads

3860 REAGENTS – LIMESTONE EQUIPMENT platform foundation **t**he primary load values are taken from the steel structure design calculation **Doc. No. : MD-216-3850-EG-ST-CAL-2002.**

1 DEAD LOAD
2 LIVE LOAD
3 WIND LOAD (+Z)
4 WIND LOAD (-Z)
5 WIND LOAD (+X)
6 WIND LOAD (-X)
7 TL1 (TEMP LOAD +25)
8 TL2 (TEMP LOAD -25)
9 EL (+X)
10 EL (+Z)

5.2 Load Combinations for Stress Calculation

L/C	LOAD COMBINATION
101	DL
102	DL + LL + TL1
103	DL + LL + TL2
104	DL + 0.75*WL(+Z) + 0.75*LL
105	DL + 0.75*WL(-Z) + 0.75*LL
106	DL + 0.75*WL(+X) + 0.75*LL
107	DL + 0.75*WL(-X) + 0.75*LL
108	DL + 0.525*EL(+X) + 0.75*LL
109	DL + 0.525*EL(-X) + 0.75*LL
110	DL + 0.525*EL(+Z) + 0.75*LL
111	DL + 0.525*EL(-Z) + 0.75*LL
112	0.6*DL + WL(+Z)
113	0.6*DL + WL(-Z)
114	0.6*DL + WL(+X)
115	0.6*DL + WL(-X)
116	0.6*DL + 0.7*EL(+X)
117	0.6*DL - 0.7*EL(+X)
118	0.6*DL + 0.7*EL(+Z)
119	0.6*DL - 0.7*EL(+Z)

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5.3 Load Combinations for Strength Calculation

L/C	LOAD COMBINATION
201	1.4*DL
202	1.4*(DL + TL1) + 1.7*LL
203	1.4*(DL + TL2) + 1.7*LL
204	1.2*DL + 1.6*WL(+Z) + 0.5*LL
205	1.2*DL + 1.6*WL(-Z) + 0.5*LL
206	1.2*DL + 1.6*WL(+Z) + 0.5*LL
207	1.2*DL + 1.6*WL(-Z) + 0.5*LL
208	1.2*DL + 1.0*EL(+X) + 0.5*LL
209	1.2*DL - 1.0*EL(+X) + 0.5*LL
210	1.2*DL + 1.0*EL(+Z) + 0.5*LL
211	1.2*DL - 1.0*EL(+Z) + 0.5*LL
212	0.9*DL + 1.6*WL(+Z)
213	0.9*DL + 1.6*WL(-Z)
214	0.9*DL + 1.6*WL(+X)
215	0.9*DL + 1.6*WL(-X)
216	0.9*DL + 1.0*EL(+X)
217	0.9*DL - 1.0*EL(+X)
218	0.9*DL + 1.0*EL(+Z)
219	0.9*DL - 1.0*EL(+Z)

6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0 for Permanent L/C
For foundation design a minimum factor of safety against overturning is consider 1.5 for Temporary L/C

6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m^2 at 1.2m depth below FGL
The neareast Bore Hole of **3860 REAGENTS – LIMESTONE EQUIPMENT** area is **BH-12 to BH-15**. The Net Safe Bearing Capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008-A02) is as mention below.

(i) 2m X 2m Net SBC at 1.5 below FGL is 980 KN/SQM
(ii) 3m X 3m Net SBC at 1.5 below FGL is 1120 KN/SQM
(iii) 4m X 4m Net SBC at 1.5 below FGL is 1250 KN/SQM
(iv) 5m X 5m Net SBC at 1.5 below FGL is 1200 KN/SQM

However for safe design purpose we have considered the Net Safe Bearing Capacity at founding level as **670 KN/m2**.

Hence, the Vertical Modulus of Subgrade Reaction (Ks) is consider as 67*3/0.025 = 8040 Mton/m2/m

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8.0 DESIGN OF FOUNDATION

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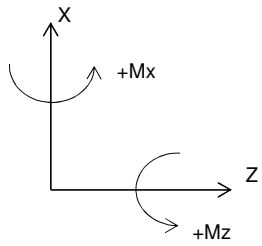
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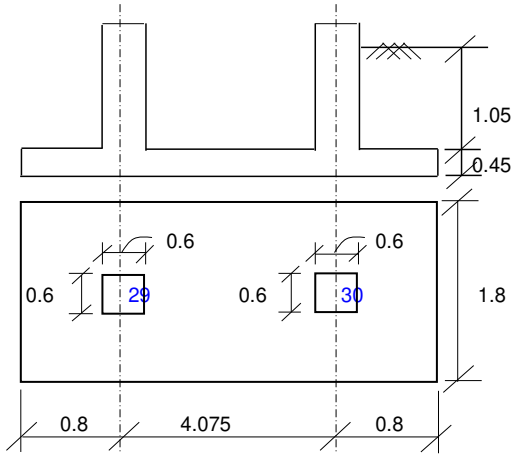
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8.1 DESIGN OF FOUNDATION MKD. F1 :

(For Staad Node. No. - 29 & 30)

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Projection from Node No. 29 =	0.8	m
Projection from Node No. 30 =	0.8	m
C/C Distance between Column =	4.075	m
Length of Foundation (Lx) =	5.675	m
Width of Foundation (Lz) =	1.8	m
Thickness of Raft =	0.45	m
Column size at Node No. 29 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.050	m
Column size at Node No. 30 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.050	m
Area , A = Lx*Lz =	10.215	m^2
Zxx = Lx*Lz^2/6 =	3.065	m^3
Zzz = Lz*Lx^2/6 =	9.662	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	5.675X1.8X0.45X2.5 =	11.492
Column =	(0.6X0.6+0.6X0.6)X1.05X2.5 =	1.89
Soil =	((5.675X1.8)-(0.6X0.6)-(0.6X0.6))X1.05X1.9 =	18.943
	Total =	32.325

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LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	29	0.399	9.729	0.695	1.821	-1.045	30	0.248	9.556	-0.116	-0.304	-0.65
102	29	1.114	17.265	-1.504	-3.94	-2.919	30	0.513	23.846	1.755	4.598	-1.344
103	29	-0.344	17.715	4.458	11.68	0.901	30	-0.135	18.878	-3.609	-9.455	0.354
104	29	0.454	15.173	1.983	5.195	-1.189	30	0.167	19.044	-0.023	-0.06	-0.438
105	29	0.323	15.926	0.58	1.52	-0.847	30	0.24	17.777	-1.426	-3.735	-0.63
106	29	0.754	16.412	1.321	3.462	-1.975	30	0.657	18.712	-0.691	-1.809	-1.721
107	29	0.023	14.688	1.242	3.253	-0.061	30	-0.249	18.109	-0.758	-1.986	0.653
108	29	0.452	15.847	1.29	3.381	-1.183	30	0.287	18.587	-0.717	-1.878	-0.753
109	29	0.325	15.253	1.273	3.334	-0.853	30	0.12	18.234	-0.732	-1.917	-0.315
110	29	0.405	15.469	1.358	3.557	-1.062	30	0.205	18.582	-0.649	-1.699	-0.537
111	29	0.372	15.631	1.205	3.158	-0.974	30	0.203	18.239	-0.8	-2.096	-0.531
112	29	0.326	5.335	1.352	3.542	-0.855	30	0.1	6.579	0.865	2.267	-0.261
113	29	0.152	6.339	-0.518	-1.357	-0.399	30	0.198	4.889	-1.005	-2.632	-0.518
114	29	0.726	6.986	0.47	1.231	-1.903	30	0.753	6.136	-0.025	-0.064	-1.972
115	29	-0.248	4.688	0.364	0.954	0.649	30	-0.455	5.332	-0.115	-0.3	1.193
116	29	0.323	6.234	0.429	1.124	-0.847	30	0.26	5.969	-0.06	-0.157	-0.681
117	29	0.155	5.441	0.405	1.061	-0.407	30	0.038	5.498	-0.079	-0.208	-0.098
118	29	0.262	5.73	0.519	1.358	-0.686	30	0.15	5.962	0.031	0.082	-0.394
119	29	0.217	5.945	0.316	0.827	-0.569	30	0.147	5.505	-0.17	-0.446	-0.386

CHECK FOR OVER TURNING

L/C : 115 *(Max. Tension at Node No. 29)*

O.T.M. = (0 X 4.875) + (0.954 + 0.3) = 1.254 Mton-m

R.M. = (5.332X0.8) + (32.325X0.9X2.8375) = 86.816 Mton-m

F.O.S. = 69.230 > 2 **SAFE**

L/C : 113 *(Max. Tension at Node No. 30)*

O.T.M. = (0 X 4.875) + (1.357 + 2.632) = 3.989

R.M. = (6.339X0.8) + (32.325X0.9X2.8375) = 87.621

F.O.S. = 21.970 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 112

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 2.258 Mton

Corresponding vertical load , ΣFy = 11.914 Mton

90% Wt. of Foundation , 0.9*Wf = 29.093 Mton

Total Downward load , V = 41.007 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 20.504 Mton

F.O.S. = 9.080 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.647	19.285	0.579	1.165	-1.695	5.726	4.379	O.K.
102	1.627	41.111	0.251	14.067	-4.263	10.036	4.342	O.K.
103	-0.479	36.593	0.849	4.595	1.255	7.632	5.862	O.K.
104	0.621	34.217	1.960	13.022	-1.627	8.393	4.636	O.K.
105	0.563	33.703	-0.846	1.556	-1.477	7.107	5.821	O.K.
106	1.411	35.124	0.630	6.339	-3.696	8.465	4.741	O.K.
107	-0.226	32.797	0.484	8.237	0.592	7.421	5.329	O.K.
108	0.739	34.434	0.573	7.086	-1.936	7.9	5.17	O.K.
109	0.445	33.487	0.541	7.491	-1.168	7.599	5.286	O.K.
110	0.610	34.051	0.709	8.201	-1.599	7.868	5.127	O.K.
111	0.575	33.870	0.405	6.376	-1.505	7.631	5.329	O.K.
112	0.426	11.914	2.217	8.344	-1.116	5.558	3.103	O.K.
113	0.350	11.228	-1.523	-6.943	-0.917	5.281	3.246	O.K.
114	1.479	13.122	0.445	-0.565	-3.875	5.772	3.126	O.K.
115	-0.703	10.020	0.249	1.966	1.842	4.95	3.341	O.K.
116	0.583	12.203	0.369	0.427	-1.528	4.902	3.816	O.K.
117	0.193	10.939	0.326	0.969	-0.505	4.5	3.97	O.K.
118	0.412	11.692	0.550	1.913	-1.080	4.859	3.759	O.K.
119	0.364	11.450	0.146	-0.516	-0.955	4.65	3.92	O.K.
				MAX. BASE PRESSURE FOR , L/C :	102	10.036		
				MIN. BASE PRESSURE FOR , L/C :	112		3.103	

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LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	29	0.559	13.621	0.973	2.549	-1.464	30	0.347	13.379	-0.162	-0.425	-0.91
202	29	1.555	26.499	-1.871	-4.902	-4.075	30	0.701	36.926	2.214	5.8	-1.835
203	29	-0.486	27.129	6.476	16.966	1.273	30	-0.207	29.971	-5.296	-13.875	0.542
204	29	0.611	14.752	2.721	7.129	-1.601	30	0.19	18.722	0.951	2.492	-0.497
205	29	0.333	16.359	-0.271	-0.71	-0.871	30	0.347	16.018	-2.041	-5.347	-0.908
206	29	1.251	17.394	1.31	3.432	-3.278	30	1.235	18.014	-0.473	-1.238	-3.234
207	29	-0.307	13.717	1.14	2.987	0.805	30	-0.698	16.727	-0.617	-1.616	1.83
208	29	0.592	16.121	1.242	3.254	-1.55	30	0.427	17.706	-0.531	-1.39	-1.119
209	29	0.352	14.989	1.208	3.165	-0.922	30	0.109	17.034	-0.559	-1.464	-0.286
210	29	0.504	15.401	1.37	3.589	-1.32	30	0.27	17.696	-0.401	-1.05	-0.708
211	29	0.44	15.709	1.08	2.83	-1.152	30	0.266	17.044	-0.689	-1.804	-0.697
212	29	0.498	7.953	2.122	5.558	-1.306	30	0.145	9.953	1.392	3.646	-0.379
213	29	0.22	9.559	-0.87	-2.281	-0.576	30	0.302	7.249	-1.6	-4.193	-0.79
214	29	1.138	10.595	0.71	1.861	-2.982	30	1.19	9.244	-0.032	-0.085	-3.117
215	29	-0.42	6.918	0.541	1.417	1.101	30	-0.743	7.957	-0.176	-0.462	1.947
216	29	0.479	9.322	0.643	1.683	-1.255	30	0.382	8.937	-0.09	-0.237	-1.001
217	29	0.239	8.19	0.609	1.594	-0.626	30	0.064	8.265	-0.118	-0.31	-0.168
218	29	0.391	8.602	0.77	2.019	-1.025	30	0.225	8.927	0.04	0.104	-0.59
219	29	0.327	8.91	0.481	1.259	-0.857	30	0.221	8.275	-0.248	-0.651	-0.58

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.906	27	0.811	1.631	-2.374	6.751	4.864
202	2.256	63.425	0.343	22.143	-5.91	13.593	5.153
203	-0.693	57.1	1.18	8.882	1.815	10.266	7.243
204	0.801	33.474	3.672	17.71	-2.098	8.959	3.924
205	0.68	32.377	-2.312	-6.752	-1.779	7.613	5.055
206	2.486	35.408	0.837	3.457	-6.512	9.113	4.148
207	-1.005	30.444	0.523	7.504	2.635	7.781	4.508
208	1.019	33.827	0.711	5.093	-2.669	7.874	5.078
209	0.461	32.023	0.649	5.868	-1.208	7.301	5.298
210	0.774	33.097	0.969	7.215	-2.028	7.813	4.996
211	0.706	32.753	0.391	3.746	-1.849	7.362	5.38
212	0.643	17.906	3.514	13.279	-1.685	6.841	2.993
213	0.522	16.808	-2.47	-11.181	-1.366	6.413	3.207
214	2.328	19.839	0.678	-0.977	-6.099	7.198	3.016
215	-1.163	14.875	0.365	3.072	3.048	5.933	3.308
216	0.861	18.259	0.553	0.662	-2.256	5.757	4.147
217	0.303	16.455	0.491	1.437	-0.794	5.183	4.368
218	0.616	17.529	0.81	2.785	-1.615	5.696	4.065
219	0.548	17.185	0.233	-0.686	-1.437	5.387	4.307

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	6.751	3.631	5.481	4.886	0.454	0.392	0.654	0.755
202	13.593	10.473	15.809	14.091	1.309	1.131	1.885	2.178
203	10.266	7.146	10.787	9.615	0.893	0.772	1.286	1.486
204	8.959	5.839	8.814	7.856	0.73	0.631	1.051	1.215
205	7.613	4.493	6.782	6.045	0.562	0.485	0.809	0.935
206	9.113	5.993	9.046	8.064	0.749	0.647	1.079	1.247
207	7.781	4.661	7.036	6.271	0.583	0.503	0.839	0.969
208	7.874	4.754	7.176	6.397	0.594	0.513	0.856	0.989
209	7.301	4.181	6.311	5.626	0.523	0.452	0.753	0.87
210	7.813	4.693	7.084	6.314	0.587	0.507	0.845	0.976
211	7.362	4.242	6.403	5.708	0.53	0.458	0.764	0.882
212	6.841	3.721	5.617	5.007	0.465	0.402	0.67	0.774
213	6.413	3.293	4.971	4.431	0.412	0.356	0.593	0.685
214	7.198	4.078	6.156	5.487	0.51	0.44	0.734	0.848
215	5.933	2.813	4.246	3.785	0.352	0.304	0.506	0.585
216	5.757	2.637	3.98	3.548	0.33	0.285	0.475	0.548
217	5.183	2.063	3.114	2.776	0.258	0.223	0.371	0.429
218	5.696	2.576	3.888	3.466	0.322	0.278	0.464	0.536
219	5.387	2.267	3.422	3.05	0.283	0.245	0.408	0.472
		MAX. =	**15.809**	**14.091**	**1.309**	**1.131**	**1.885**	**2.178**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 15.809 Mton-m

Φ*Kn = Mu/bd^2 deff. = 392 mm

Mu/bd^2 = 15.809X10^7/(1000X392^2) = 1.029 N/mm^2 < 6.9 N/mm^2

Pt = 0.0028

Min. Pt = 0.0018 Reqd. Ast = 0.0028X1000X392/100 = 10.98 cm^2

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.0034

ONE-WAY SHEAR CHECK

Vu = 14.091 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0034X14.091X10^4X392/15.809/10^7)X1000X392/10^4

= 37.897 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0034)X1000X392/10^4

= 39.371 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X392/10^4

= 67.254 Mton

HENCE, Vc = 37.897 Mton

Φt*Vc = 0.75X37.897 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 28.423 Mton **> 14.091Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(37.897X10^4+0.66X35^0.5X1000X392)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 117.638 Mton **> Vu= 14.091Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 1.309 Mton-m
Φ*Kn = Mu/bd^2 deff. = 392 mm
Mu/bd^2 = 1.309X10^7/(1000X392^2) = 0.085 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X392/100 = 7.06 cm^2
Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0034

ONE-WAY SHEAR CHECK

Vu = 1.131 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0034X1.131X10^4X392/1.309/10^7)X1000X392/10^4
= 37.873 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0034)X1000X392/10^4
= 39.371 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X392/10^4
= 67.254 Mton
HENCE, Vc = 37.873 Mton
Φt*Vc = 0.75X37.873 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 28.405 Mton **> 1.131Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(37.873X10^4+0.66X35^0.5X1000X392)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 117.636 Mton **> Vu= 1.131Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 1.885 Mton-m
Φ*Kn = Mu/bd^2 deff. = 392 mm
Mu/bd^2 = 1.885X10^7/(1000X392^2) = 0.123 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X392/100 = 7.06 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0026

ONE-WAY SHEAR CHECK

Vu = 2.178 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X2.178X10^4X392/1.885/10^7)X1000X392/10^4
= 37.89 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X392/10^4
= 38.838 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X392/10^4
= 67.254 Mton
HENCE, Vc = 37.89 Mton
Φt*Vc = 0.75X37.89 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 28.418 Mton **> 2.178Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(37.89X10^4+0.66X35^0.5X1000X392)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 117.637 Mton **> Vu= 2.178Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 10.215 m^2

Perimeter (bo) = 2 X ((0.6+0.392) + (0.6+0.392) + (0.6+0.392) + (0.6 + 0.392)) = 7.936 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 10.036 Mton/m^2

Vu = ((10.215 - (0.6X0.6) - (0.6X0.6))X10.036

= 95.292 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X7.936X10^3X392/10^4

= 607.345 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X7.936X10^3X392/10^4

= 938.625 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X392/(7.936X10^3))X1X35^0.5X(7.936X10^3)X392/10^4

= 531.876 Mton

HENCE, Vc = 531.876 Mton

Φt*Vn = Φt*Vc = 0.75X531.876 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

398.907 Mton **> 95.292Mton , Hence O.K.**

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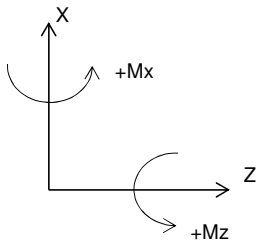
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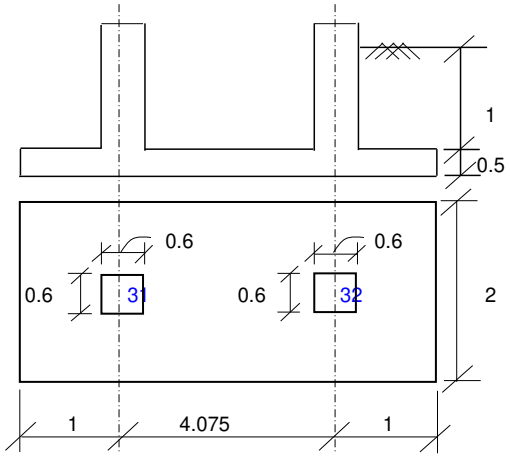
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8.2 DESIGN OF FOUNDATION MKD. F2 : *(For Staad Node. No. - 31 & 32)*

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Projection from Node No. 31 =	1	m
Projection from Node No. 32 =	1	m
C/C Distance between Column =	4.075	m
Length of Foundation (Lx) =	6.075	m
Width of Foundation (Lz) =	2	m
Thickness of Raft =	0.5	m
Column size at Node No. 31 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Column size at Node No. 32 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Area , A = Lx*Lz =	12.15	m^2
Zxx = Lx*Lz^2/6 =	4.05	m^3
Zzz = Lz*Lx^2/6 =	12.302	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	6.075X2X0.5X2.5 =	15.188
Column =	(0.6X0.6+0.6X0.6)X1X2.5 =	1.8
Soil =	((6.075X2)-(0.6X0.6)-(0.6X0.6))X1X1.9 =	21.717
	Total =	38.705

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L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	31	1.279	5.096	0.52	1.362	-3.351	32	0.139	7.449	0.133	0.348	-0.364
102	31	7.658	11.698	-1.587	-4.158	-20.064	32	0.983	20.14	2.15	5.633	-2.575
103	31	-3.05	11.358	4.313	11.3	7.991	32	-0.687	20.48	-3.766	-9.867	1.8
104	31	2.199	9.142	1.889	4.95	-5.76	32	0.129	17.873	0.143	0.374	-0.339
105	31	1.897	10.699	0.415	1.087	-4.97	32	0.162	16.316	-1.288	-3.375	-0.425
106	31	2.633	9.845	1.179	3.09	-6.898	32	-0.005	17.17	-0.556	-1.457	0.013
107	31	1.463	9.995	1.125	2.948	-3.832	32	0.297	17.02	-0.589	-1.544	-0.777
108	31	2.185	9.904	1.157	3.031	-5.724	32	0.184	17.111	-0.569	-1.49	-0.482
109	31	1.911	9.936	1.148	3.006	-5.006	32	0.107	17.079	-0.577	-1.512	-0.281
110	31	2.087	9.741	1.232	3.228	-5.467	32	0.146	17.274	-0.495	-1.297	-0.383
111	31	2.009	10.099	1.072	2.81	-5.263	32	0.145	16.916	-0.65	-1.704	-0.38
112	31	0.968	2.02	1.295	3.393	-2.537	32	0.061	5.507	1.034	2.709	-0.161
113	31	0.566	4.096	-0.671	-1.758	-1.484	32	0.105	3.431	-0.874	-2.29	-0.276
114	31	1.547	2.958	0.348	0.912	-4.054	32	-0.118	4.569	0.102	0.267	0.308
115	31	-0.013	3.158	0.276	0.723	0.033	32	0.284	4.369	0.058	0.151	-0.745
116	31	0.95	3.036	0.318	0.834	-2.489	32	0.134	4.491	0.085	0.224	-0.352
117	31	0.585	3.079	0.306	0.801	-1.532	32	0.032	4.448	0.074	0.194	-0.085
118	31	0.819	2.819	0.418	1.096	-2.146	32	0.084	4.708	0.183	0.481	-0.22
119	31	0.716	3.296	0.206	0.539	-1.875	32	0.083	4.231	-0.024	-0.062	-0.217

CHECK FOR OVER TURNING

L/C : 112 *(Max. Tension at Node No. 31)*

O.T.M. = (0 X 5.075) + (3.393 + 2.709) = 6.102 Mton-m

R.M. = (5.507X1) + (38.705X0.9X3.0375) = 111.317 Mton-m

F.O.S. = 18.240 > 2 **SAFE**

L/C : 113 *(Max. Tension at Node No. 32)*

O.T.M. = (0 X 5.075) + (1.758 + 2.29) = 4.048

R.M. = (4.096X1) + (38.705X0.9X3.0375) = 109.906

F.O.S. = 27.150 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 102

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 8.659 Mton

Corresponding vertical load , ΣFy = 31.838 Mton

90% Wt. of Foundation , 0.9*Wf = 34.835 Mton

Total Downward load , V = 66.673 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 33.337 Mton

F.O.S. = 3.850 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	1.418	12.545	0.653	6.504	-3.715	5.664	2.772	O.K.
102	8.641	31.838	0.563	18.676	-22.639	12.914	-1.302	O.K.
103	-3.737	31.838	0.547	20.019	9.791	9.851	1.761	O.K.
104	2.328	27.015	2.032	23.113	-6.099	8.794	2.024	O.K.
105	2.059	27.015	-0.873	9.157	-5.395	7.486	3.333	O.K.
106	2.628	27.015	0.623	16.558	-6.885	8.455	2.363	O.K.
107	1.760	27.015	0.536	15.717	-4.609	7.825	2.993	O.K.
108	2.369	27.015	0.588	16.225	-6.206	8.26	2.558	O.K.
109	2.018	27.015	0.571	16.048	-5.287	8.019	2.799	O.K.
110	2.233	27.015	0.737	17.279	-5.850	8.258	2.56	O.K.
111	2.154	27.015	0.422	14.996	-5.643	8.021	2.797	O.K.
112	1.029	7.527	2.329	13.207	-2.698	5.545	2.065	O.K.
113	0.671	7.527	-1.545	-5.403	-1.760	4.679	2.931	O.K.
114	1.429	7.527	0.450	4.461	-3.746	5.093	2.518	O.K.
115	0.271	7.527	0.334	3.341	-0.712	4.252	3.358	O.K.
116	1.084	7.527	0.403	4.023	-2.841	4.834	2.777	O.K.
117	0.617	7.527	0.380	3.784	-1.617	4.512	3.098	O.K.
118	0.903	7.527	0.601	5.426	-2.366	4.83	2.78	O.K.
119	0.799	7.527	0.182	2.382	-2.092	4.515	3.095	O.K.
				MAX. BASE PRESSURE FOR , L/C :	102	12.914		
				MIN. BASE PRESSURE FOR , L/C :	102		-1.302	

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LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	31	1.791	7.135	0.728	1.907	-4.691	32	0.195	10.429	0.186	0.488	-0.51
202	31	11.029	18.307	-1.969	-5.158	-28.895	32	1.379	32.054	2.728	7.146	-3.613
203	31	-3.962	17.831	6.291	16.482	10.382	32	-0.959	32.53	-5.555	-14.553	2.513
204	31	2.369	7.671	2.618	6.86	-6.206	32	0.136	17.03	1.216	3.185	-0.357
205	31	1.726	10.992	-0.527	-1.382	-4.521	32	0.207	13.709	-1.837	-4.814	-0.541
206	31	3.295	9.171	1.103	2.89	-8.634	32	-0.15	15.529	-0.276	-0.722	0.394
207	31	0.799	9.491	0.988	2.588	-2.094	32	0.493	15.209	-0.346	-0.907	-1.291
208	31	2.308	9.3	1.055	2.763	-6.048	32	0.244	15.4	-0.303	-0.794	-0.64
209	31	1.786	9.362	1.037	2.716	-4.68	32	0.098	15.338	-0.319	-0.836	-0.258
210	31	2.121	8.99	1.198	3.137	-5.558	32	0.172	15.71	-0.163	-0.427	-0.451
211	31	1.973	9.672	0.894	2.341	-5.17	32	0.17	15.028	-0.459	-1.202	-0.446
212	31	1.473	2.926	2.041	5.347	-3.858	32	0.09	8.365	1.646	4.313	-0.236
213	31	0.83	6.247	-1.105	-2.895	-2.173	32	0.16	5.043	-1.407	-3.686	-0.42
214	31	2.399	4.427	0.526	1.377	-6.286	32	-0.197	6.864	0.155	0.406	0.515
215	31	-0.097	4.747	0.41	1.075	0.254	32	0.447	6.544	0.084	0.221	-1.17
216	31	1.412	4.556	0.477	1.25	-3.7	32	0.198	6.735	0.128	0.335	-0.519
217	31	0.89	4.618	0.459	1.203	-2.332	32	0.052	6.673	0.112	0.293	-0.137
218	31	1.225	4.246	0.62	1.624	-3.21	32	0.126	7.045	0.268	0.701	-0.33
219	31	1.077	4.928	0.316	0.828	-2.822	32	0.124	6.363	-0.028	-0.074	-0.325

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	1.986	17.564	0.914	9.107	-5.201	6.656	2.607
202	12.408	50.361	0.759	29.998	-32.508	17.796	-3.135
203	-4.921	50.361	0.736	31.878	12.895	13.106	1.555
204	2.505	24.701	3.834	29.114	-6.563	9.206	1.231
205	1.933	24.701	-2.364	-0.66	-5.062	6.522	3.915
206	3.145	24.7	0.827	15.122	-8.24	8.482	1.955
207	1.292	24.7	0.642	13.331	-3.385	7.138	3.299
208	2.552	24.7	0.752	14.398	-6.688	8.04	2.397
209	1.884	24.7	0.718	14.056	-4.938	7.58	2.857
210	2.293	24.7	1.035	16.402	-6.009	8.036	2.402
211	2.143	24.7	0.435	12.052	-5.616	7.585	2.852
212	1.563	11.291	3.687	20.742	-4.094	6.812	1.418
213	0.99	11.29	-2.512	-9.034	-2.593	5.489	2.74
214	2.202	11.291	0.681	6.748	-5.771	6.088	2.141
215	0.35	11.291	0.494	4.957	-0.916	4.744	3.486
216	1.61	11.291	0.605	6.025	-4.219	5.646	2.583
217	0.942	11.291	0.571	5.683	-2.469	5.186	3.043
218	1.351	11.291	0.888	8.028	-3.54	5.642	2.588
219	1.201	11.291	0.288	3.678	-3.147	5.191	3.039

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	6.656	3.506	5.292	4.549	0.859	0.905	0.859	0.905
202	17.796	14.646	22.107	19.003	3.588	3.779	3.588	3.779
203	13.106	9.956	15.028	12.918	2.439	2.569	2.439	2.569
204	9.206	6.056	9.141	7.858	1.484	1.562	1.484	1.562
205	6.522	3.372	5.09	4.375	0.826	0.87	0.826	0.87
206	8.482	5.332	8.048	6.918	1.306	1.376	1.306	1.376
207	7.138	3.988	6.02	5.174	0.977	1.029	0.977	1.029
208	8.04	4.89	7.381	6.345	1.198	1.262	1.198	1.262
209	7.58	4.43	6.687	5.748	1.085	1.143	1.085	1.143
210	8.036	4.886	7.375	6.34	1.197	1.261	1.197	1.261
211	7.585	4.435	6.694	5.754	1.087	1.144	1.087	1.144
212	6.812	3.662	5.528	4.751	0.897	0.945	0.897	0.945
213	5.489	2.339	3.531	3.035	0.573	0.603	0.573	0.603
214	6.088	2.938	4.435	3.812	0.72	0.758	0.72	0.758
215	4.744	1.594	2.406	2.068	0.391	0.411	0.391	0.411
216	5.646	2.496	3.768	3.239	0.612	0.644	0.612	0.644
217	5.186	2.036	3.073	2.642	0.499	0.525	0.499	0.525
218	5.642	2.492	3.762	3.233	0.611	0.643	0.611	0.643
219	5.191	2.041	3.081	2.648	0.5	0.527	0.5	0.527
		MAX. =	**22.107**	**19.003**	**3.588**	**3.779**	**3.588**	**3.779**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 22.107 Mton-m

Φ*Kn = Mu/bd^2 deff. = 440 mm

Mu/bd^2 = 22.107X10^7/(1000X440^2) = 1.142 N/mm^2 < 6.9 N/mm^2

Pt = 0.0031

Min. Pt = 0.0018 Reqd. Ast = 0.0031X1000X440/100 = 13.64 cm^2

Provided **20** mm @ **200** C/C Provide Ast = 15.71 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.0036

ONE-WAY SHEAR CHECK

Vu = 19.003 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0036X19.003X10^4X440/22.107/10^7)X1000X440/10^4

= 42.668 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0036)X1000X440/10^4

= 44.342 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X440/10^4

= 75.489 Mton

HENCE, Vc = 42.668 Mton

Φt*Vc = 0.75X42.668 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 32.001 Mton **> 19.003Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.668X10^4+0.66X35^0.5X1000X440)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 132.052 Mton **> Vu= 19.003Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 3.588 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 3.588X10^7/(1000X442^2) = 0.184 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.0023

ONE-WAY SHEAR CHECK

Vu = 3.779 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0023X3.779X10^4X442/3.588/10^7)X1000X442/10^4
= 42.643 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0023)X1000X442/10^4
= 43.567 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.643 Mton
Φt*Vc = 0.75X42.643 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.982 Mton **> 3.779Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.643X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.636 Mton **> Vu= 3.779Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 3.588 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 3.588X10^7/(1000X442^2) = 0.184 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0023

ONE-WAY SHEAR CHECK

Vu = 3.779 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0023X3.779X10^4X442/3.588/10^7)X1000X442/10^4
= 42.643 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0023)X1000X442/10^4
= 43.567 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.643 Mton
Φt*Vc = 0.75X42.643 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.982 Mton **> 3.779Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.643X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.636 Mton **> Vu= 3.779Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 12.15 m^2

Perimeter (bo) = 2 X ((0.6+0.442) + (0.6+0.442) + (0.6+0.442) + (0.6 + 0.442)) = 8.336 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 12.914 Mton/m^2

Vu = ((12.15 - (0.6X0.6) - (0.6X0.6))X12.914

= 147.607 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X8.336X10^3X442/10^4

= 719.33 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X8.336X10^3X442/10^4

= 1111.691 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X442/(8.336X10^3))X1X35^0.5X(8.336X10^3)X442/10^4

= 649.636 Mton

HENCE, Vc = 649.636 Mton

Φt*Vn = Φt*Vc = 0.75X649.636 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

487.227 Mton **> 147.607Mton , Hence O.K.**

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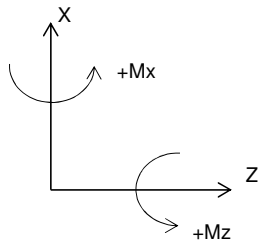
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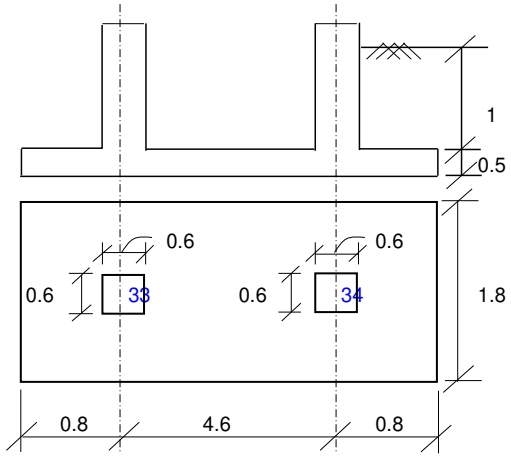
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8.3 DESIGN OF FOUNDATION MKD. F3 : *(For Staad Node. No. - 33 & 34)*

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Projection from Node No. 33 =	0.8	m
Projection from Node No. 34 =	0.8	m
C/C Distance between Column =	4.6	m
Length of Foundation (Lx) =	6.2	m
Width of Foundation (Lz) =	1.8	m
Thickness of Raft =	0.5	m
Column size at Node No. 33 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Column size at Node No. 34 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Area , A = Lx*Lz =	11.16	m^2
Zxx = Lx*Lz^2/6 =	3.348	m^3
Zzz = Lz*Lx^2/6 =	11.532	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	6.2X1.8X0.5X2.5 =	13.95
Column =	(0.6X0.6+0.6X0.6)X1X2.5 =	1.8
Soil =	((6.2X1.8)-(0.6X0.6)-(0.6X0.6))X1X1.9 =	19.836
	Total =	35.586

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LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	33	0.274	5.71	0.499	1.307	-0.718	34	0.259	6.967	-0.065	-0.17	-0.679
102	33	2.825	13.347	-1.991	-5.216	-7.401	34	0.302	17.868	1.805	4.729	-0.791
103	33	-2.215	19.081	5.115	13.401	5.803	34	0.29	17.9	-3.859	-10.11	-0.76
104	33	0.264	13.063	1.749	4.583	-0.692	34	0.33	15.697	-0.313	-0.819	-0.865
105	33	0.33	14.113	0.843	2.209	-0.865	34	0.243	14.613	-1.261	-3.302	-0.637
106	33	0.699	13.199	1.303	3.414	-1.83	34	0.823	15.228	-0.767	-2.01	-2.156
107	33	-0.104	13.977	1.29	3.378	0.272	34	-0.25	15.082	-0.806	-2.112	0.654
108	33	0.375	13.434	1.294	3.391	-0.982	34	0.322	15.158	-0.785	-2.058	-0.843
109	33	0.22	13.742	1.298	3.402	-0.575	34	0.252	15.152	-0.788	-2.063	-0.659
110	33	0.298	13.461	1.356	3.553	-0.782	34	0.309	15.266	-0.723	-1.894	-0.81
111	33	0.296	13.715	1.236	3.239	-0.776	34	0.264	15.044	-0.85	-2.227	-0.692
112	33	0.12	2.726	0.903	2.367	-0.315	34	0.213	4.903	0.593	1.554	-0.559
113	33	0.208	4.126	-0.305	-0.798	-0.546	34	0.097	3.457	-0.671	-1.758	-0.255
114	33	0.699	2.907	0.308	0.808	-1.832	34	0.87	4.277	-0.013	-0.034	-2.28
115	33	-0.371	3.945	0.29	0.761	0.971	34	-0.56	4.083	-0.065	-0.17	1.466
116	33	0.268	3.221	0.297	0.777	-0.702	34	0.202	4.184	-0.038	-0.099	-0.53
117	33	0.061	3.631	0.302	0.792	-0.159	34	0.109	4.176	-0.04	-0.106	-0.284
118	33	0.166	3.257	0.379	0.993	-0.434	34	0.186	4.329	0.046	0.12	-0.486
119	33	0.163	3.595	0.22	0.575	-0.427	34	0.125	4.032	-0.124	-0.324	-0.328

CHECK FOR OVER TURNING

L/C : 112 *(Max. Tension at Node No. 33)*

O.T.M. = (0 X 5.4) + (2.367 + 1.554) = 3.921 Mton-m

R.M. = (4.903X0.8) + (35.586X0.9X3.1) = 103.207 Mton-m

F.O.S. = 26.320 > 2 **SAFE**

L/C : 113 *(Max. Tension at Node No. 34)*

O.T.M. = (0 X 5.4) + (0.798 + 1.758) = 2.556

R.M. = (4.126X0.8) + (35.586X0.9X3.1) = 102.586

F.O.S. = 40.140 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 102

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 3.133 Mton

Corresponding vertical load , ΣFy = 31.215 Mton

90% Wt. of Foundation , 0.9*Wf = 32.027 Mton

Total Downward load , V = 63.242 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 31.621 Mton

F.O.S. = 10.090 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.533	12.677	0.434	4.028	-1.397	5.091	3.558	O.K.
102	3.127	31.215	-0.186	9.911	-8.192	9.292	2.679	O.K.
103	-1.925	36.981	1.256	0.575	5.043	8.059	4.946	O.K.
104	0.594	28.760	1.436	9.822	-1.557	7.083	4.449	O.K.
105	0.573	28.726	-0.418	0.057	-1.502	6.216	5.309	O.K.
106	1.522	28.427	0.536	6.071	-3.986	7.453	4.019	O.K.
107	-0.354	29.059	0.484	3.808	0.926	6.399	5.186	O.K.
108	0.697	28.592	0.509	5.298	-1.825	6.755	4.746	O.K.
109	0.472	28.894	0.510	4.582	-1.234	6.544	5.012	O.K.
110	0.607	28.727	0.633	5.811	-1.592	6.742	4.783	O.K.
111	0.560	28.759	0.386	4.069	-1.468	6.557	4.974	O.K.
112	0.333	7.629	1.496	8.928	-0.874	4.908	2.837	O.K.
113	0.305	7.583	-0.976	-4.095	-0.801	4.463	3.274	O.K.
114	1.569	7.184	0.295	3.925	-4.112	5.401	2.264	O.K.
115	-0.931	8.028	0.225	0.908	2.437	4.715	3.101	O.K.
116	0.470	7.405	0.259	2.893	-1.232	4.471	3.233	O.K.
117	0.170	7.807	0.262	1.940	-0.443	4.189	3.588	O.K.
118	0.352	7.586	0.425	3.579	-0.920	4.454	3.283	O.K.
119	0.288	7.627	0.096	1.256	-0.755	4.207	3.538	O.K.
MAX. BASE PRESSURE FOR , L/C : 102						9.292		
MIN. BASE PRESSURE FOR , L/C : 114							2.264	

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LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	33	0.384	7.994	0.699	1.83	-1.005	34	0.363	9.754	-0.091	-0.238	-0.95
202	33	3.964	21.837	-2.468	-6.467	-10.386	34	0.434	28.29	2.238	5.865	-1.137
203	33	-3.092	29.865	7.48	19.597	8.1	34	0.417	28.335	-5.691	-14.911	-1.093
204	33	0.274	10.984	2.097	5.493	-0.718	34	0.422	14.976	0.452	1.185	-1.106
205	33	0.415	13.224	0.164	0.429	-1.087	34	0.237	12.662	-1.57	-4.114	-0.62
206	33	1.2	11.274	1.145	2.999	-3.145	34	1.473	13.974	-0.517	-1.356	-3.86
207	33	-0.512	12.935	1.116	2.924	1.341	34	-0.815	13.664	-0.601	-1.574	2.134
208	33	0.492	11.811	1.126	2.951	-1.29	34	0.396	13.825	-0.557	-1.459	-1.038
209	33	0.196	12.397	1.134	2.972	-0.514	34	0.262	13.813	-0.561	-1.47	-0.687
210	33	0.346	11.862	1.244	3.26	-0.907	34	0.372	14.031	-0.438	-1.148	-0.975
211	33	0.342	12.346	1.016	2.663	-0.897	34	0.286	13.607	-0.68	-1.782	-0.75
212	33	0.176	4.019	1.416	3.709	-0.462	34	0.326	7.427	0.953	2.496	-0.854
213	33	0.317	6.259	-0.517	-1.355	-0.831	34	0.14	5.114	-1.07	-2.803	-0.368
214	33	1.103	4.309	0.464	1.214	-2.889	34	1.377	6.426	-0.017	-0.044	-3.608
215	33	-0.609	5.97	0.435	1.139	1.597	34	-0.911	6.115	-0.1	-0.262	2.387
216	33	0.395	4.846	0.445	1.166	-1.034	34	0.3	6.276	-0.056	-0.148	-0.786
217	33	0.099	5.432	0.453	1.187	-0.258	34	0.166	6.264	-0.06	-0.159	-0.435
218	33	0.249	4.897	0.563	1.475	-0.651	34	0.276	6.482	0.063	0.164	-0.723
219	33	0.245	5.381	0.335	0.878	-0.641	34	0.19	6.058	-0.18	-0.47	-0.498

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.747	17.748	0.608	5.64	-1.955	5.852	3.706
202	4.398	50.127	-0.23	14.24	-11.523	12.357	3.004
203	-2.675	58.2	1.789	1.167	7.007	10.598	6.21
204	0.696	25.96	2.549	15.86	-1.824	7.435	3.595
205	0.652	25.886	-1.406	-4.978	-1.707	6.45	4.567
206	2.673	25.248	0.628	7.853	-7.005	8.224	2.678
207	-1.327	26.599	0.515	3.027	3.475	6.873	4.272
208	0.888	25.636	0.569	6.124	-2.328	6.712	4.259
209	0.458	26.21	0.573	4.759	-1.201	6.309	4.766
210	0.718	25.893	0.806	7.101	-1.882	6.687	4.331
211	0.628	25.953	0.336	3.781	-1.647	6.334	4.694
212	0.502	11.446	2.369	14.043	-1.316	5.825	2.604
213	0.457	11.373	-1.587	-6.792	-1.199	5.155	3.261
214	2.48	10.735	0.447	6.039	-6.497	6.615	1.686
215	-1.52	12.085	0.335	1.211	3.984	5.567	2.977
216	0.695	11.122	0.389	4.307	-1.82	5.102	3.268
217	0.265	11.696	0.393	2.942	-0.693	4.699	3.775
218	0.525	11.379	0.626	5.285	-1.374	5.077	3.34
219	0.435	11.439	0.155	1.965	-1.139	4.724	3.703

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	5.852	2.702	5.404	4.21	0.338	0.157	0.486	0.427
202	12.357	9.207	18.414	14.345	1.151	0.534	1.657	1.455
203	10.598	7.448	14.896	11.604	0.931	0.432	1.341	1.177
204	7.435	4.285	8.57	6.676	0.536	0.249	0.771	0.677
205	6.45	3.3	6.6	5.141	0.413	0.191	0.594	0.521
206	8.224	5.074	10.148	7.905	0.634	0.294	0.913	0.802
207	6.873	3.723	7.446	5.8	0.465	0.216	0.67	0.588
208	6.712	3.562	7.124	5.55	0.445	0.207	0.641	0.563
209	6.309	3.159	6.318	4.922	0.395	0.183	0.569	0.499
210	6.687	3.537	7.074	5.511	0.442	0.205	0.637	0.559
211	6.334	3.184	6.368	4.961	0.398	0.185	0.573	0.503
212	5.825	2.675	5.35	4.168	0.334	0.155	0.482	0.423
213	5.155	2.005	4.01	3.124	0.251	0.116	0.361	0.317
214	6.615	3.465	6.93	5.398	0.433	0.201	0.624	0.547
215	5.567	2.417	4.834	3.766	0.302	0.14	0.435	0.382
216	5.102	1.952	3.904	3.041	0.244	0.113	0.351	0.308
217	4.699	1.549	3.098	2.413	0.194	0.09	0.279	0.245
218	5.077	1.927	3.854	3.002	0.241	0.112	0.347	0.304
219	4.724	1.574	3.148	2.452	0.197	0.091	0.283	0.249
		MAX. =	**18.414**	**14.345**	**1.151**	**0.534**	**1.657**	**1.455**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 18.414 Mton-m

Φ*Kn = Mu/bd^2 deff. = 442 mm

Mu/bd^2 = 18.414X10^7/(1000X442^2) = 0.943 N/mm^2 < 6.9 N/mm^2

Pt = 0.0025

Min. Pt = 0.0018 Reqd. Ast = 0.0025X1000X442/100 = 11.05 cm^2

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.003

ONE-WAY SHEAR CHECK

Vu = 14.345 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.003X14.345X10^4X442/18.414/10^7)X1000X442/10^4

= 42.615 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.003)X1000X442/10^4

= 44.093 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X442/10^4

= 75.832 Mton

HENCE, Vc = 42.615 Mton

Φt*Vc = 0.75X42.615 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 31.961 Mton **> 14.345Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.615X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 132.634 Mton **> Vu= 14.345Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 1.151 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 1.151X10^7/(1000X442^2) = 0.059 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.003

ONE-WAY SHEAR CHECK

Vu = 0.534 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.003X0.534X10^4X442/1.151/10^7)X1000X442/10^4
= 42.301 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.003)X1000X442/10^4
= 44.093 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.301 Mton
Φt*Vc = 0.75X42.301 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.726 Mton **> 0.534Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.301X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.61 Mton **> Vu= 0.534Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 1.657 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 1.657X10^7/(1000X442^2) = 0.085 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0023

ONE-WAY SHEAR CHECK

Vu = 1.455 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0023X1.455X10^4X442/1.657/10^7)X1000X442/10^4
= 42.509 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0023)X1000X442/10^4
= 43.567 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.509 Mton
Φt*Vc = 0.75X42.509 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.882 Mton **> 1.455Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.509X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.626 Mton **> Vu= 1.455Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 11.16 m^2

Perimeter (bo) = 2 X ((0.6+0.442) + (0.6+0.442) + (0.6+0.442) + (0.6 + 0.442)) = 8.336 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 9.292 Mton/m^2

Vu = ((11.16 - (0.6X0.6) - (0.6X0.6))X9.292

= 97.008 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X8.336X10^3X442/10^4

= 719.33 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X8.336X10^3X442/10^4

= 1111.691 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X442/(8.336X10^3))X1X35^0.5X(8.336X10^3)X442/10^4

= 649.636 Mton

HENCE, Vc = 649.636 Mton

Φt*Vn = Φt*Vc = 0.75X649.636 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

487.227 Mton **> 97.008Mton , Hence O.K.**

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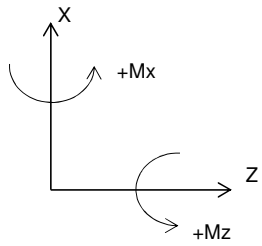
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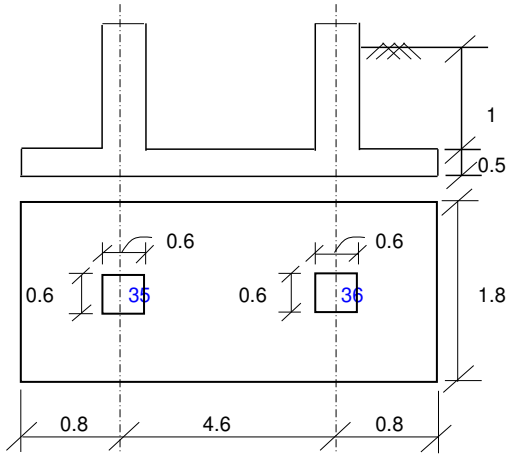
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8.4 DESIGN OF FOUNDATION MKD. F4 : *(For Staad Node. No. - 35 & 36)*

Net Safe Bearing Capacity of Soil =	67	Mton/m^2	*(For BH-12 to BH-15)*
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	67 + 1.5 X 1.9 =	69.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Projection from Node No. 35 =	0.8	m
Projection from Node No. 36 =	0.8	m
C/C Distance between Column =	4.6	m
Length of Foundation (Lx) =	6.2	m
Width of Foundation (Lz) =	1.8	m
Thickness of Raft =	0.5	m
Column size at Node No. 35 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Column size at Node No. 36 :		
Length along X-Direction =	0.6	m
Length along Z-Direction =	0.6	m
Height =	1.000	m
Area , A = Lx*Lz =	11.16	m^2
Zxx = Lx*Lz^2/6 =	3.348	m^3
Zzz = Lz*Lx^2/6 =	11.532	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	6.2X1.8X0.5X2.5 =	13.95
Column =	(0.6X0.6+0.6X0.6)X1X2.5 =	1.8
Soil =	((6.2X1.8)-(0.6X0.6)-(0.6X0.6))X1X1.9 =	19.836
	Total =	35.586

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LOAD FROM STAAD OUTPUT FOR STABILITY & STRESS CHECK : (Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
101	35	0.275	4.417	0.046	0.121	-0.72	36	-0.064	3.556	-0.091	-0.238	0.168
102	35	4.248	12.873	-1.928	-5.051	-11.13	36	0.924	9.258	2.628	6.885	-2.421
103	35	-3.71	7.137	4.156	10.889	9.72	36	-0.944	9.228	-4.894	-12.822	2.473
104	35	0.238	8.186	1.208	3.164	-0.622	36	-0.038	8.226	-0.519	-1.36	0.099
105	35	0.304	9.03	0.486	1.274	-0.795	36	-0.009	7.416	-1.226	-3.211	0.024
106	35	0.673	9.164	0.737	1.932	-1.762	36	-0.137	7.582	-0.948	-2.484	0.358
107	35	-0.132	8.052	0.956	2.506	0.345	36	0.09	8.061	-0.797	-2.087	-0.235
108	35	0.349	8.792	0.827	2.167	-0.915	36	-0.024	7.788	-0.88	-2.305	0.062
109	35	0.192	8.424	0.867	2.271	-0.502	36	-0.024	7.854	-0.865	-2.267	0.062
110	35	0.272	8.569	0.867	2.271	-0.711	36	-0.029	7.876	-0.852	-2.231	0.075
111	35	0.269	8.647	0.827	2.167	-0.706	36	-0.018	7.766	-0.894	-2.341	0.048
112	35	0.121	2.087	0.509	1.333	-0.317	36	-0.057	2.674	0.416	1.091	0.15
113	35	0.209	3.213	-0.453	-1.188	-0.548	36	-0.019	1.594	-0.526	-1.377	0.051
114	35	0.701	3.391	-0.118	-0.31	-1.837	36	-0.189	1.815	-0.156	-0.408	0.496
115	35	-0.371	1.909	0.174	0.455	0.972	36	0.113	2.453	0.046	0.122	-0.295
116	35	0.27	2.895	0.001	0.003	-0.707	36	-0.038	2.09	-0.064	-0.169	0.101
117	35	0.06	2.405	0.054	0.142	-0.157	36	-0.038	2.178	-0.045	-0.117	0.101
118	35	0.166	2.598	0.054	0.142	-0.436	36	-0.045	2.207	-0.027	-0.07	0.119
119	35	0.164	2.703	0.001	0.003	-0.429	36	-0.031	2.06	-0.083	-0.216	0.082

CHECK FOR OVER TURNING

L/C : 115 *(Max. Tension at Node No. 35)*

O.T.M. = (0 X 5.4) + (0.455 + 0.122) = 0.577 Mton-m

R.M. = (2.453X0.8) + (35.586X0.9X3.1) = 101.247 Mton-m

F.O.S. = 175.470 > 2 **SAFE**

L/C : 113 *(Max. Tension at Node No. 36)*

O.T.M. = (0 X 5.4) + (1.188 + 1.377) = 2.565

R.M. = (3.213X0.8) + (35.586X0.9X3.1) = 101.855

F.O.S. = 39.710 > 2 **SAFE**

CHECK FOR SLIDING (BUOYANT CONDITION)

L/C : 102

Max. Horizontal shear , ΣFx^2+ΣFz^2 = 5.219 Mton

Corresponding vertical load , ΣFy = 22.131 Mton

90% Wt. of Foundation , 0.9*Wf = 32.027 Mton

Total Downward load , V = 54.158 Mton

Horizontal frictional coeff. μ = 0.5 *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.3.2)*

Resisting shear force = μ * V = 27.079 Mton

F.O.S. = 5.190 > 1.5 **SAFE** *(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.2)*

BASE PRESSURE :

						BASE PRESSURE		
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.	REMARKS
101	0.211	7.973	-0.045	-2.097	-0.552	4.25	3.556	O.K.
102	5.172	22.131	0.700	-6.481	-13.551	9.781	0.562	O.K.
103	-4.654	16.365	-0.738	2.876	12.193	8.546	0.764	O.K.
104	0.200	16.412	0.689	1.896	-0.523	4.98	4.339	O.K.
105	0.295	16.446	-0.740	-5.649	-0.771	5.383	3.942	O.K.
106	0.536	16.746	-0.211	-4.191	-1.404	5.472	3.906	O.K.
107	-0.042	16.113	0.159	0.440	0.110	4.704	4.562	O.K.
108	0.325	16.580	-0.053	-2.447	-0.853	5.141	4.207	O.K.
109	0.168	16.278	0.002	-1.307	-0.440	4.892	4.403	O.K.
110	0.243	16.445	0.015	-1.554	-0.636	4.987	4.338	O.K.
111	0.251	16.413	-0.067	-2.200	-0.658	5.047	4.272	O.K.
112	0.064	4.761	0.925	3.774	-0.167	3.992	3.238	O.K.
113	0.190	4.807	-0.979	-6.289	-0.497	4.313	2.926	O.K.
114	0.512	5.206	-0.274	-4.343	-1.341	4.432	2.878	O.K.
115	-0.258	4.362	0.220	1.828	0.677	3.94	3.219	O.K.
116	0.232	4.985	-0.063	-2.018	-0.606	3.991	3.279	O.K.
117	0.022	4.583	0.009	-0.497	-0.056	3.659	3.54	O.K.
118	0.121	4.805	0.027	-0.827	-0.317	3.786	3.453	O.K.
119	0.133	4.763	-0.082	-1.692	-0.347	3.866	3.365	O.K.
				MAX. BASE PRESSURE FOR , L/C :	102	9.781		
				MIN. BASE PRESSURE FOR , L/C :	102		0.562	

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LOAD FROM STAAD OUTPUT FOR STRENGTH DESIGN : (Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

L/C	NODE	Fx	Fy	Fz	Mx	Mz	NODE	Fx	Fy	Fz	Mx	Mz
201	35	0.385	6.184	0.064	0.169	-1.009	36	-0.09	4.979	-0.127	-0.334	0.235
202	35	5.945	19.699	-2.379	-6.232	-15.577	36	1.31	14.667	3.367	8.82	-3.432
203	35	-5.196	11.668	6.139	16.083	13.613	36	-1.305	14.625	-7.164	-18.77	3.42
204	35	0.257	7.194	1.359	3.56	-0.672	36	-0.08	7.975	0.123	0.323	0.21
205	35	0.397	8.995	-0.18	-0.473	-1.041	36	-0.019	6.247	-1.384	-3.626	0.051
206	35	1.185	9.28	0.356	0.932	-3.104	36	-0.291	6.6	-0.792	-2.074	0.763
207	35	-0.531	6.909	0.823	2.156	1.39	36	0.192	7.621	-0.469	-1.228	-0.503
208	35	0.477	8.445	0.551	1.444	-1.25	36	-0.05	7.048	-0.644	-1.688	0.13
209	35	0.177	7.745	0.627	1.643	-0.464	36	-0.05	7.174	-0.616	-1.614	0.13
210	35	0.329	8.02	0.627	1.643	-0.862	36	-0.06	7.216	-0.59	-1.546	0.157
211	35	0.325	8.17	0.551	1.444	-0.851	36	-0.04	7.006	-0.67	-1.756	0.104
212	35	0.177	3.075	0.811	2.125	-0.464	36	-0.088	4.065	0.672	1.76	0.231
213	35	0.318	4.876	-0.728	-1.908	-0.833	36	-0.027	2.337	-0.836	-2.189	0.071
214	35	1.105	5.161	-0.192	-0.504	-2.895	36	-0.299	2.69	-0.244	-0.638	0.784
215	35	-0.61	2.79	0.275	0.72	1.598	36	0.184	3.711	0.08	0.209	-0.482
216	35	0.397	4.325	0.003	0.009	-1.041	36	-0.058	3.138	-0.096	-0.251	0.151
217	35	0.097	3.625	0.079	0.208	-0.255	36	-0.058	3.264	-0.068	-0.178	0.151
218	35	0.25	3.9	0.079	0.208	-0.654	36	-0.068	3.306	-0.042	-0.11	0.177
219	35	0.245	4.05	0.003	0.009	-0.643	36	-0.048	3.096	-0.122	-0.319	0.125

STRESS CALCULATION FOR STRENGTH DESIGN :

						BASE PRESSURE	
L/C	Σ Fx	Σ Fy	Σ Fz	Σ Mx	Σ Mz	P max.	P min.
201	0.295	11.163	-0.063	-2.937	-0.774	4.675	3.703
202	7.255	34.366	0.988	-8.986	-19.009	12.725	-0.189
203	-6.501	26.293	-1.025	4.114	17.033	10.989	0.1
204	0.177	15.169	1.482	5.679	-0.462	5.178	3.917
205	0.378	15.242	-1.564	-10.419	-0.99	5.754	3.355
206	0.894	15.88	-0.436	-7.306	-2.341	5.944	3.279
207	-0.339	14.53	0.354	2.566	0.887	4.978	4.003
208	0.427	15.493	-0.093	-3.457	-1.12	5.211	3.943
209	0.127	14.919	0.011	-1.284	-0.334	4.737	4.314
210	0.269	15.236	0.037	-1.752	-0.705	4.916	4.191
211	0.285	15.176	-0.119	-2.989	-0.747	5.031	4.066
212	0.089	7.14	1.483	6.162	-0.233	4.432	3.225
213	0.291	7.213	-1.564	-9.937	-0.762	4.924	2.746
214	0.806	7.851	-0.436	-6.825	-2.111	5.115	2.67
215	-0.426	6.501	0.355	3.047	1.116	4.369	3.174
216	0.339	7.463	-0.093	-2.972	-0.89	4.381	3.334
217	0.039	6.889	0.011	-0.8	-0.104	3.906	3.706
218	0.182	7.206	0.037	-1.268	-0.477	4.087	3.582
219	0.197	7.146	-0.119	-2.504	-0.518	4.201	3.457

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DESIGN OF BASE SLAB :

			SPAN PORTION		CANTILEVER PORTION (FOR BOTTOM R/F)			
	PRESSURE (Mton/m^2)		(FOR TOP REINF.)		LONG DIRECTION		CROSS DIRECTION	
L/C	GROSS	NET	B.M.	S.F.	B.M.	S.F.	B.M.	S.F.
201	4.675	1.525	3.05	2.376	0.191	0.088	0.275	0.241
202	12.725	9.575	19.15	14.918	1.197	0.555	1.724	1.513
203	10.989	7.839	15.678	12.213	0.98	0.455	1.411	1.239
204	5.178	2.028	4.056	3.16	0.254	0.118	0.365	0.32
205	5.754	2.604	5.208	4.057	0.326	0.151	0.469	0.411
206	5.944	2.794	5.588	4.353	0.349	0.162	0.503	0.441
207	4.978	1.828	3.656	2.848	0.229	0.106	0.329	0.289
208	5.211	2.061	4.122	3.211	0.258	0.12	0.371	0.326
209	4.737	1.587	3.174	2.473	0.198	0.092	0.286	0.251
210	4.916	1.766	3.532	2.751	0.221	0.102	0.318	0.279
211	5.031	1.881	3.762	2.931	0.235	0.109	0.339	0.297
212	4.432	1.282	2.564	1.997	0.16	0.074	0.231	0.203
213	4.924	1.774	3.548	2.764	0.222	0.103	0.319	0.28
214	5.115	1.965	3.93	3.061	0.246	0.114	0.354	0.31
215	4.369	1.219	2.438	1.899	0.152	0.071	0.219	0.193
216	4.381	1.231	2.462	1.918	0.154	0.071	0.222	0.194
217	3.906	0.756	1.512	1.178	0.095	0.044	0.136	0.119
218	4.087	0.937	1.874	1.46	0.117	0.054	0.169	0.148
219	4.201	1.051	2.102	1.637	0.131	0.061	0.189	0.166
		MAX. =	**19.15**	**14.918**	**1.197**	**0.555**	**1.724**	**1.513**

TOP LONGITUDINAL REINFORCEMENT :

Mu = 19.15 Mton-m

Φ*Kn = Mu/bd^2 deff. = 442 mm

Mu/bd^2 = 19.15X10^7/(1000X442^2) = 0.98 N/mm^2 < 6.9 N/mm^2

Pt = 0.0026

Min. Pt = 0.0018 Reqd. Ast = 0.0026X1000X442/100 = 11.49 cm^2

Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Top - Longitudinal **O.K.**

Total Provide Pt = 0.003

ONE-WAY SHEAR CHECK

Vu = 14.918 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.003X14.918X10^4X442/19.15/10^7)X1000X442/10^4

= 42.615 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.003)X1000X442/10^4

= 44.093 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X442/10^4

= 75.832 Mton

HENCE, Vc = 42.615 Mton

Φt*Vc = 0.75X42.615 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 31.961 Mton **> 14.918Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.615X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 132.634 Mton **> Vu= 14.918Mton , CROSS SECTION O.K.**

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BOTTOM LONGITUDINAL REINFORCEMENT :

Mu = 1.197 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 1.197X10^7/(1000X442^2) = 0.061 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 150 C/C Provide Ast = 13.4 cm^2 at Bottom - Longitudinal **O.K.**
Total Provide Pt = 0.003

ONE-WAY SHEAR CHECK

Vu = 0.555 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.003X0.555X10^4X442/1.197/10^7)X1000X442/10^4
= 42.3 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.003)X1000X442/10^4
= 44.093 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.3 Mton
Φt*Vc = 0.75X42.3 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.725 Mton **> 0.555Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.3X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.61 Mton **> Vu= 0.555Mton , CROSS SECTION O.K.**

BOTTOM CROSS REINFORCEMENT :

Mu = 1.724 Mton-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 1.724X10^7/(1000X442^2) = 0.088 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002
Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X442/100 = 7.96 cm^2
Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom - Cross **O.K.**
Total Provide Pt = 0.0023

ONE-WAY SHEAR CHECK

Vu = 1.513 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0023X1.513X10^4X442/1.724/10^7)X1000X442/10^4
= 42.509 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0023)X1000X442/10^4
= 43.567 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X442/10^4
= 75.832 Mton
HENCE, Vc = 42.509 Mton
Φt*Vc = 0.75X42.509 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 31.882 Mton **> 1.513Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(42.509X10^4+0.66X35^0.5X1000X442)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 132.626 Mton **> Vu= 1.513Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Area (A) = 11.16 m^2

Perimeter (bo) = 2 X ((0.6+0.442) + (0.6+0.442) + (0.6+0.442) + (0.6 + 0.442)) = 8.336 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 9.781 Mton/m^2

Vu = ((11.16 - (0.6X0.6) - (0.6X0.6))X9.781

= 102.114 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X8.336X10^3X442/10^4

= 719.33 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X8.336X10^3X442/10^4

= 1111.691 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X442/(8.336X10^3))X1X35^0.5X(8.336X10^3)X442/10^4

= 649.636 Mton

HENCE, Vc = 649.636 Mton

Φt*Vn = Φt*Vc = 0.75X649.636 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

487.227 Mton **> 102.114Mton , Hence O.K.**

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8.5 DESIGN OF FOUNDATION MKD. F5 :

(STAAD File : 3860_Lime Stone Silo_Plate Mat_A01)

BASE PRESSURE CHECK :

Net Safe Bearing Capacity of Soil = 67 Mton/m^2 *(For BH-12 to BH-15)*
Depth Of Foundation = 2.5 m
Gross Allowable Bearing Capacity = 67 + 2.5 X 1.9 = 71.75 Mton/m^2

STAAD OUTPUT FROM BASE PRESSURE SUMMARY
(Staad Group : F5_BP)

	NODE	L/C	FX Mton/m^2	FY Mton/m^2	FZ Mton/m^2	
Max Px	1	101 SWDL	0	18.261	0	
Min Px	1	101 SWDL	0	18.261	0	
Max Py	7	104 SWDL	0	22.828	0	< 71.75 T/m2 , O.K.
Min Py	1	118 0.6*(SV	0	0	0	
Max Pz	1	101 SWDL	0	18.261	0	
Min Pz	1	101 SWDL	0	18.261	0	

STRENGTH CALCULATION :

Slab Thickness = 650 mm
Width = 1000 mm
Clear Cover = 50 mm
Compressive Strength of Concrete , fc' = 35 N/mm^2
Yield Strength of Main Reinforcement , fy = 420 N/mm^2
Yield Strength of Shear Reinforcement , fyt = 420 N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ = 0.9
Strength Reduction Factor of Shear Reinforcement , Φt = 0.75

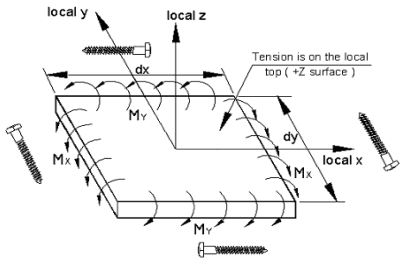
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : F5_BM)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	17.011	Mton-m	Corresponding Mxy	4.099	Mton-m	*(Plate No. - 1115 , L/C No. - 225)*
Min. Mx	-27.721	Mton-m	Corresponding Mxy	2.801	Mton-m	*(Plate No. - 1064 , L/C No. - 223)*
Max. My	16.786	Mton-m	Corresponding Mxy	2.509	Mton-m	*(Plate No. - 1132 , L/C No. - 233)*
Min. My	-42.288	Mton-m	Corresponding Mxy	0.122	Mton-m	*(Plate No. - 1303 , L/C No. - 221)*

FOR TOP REINFORCEMRNT :

Mx1 = Mx + abs(Mxy) = 17.011 + 4.099 = 21.11 Mton-m
My1 = My + abs(Mxy) = 16.786 + 2.509 = 19.295 Mton-m
Mx2 = Mx+abs(Mxy^2/My) = 17.011 + 1.001 = 18.012 Mton-m
My2 = My+abs(Mxy^2/Mx) = 16.786 + 0.37 = 17.156 Mton-m



DESIGN MOMENT CALCULATION FOR TOP REINFORCEMENT :

Mxd = 21.11 Mton-m
Myd = 19.295 Mton-m

ALONG X - DIRECTION :

Mu = 21.11 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 21.11X10^7/(1000X592^2) deff. = 592 mm
= 0.602 N/mm^2 < 6.9 N/mm^2
Pt = 0.0016
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0018X1000X592/100
= 10.66 cm^2

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Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Top
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Top
Total Provide Ast = 13.4 cm2 at Top **O.K.**
Total Provide Pt = 0.0023

ALONG Y - DIRECTION :

Mu = 19.295 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 19.295X10^7/(1000X576^2) deff. = 576 mm
= 0.551 N/mm^2 < 6.9 N/mm^2
Pt = 0.0015
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0018X1000X592/100
= 10.37 cm^2

Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Top
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Top
Total Provide Ast = 13.4 cm2 at Top **O.K.**
Total Provide Pt = 0.0023

FOR BOTTOM REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -27.721 - 2.801 = -30.522 Mton-m
My1 = My - abs(Mxy) = -42.288 - 0.122 = -42.41 Mton-m
Mx2 = Mx-abs(Mxy^2/My) = -27.721 - 0.186 = -27.907 Mton-m
My2 = My-abs(Mxy^2/Mx) = -42.288 - 0.001 = -42.287 Mton-m

DESIGN MOMENT CALCULATION FOR BOTTOM REINFORCEMENT :

Mxd = -30.522 Mton-m
Myd = -42.41 Mton-m

ALONG X - DIRECTION :

Mu = 30.522 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 30.522X10^7/(1000X590^2) deff. = 590 mm
= 0.877 N/mm^2 < 6.9 N/mm^2
Pt = 0.0024
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0024X1000X590/100
= 14.16 cm^2

Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Bottom
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Bottom
Total Provide Ast = 20.94 cm2) at Bottom **O.K.**
Total Provide Pt = 0.0035

ALONG Y - DIRECTION :

Mu = 42.41 Mton-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 42.41X10^7/(1000X570^2) deff. = 570 mm
= 1.305 N/mm^2 < 6.9 N/mm^2
Pt = 0.0035
Min. Pt = 0.0018 *[SBC:304-CR-18, Table 7.6.1.1, Page-45]*
Reqd. Ast = 0.0035X1000X570/100
= 19.95 cm^2

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Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Bottom
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Bottom
Total Provide Ast = 20.94 cm2) at Bottom **O.K.**
Total Provide Pt = 0.0037

SHEAR CHECK
(Staad Group : F8_SF)
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :
SQx = 0.36 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1280 , L/C No. - 220)*
Vu = 23.4 Mton

Provided Pt = 0.0035
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0035X23.4X10^4X590/30.522/10^7)X1000X590/10^4
= 57.436 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0035)X1000X590/10^4
= 59.358 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X590/10^4
= 101.224 Mton
HENCE, Vc = 57.436 Mton
Φt*Vc = 0.75X57.436 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 43.077 Mton **> 23.4Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(57.436X10^4+0.66X35^0.5X1000X590)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 177.087 Mton **> Vu= 23.4Mton , CROSS SECTION O.K.**

ALONG Y - DIRECTION :
SQy = 0.419 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 1113 , L/C No. - 220)*
Vu = 27.235 Mton

Pt provided = 0.0037
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0037X27.235X10^4X570/42.41/10^7)X1000X570/10^4
= 55.267 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0037)X1000X570/10^4
= 57.54 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X570/10^4
= 97.793 Mton
HENCE, Vc = 55.267 Mton
Φt*Vc = 0.75X55.267 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 41.45 Mton **> 27.235Mton , Hence O.K.**
Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(55.267X10^4+0.66X35^0.5X1000X570)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*
= 171.067 Mton **> Vu= 27.235Mton , CROSS SECTION O.K.**

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TWO-WAY SHEAR CHECK

Length of Foundation = 10.807 m
Width of Foundation = 6.107 m
Length of Column = 1 m
Width of Column = 1 m
No. of Column = 8
Area (A) = 65.998 m^2
Perimeter (bo) = 8 X 2 X ((1 + 0.59) + (1 + 0.57)) = 50.56 m
β = 1
αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*
Pmax = 22.828 Mton/m^2
Vu = ((65.998 - (1X1X8))X22.828
= 1323.978 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*
[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1
= 0.33X1X35^0.5X50.56X10^3X590/10^4
= 5823.808 Mton
[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff
= 0.17X(1+2/1)X1X35^0.5X50.56X10^3X590/10^4
= 9000.43 Mton
[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff
= 0.083X(2+30X590/(50.56X10^3))X1X35^0.5X(50.56X10^3)X590/10^4
= 3442.339 Mton
HENCE, Vc = 3442.339 Mton
Φt*Vn = Φt*Vc = 0.75X3442.339 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
2581.754 Mton **> 1323.978Mton , Hence O.K.**

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9.0 DESIGN OF COLUMN

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Beam Force Detail Summary

(COLUMN MKD. C1)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (MTon-m)	My (MTon-m)	Mz (MTon-m)
Max Fx	10	203:1.4*(DL + -	0	**40.286**	2.137	5.422	0	-14.205	5.600
Min Fx	3	202:1.4*(DL + -	2.620	**-3.585**	-5.337	2.618	-0	0.000	-0.000
Max Fy	13	202:1.4*(DL + -	0	18.307	**11.029**	1.969	0	-5.158	28.895
Min Fy	3	202:1.4*(DL + -	0	-0.412	**-5.337**	2.618	0	-6.860	-13.982
Max Fz	18	203:1.4*(DL + -	0	14.625	-1.305	**7.164**	0	-18.770	-3.420
Min Fz	15	203:1.4*(DL + -	0	29.865	-3.092	**-7.480**	0	19.597	-8.100
Max Mx	3	201:1.4*DL	0	5.520	0.405	-0.308	**0**	0.807	1.060
Min Mx	3	201:1.4*DL	0	5.520	0.405	-0.308	**0**	0.807	1.060
Max My	15	203:1.4*(DL + -	0	29.865	-3.092	-7.480	0	**19.597**	-8.100
Min My	18	203:1.4*(DL + -	0	14.625	-1.305	7.164	0	**-18.770**	-3.420
Max Mz	13	202:1.4*(DL + -	0	18.307	11.029	1.969	0	-5.158	**28.895**
Min Mz	3	202:1.4*(DL + -	0	-0.412	-5.337	2.618	0	-6.860	**-13.982**

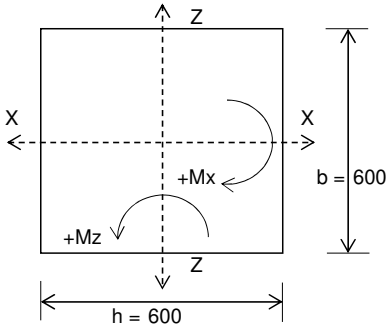
MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO	
MD-216-3860-EG-CV-CAL-1434	Mansourah Massarah Gold Project	Page	OF
900707_GAU01_AA23_CED01_38602_ OU100695883_R2	CALCULATION REAGENTS – LIME STONE EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date	30.11.2023
O19045_450_C_K_24169_002_R2	AS BUILT	Rev.	D01

9.1 DESIGN OF COLUMN MKD. C1

(Staad File : 3850_3860 - Lime Slaking Platform Supporting Foundation_A01)

Design Data :

Depth Parallel to X-X axis, h =	600	mm
Depth Parallel to Z-Z axis, b =	600	mm
Clear Cover =	50	mm
d' =	70	mm
Y'_h = (h-d')/h =	0.8	mm
Y'_b = (b-d')/b =	0.8	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	1.6	M, Hence	k*lux/b =	2.667 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	1.6	M, Hence	k*luz/h =	2.667 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	12	Nos	20	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	20	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	37.70	cm^2		
Total Provide Pt =	0.01	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 975 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^α$ + $(Mnz/Mnz1)^α$
203	40.286	-14.205	5.6	61.978	21.854	8.615	0.049	**0.07**	52.92	0.07	52.92	0.29
202	-3.585	0	0	-5.515	0	0	-0.004	0.07	52.92	0.07	52.92	0
202	18.307	-5.158	28.895	28.165	7.935	44.454	0.022	0.07	52.92	0.07	52.92	0.8
202	-0.412	-6.86	-13.982	-0.634	10.554	21.511	-0.001	0.07	52.92	0.07	52.92	0.31
203	14.625	-18.77	-3.42	22.5	28.877	5.262	0.018	0.07	52.92	0.07	52.92	0.4
203	29.865	19.597	-8.1	45.946	30.149	12.462	0.036	0.07	52.92	0.07	52.92	0.5
201	5.52	0.807	1.06	8.492	1.242	1.631	0.007	0.07	52.92	0.07	52.92	0.01
201	5.52	0.807	1.06	8.492	1.242	1.631	0.007	0.07	52.92	0.07	52.92	0.01
203	29.865	19.597	-8.1	45.946	30.149	12.462	0.036	0.07	52.92	0.07	52.92	0.5
203	14.625	-18.77	-3.42	22.5	28.877	5.262	0.018	0.07	52.92	0.07	52.92	0.4
202	18.307	-5.158	28.895	28.165	7.935	44.454	0.022	0.07	52.92	0.07	52.92	0.8
202	-0.412	-6.86	-13.982	-0.634	10.554	21.511	-0.001	0.07	52.92	0.07	52.92	0.31
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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Beam Force Detail Summary

(COLUMN MKD. C2)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (MTon-m)	My (MTon-m)	Mz (MTon-m)
Max Fx	8	204:1.2*(SWDl	0	**213.783**	10.410	-9.890	0	42.102	44.315
Min Fx	1	220:0.9*(SWDl	4.257	**-119.650**	7.800	-7.300	-0	0.000	-0
Max Fy	4	203:1.2*(SWDl	0	163.833	**10.440**	5.190	0	-22.094	44.443
Min Fy	1	207:1.2*(SWDl	0	163.833	**-10.440**	5.360	0	-22.817	-44.443
Max Fz	2	217:1.2*(SWDl	0	115.133	2.830	**36.170**	0	-153.975	12.047
Min Fz	5	205:1.2*(SWDl	0	175.223	-5.770	**-10.370**	0	44.145	-24.563
Max Mx	1	201:1.4*(SWDl	0	29.299	-0.560	0.840	**0**	-3.576	-2.384
Min Mx	1	201:1.4*(SWDl	0	29.299	-0.560	0.840	**0**	-3.576	-2.384
Max My	5	205:1.2*(SWDl	0	175.223	-5.770	-10.370	0	**44.145**	-24.563
Min My	2	217:1.2*(SWDl	0	115.133	2.830	36.170	0	**-153.975**	12.047
Max Mz	4	203:1.2*(SWDl	0	163.833	10.440	5.190	0	-22.094	**44.443**
Min Mz	1	207:1.2*(SWDl	0	163.833	-10.440	5.360	0	-22.817	**-44.443**

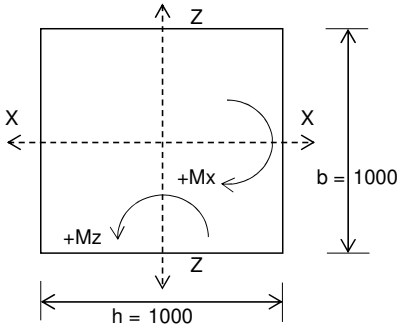
MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3860-EG-CV-CAL-1434	Mansourah Massarah Gold Project	Page OF
900707_GAU01_AA23_CED01_38602_ OU100695883_R2	CALCULATION REAGENTS – LIME STONE EQUIPMENT DESIGN CALCULATION OF FOUNDATION	Date 30.11.2023
O19045_450_C_K_24169_002_R2	AS BUILT	Rev. D01

9.2 DESIGN OF COLUMN MKD. C2

(STAAD File : 3860_Lime Slaking Silo_Plate Mat_A01)

Design Data :

Depth Parallel to X-X axis, h =	1000	mm
Depth Parallel to Z-Z axis, b =	1000	mm
Clear Cover =	50	mm
d' =	72.5	mm
Y'_h = (h-d')/h =	0.9	mm
Y'_b = (b-d')/b =	0.9	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2



Assumptions :

Strength Reduction Factor , Φ =	0.65
Effective Length factor , k =	1
β =	0.65

Use Interaction Diagram R35-420.7
[ACI DESIGN HANDBOOK-SP-17M(09)]

Unsupported Length of Column Parallel to X-X axis , lux =	3.657	M, Hence	k*lux/b =	3.657 < 22 , Neglect Slenderness Effect
Unsupported Length of Column Parallel to Z-Z axis , luz =	3.657	M, Hence	k*luz/h =	3.657 < 22 , Neglect Slenderness Effect

Main Reinforcement

Provided	28	Nos	25	Dia Bar Equally Distributed In Four Side
Provided	0	NOS	25	Dia Bar Equally Distributed In Four Side
Total Provided Ast =	137.44	cm^2		
Total Provide Pt =	0.014	SAFE		

Max. Nominal Axial Compressive Strength, Pn.max = 0.8[0.85*fc'(Ag-Ast)+fy*Ast]
= 2810 Mton SAFE

α = log0.5/logβ = 1.61

L/C	Pu (Mton)	Mux (Mton-m)	Muz (Mton-m)	Pn = Pu/ϕ (Mton)	Mnx = Mux/ϕ (Mton-m)	Mnz = Muz/ϕ (Mton-m)	Kn = Pn/fc'*b*h	Rnx = Mnx1 / fc'*b^2*h	Mnx1 (Mton-m)	Rnz = Mnz1 / fc'*h^2*b	Mnz1 (Mton-m)	$(Mnx/Mnx1)^{\alpha}$ + $(Mnz/Mnz1)^{\alpha}$
204	213.783	42.102	44.315	328.897	64.772	68.177	0.094	0.075	262.5	0.075	262.5	0.22
220	-119.65	0	0	-184.077	0	0	-0.053	0.075	262.5	0.075	262.5	0
203	163.833	-22.094	44.443	252.051	33.991	68.374	0.072	0.075	262.5	0.075	262.5	0.15
207	163.833	-22.817	-44.443	252.051	35.103	68.374	0.072	0.075	262.5	0.075	262.5	0.15
217	115.133	-153.975	12.047	177.128	236.885	18.534	0.051	0.075	262.5	0.075	262.5	0.86
205	175.223	44.145	-24.563	269.574	67.915	37.789	0.077	0.075	262.5	0.075	262.5	0.16
201	29.299	-3.576	-2.384	45.075	5.502	3.668	0.013	0.075	262.5	0.075	262.5	0
201	29.299	-3.576	-2.384	45.075	5.502	3.668	0.013	0.075	262.5	0.075	262.5	0
205	175.223	44.145	-24.563	269.574	67.915	37.789	0.077	0.075	262.5	0.075	262.5	0.16
217	115.133	-153.975	12.047	177.128	236.885	18.534	0.051	0.075	262.5	0.075	262.5	0.86
203	163.833	-22.094	44.443	252.051	33.991	68.374	0.072	0.075	262.5	0.075	262.5	0.15
207	163.833	-22.817	-44.443	252.051	35.103	68.374	0.072	0.075	262.5	0.075	262.5	0.15
												SAFE

Stirrups

Provided **10** Dia. Stirrups @ **250** mm C/C **O.K.** *(Ref. SBC:301-2007, CL. No. 25.7.2.2 & 25.7.2.1)*

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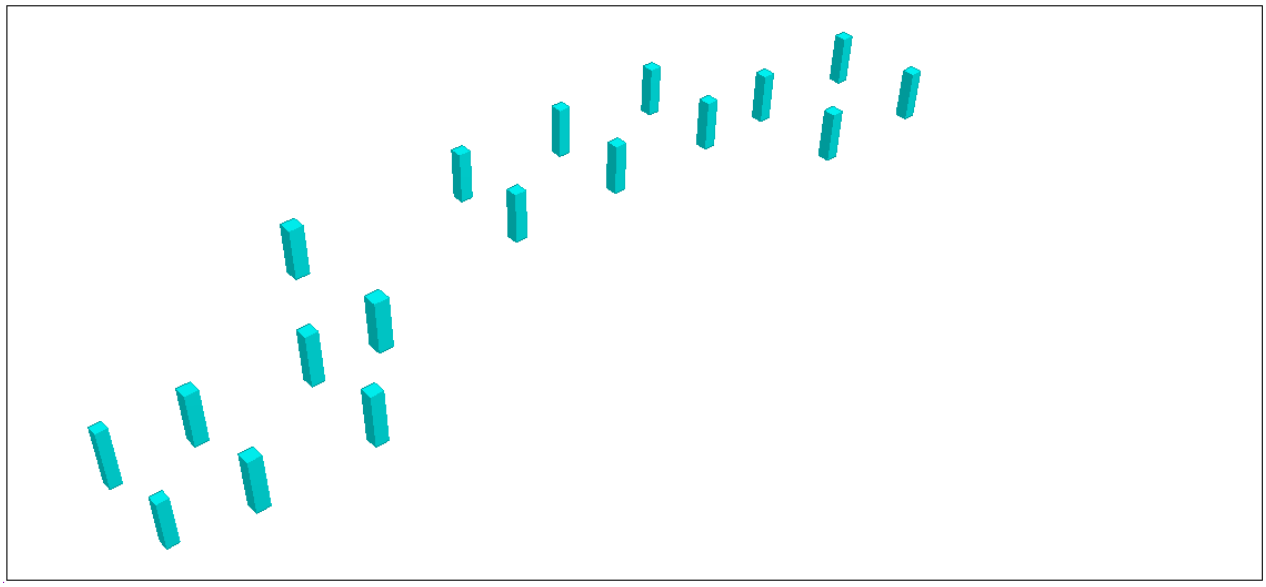
ATTACHMENT - 1 : ISOMETRIC VIEW OF STAAD MODEL

REAGENTS LIME STONE PLATFORM FOUNDATION

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3D Rendered View

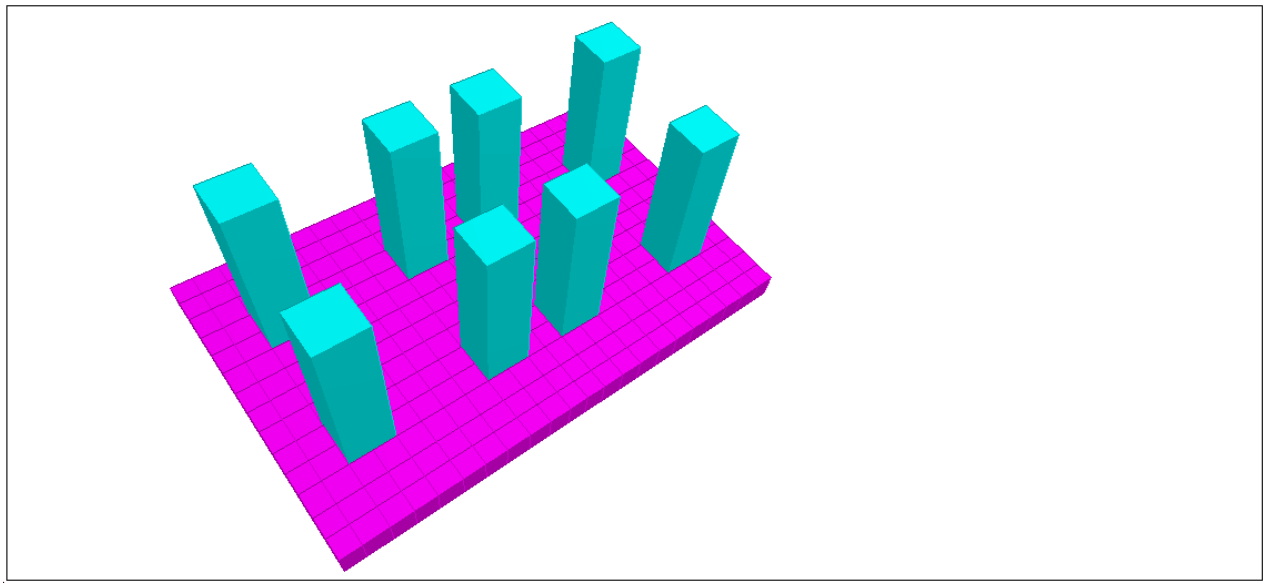
ATTACHMENT - 2 : ISOMETRIC VIEW OF STAAD MODEL

REAGENTS LIME STONE SILO FOUNDATION

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3D Rendered View

ATTACHMENT - 3 : STAAD INPUT

REAGENTS LIME STONE PLATFORM FOUNDATION

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Job Information

	Engineer	Checked	Approved
Name:	SUE	BDA	AG
Date:	9/25/2020		

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	36	Highest Node	36
Number of Elements	18	Highest Beam	18

Number of Basic Load Cases	10
Number of Combination Load Cases	38

Included in this printout are data for:

All	The Whole Structure

Included in this printout are results for load cases:

Type	L/C	Name
Primary	1	DEAD LOAD
Primary	2	LIVE LOAD
Primary	3	WIND LOAD (+Z)
Primary	4	WIND LOAD (-Z)
Primary	5	WIND LOAD (+X)
Primary	6	WIND LOAD (-X)
Primary	7	TL1 (TEMP LOAD +25)
Primary	8	TL2 (TEMP LOAD -25)
Primary	9	EL (+X)
Primary	10	EL (+Z)
Combination	101	DL
Combination	102	DL + LL + TL1
Combination	103	DL + LL + TL2
Combination	104	DL + 0.75*WL(+Z) + 0.75*LL
Combination	105	DL + 0.75*WL(-Z) + 0.75*LL
Combination	106	DL + 0.75*WL(+X) + 0.75*LL
Combination	107	DL + 0.75*WL(-X) + 0.75*LL
Combination	108	DL + 0.525*EL(+X) + 0.75*LL
Combination	109	DL + 0.525*EL(-X) + 0.75*LL
Combination	110	DL + 0.525*EL(+Z) + 0.75*LL
Combination	111	DL + 0.525*EL(-Z) + 0.75*LL
Combination	112	0.6*DL + WL(+Z)
Combination	113	0.6*DL + WL(-Z)
Combination	114	0.6*DL + WL(+X)
Combination	115	0.6*DL + WL(-X)
Combination	116	0.6*DL + 0.7*EL(+X)
Combination	117	0.6*DL - 0.7*EL(+X)
Combination	118	0.6*DL + 0.7*EL(+Z)
Combination	119	0.6*DL - 0.7*EL(+Z)

Job Information Cont...

Type	L/C	Name
Combination	204	1.2*DL + 1.6*WL(+Z) + 0.5*LL
Combination	205	1.2*DL + 1.6*WL(-Z) + 0.5*LL
Combination	206	1.2*DL + 1.6*WL(+Z) + 0.5*LL
Combination	207	1.2*DL + 1.6*WL(-Z) + 0.5*LL
Combination	208	1.2*DL + 1.0*EL(+X) + 0.5*LL
Combination	209	1.2*DL - 1.0*EL(+X) + 0.5*LL
Combination	210	1.2*DL + 1.0*EL(+Z) + 0.5*LL
Combination	211	1.2*DL - 1.0*EL(+Z) + 0.5*LL
Combination	212	0.9*DL + 1.6*WL(+Z)
Combination	213	0.9*DL + 1.6*WL(-Z)
Combination	214	0.9*DL + 1.6*WL(+X)
Combination	215	0.9*DL + 1.6*WL(-X)
Combination	216	0.9*DL + 1.0*EL(+X)
Combination	217	0.9*DL - 1.0*EL(+X)
Combination	218	0.9*DL + 1.0*EL(+Z)
Combination	219	0.9*DL - 1.0*EL(+Z)

Nodes

Node	X (m)	Y (m)	Z (m)
1	-10.950	913.700	9.350
2	-10.950	913.700	13.800
3	-7.450	913.470	9.350
4	-7.450	913.470	13.800
5	-2.250	913.470	9.350
6	-2.250	913.470	13.800
7	1.000	913.470	3.025
8	1.000	913.470	9.350
9	10.000	913.470	3.025
10	10.000	913.470	7.100
11	16.000	913.470	3.025
12	16.000	913.470	7.100
13	22.000	913.470	3.025
14	22.000	913.470	7.100
15	26.000	913.470	7.100
16	26.000	913.470	11.700
17	32.000	913.470	7.100
18	32.000	913.470	11.700
19	-10.950	910.850	9.350
20	-10.950	911.450	13.800
21	-7.450	910.850	9.350
22	-7.450	910.850	13.800
23	-2.250	910.850	9.350
24	-2.250	910.850	13.800
25	1.000	910.850	3.025
26	1.000	910.850	9.350
27	10.000	910.850	3.025

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
28	10.000	910.850	7.100
29	16.000	910.850	3.025
30	16.000	910.850	7.100
31	22.000	910.850	3.025
32	22.000	910.850	7.100
33	26.000	910.850	7.100
34	26.000	910.850	11.700
35	32.000	910.850	7.100
36	32.000	910.850	11.700

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	19	1	2.850	3	0
2	20	2	2.250	3	0
3	21	3	2.620	2	0
4	22	4	2.620	2	0
5	23	5	2.620	2	0
6	24	6	2.620	2	0
7	25	7	2.620	1	0
8	26	8	2.620	1	0
9	27	9	2.620	2	0
10	28	10	2.620	2	0
11	29	11	2.620	2	0
12	30	12	2.620	2	0
13	31	13	2.620	2	0
14	32	14	2.620	2	0
15	33	15	2.620	2	0
16	34	16	2.620	2	0
17	35	17	2.620	2	0
18	36	18	2.620	2	0

Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
1	Rect 0.70x0.70	4.9E+3	2E+6	2E+6	3.38E+6	CONCRETE
2	Rect 0.60x0.60	3.6E+3	1.08E+6	1.08E+6	1.82E+6	CONCRETE
3	Rect 0.50x0.50	2.5E+3	521E+3	521E+3	879E+3	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	205.000	0.300	7.83E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
19	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
20	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
21	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
22	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
23	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
24	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
25	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
26	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
27	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
28	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
29	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
30	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
31	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
32	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
33	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
34	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
35	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
36	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

There is no data of this type.

Primary Load Cases

Number	Name	Type
1	DEAD LOAD	None
2	LIVE LOAD	None
3	WIND LOAD (+Z)	None
4	WIND LOAD (-Z)	None
5	WIND LOAD (+X)	None
6	WIND LOAD (-X)	None
7	TL1 (TEMP LOAD +25)	None
8	TL2 (TEMP LOAD -25)	None
9	EL (+X)	None
10	EL (+Z)	None

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	DL	1	DEAD LOAD	1.00
102	DL + LL + TL1	1	DEAD LOAD	1.00
		2	LIVE LOAD	1.00
		7	TL1 (TEMP LOAD +25)	1.00
103	DL + LL + TL2	1	DEAD LOAD	1.00
		2	LIVE LOAD	1.00
		8	TL2 (TEMP LOAD -25)	1.00
104	DL + 0.75*WL(+Z) + 0.75*LL	1	DEAD LOAD	1.00
		3	WIND LOAD (+Z)	0.75
		2	LIVE LOAD	0.75
105	DL + 0.75*WL(-Z) + 0.75*LL	1	DEAD LOAD	1.00
		4	WIND LOAD (-Z)	0.75
		2	LIVE LOAD	0.75
106	DL + 0.75*WL(+X) + 0.75*LL	1	DEAD LOAD	1.00
		5	WIND LOAD (+X)	0.75
		2	LIVE LOAD	0.75
107	DL + 0.75*WL(-X) + 0.75*LL	1	DEAD LOAD	1.00
		6	WIND LOAD (-X)	0.75
		2	LIVE LOAD	0.75
108	DL + 0.525*EL(+X) + 0.75*LL	1	DEAD LOAD	1.00
		9	EL (+X)	0.52
		2	LIVE LOAD	0.75
109	DL + 0.525*EL(-X) + 0.75*LL	1	DEAD LOAD	1.00
		9	EL (+X)	-0.52
		2	LIVE LOAD	0.75
110	DL + 0.525*EL(+Z) + 0.75*LL	1	DEAD LOAD	1.00
		10	EL (+Z)	0.52
		2	LIVE LOAD	0.75
111	DL + 0.525*EL(-Z) + 0.75*LL	1	DEAD LOAD	1.00
		10	EL (+Z)	-0.52
		2	LIVE LOAD	0.75
112	0.6*DL + WL(+Z)	1	DEAD LOAD	0.60
		3	WIND LOAD (+Z)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
113	0.6*DL + WL(-Z)	1	DEAD LOAD	0.60
		4	WIND LOAD (-Z)	1.00
114	0.6*DL + WL(+X)	1	DEAD LOAD	0.60
		5	WIND LOAD (+X)	1.00
115	0.6*DL + WL(-X)	1	DEAD LOAD	0.60
		6	WIND LOAD (-X)	1.00
116	0.6*DL + 0.7*EL(+X)	1	DEAD LOAD	0.60
		9	EL (+X)	0.70
117	0.6*DL - 0.7*EL(+X)	1	DEAD LOAD	0.60
		9	EL (+X)	-0.70
118	0.6*DL + 0.7*EL(+Z)	1	DEAD LOAD	0.60
		10	EL (+Z)	0.70
119	0.6*DL - 0.7*EL(+Z)	1	DEAD LOAD	0.60
		10	EL (+Z)	-0.70
201	1.4*DL	1	DEAD LOAD	1.40
202	1.4*(DL + TL1) + 1.7*LL	1	DEAD LOAD	1.40
		7	TL1 (TEMP LOAD +25)	1.40
		2	LIVE LOAD	1.70
203	1.4*(DL + TL2) + 1.7*LL	1	DEAD LOAD	1.40
		8	TL2 (TEMP LOAD -25)	1.40
		2	LIVE LOAD	1.70
204	1.2*DL + 1.6*WL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		3	WIND LOAD (+Z)	1.60
		2	LIVE LOAD	0.50
205	1.2*DL + 1.6*WL(-Z) + 0.5*LL	1	DEAD LOAD	1.20
		4	WIND LOAD (-Z)	1.60
		2	LIVE LOAD	0.50
206	1.2*DL + 1.6*WL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		5	WIND LOAD (+X)	1.60
		2	LIVE LOAD	0.50
207	1.2*DL + 1.6*WL(-Z) + 0.5*LL	1	DEAD LOAD	1.20
		6	WIND LOAD (-X)	1.60
		2	LIVE LOAD	0.50
208	1.2*DL + 1.0*EL(+X) + 0.5*LL	1	DEAD LOAD	1.20
		9	EL (+X)	1.00
		2	LIVE LOAD	0.50
209	1.2*DL - 1.0*EL(+X) + 0.5*LL	1	DEAD LOAD	1.20
		9	EL (+X)	-1.00
		2	LIVE LOAD	0.50
210	1.2*DL + 1.0*EL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		10	EL (+Z)	1.00
		2	LIVE LOAD	0.50
211	1.2*DL - 1.0*EL(+Z) + 0.5*LL	1	DEAD LOAD	1.20
		10	EL (+Z)	-1.00
		2	LIVE LOAD	0.50
212	0.9*DL + 1.6*WL(+Z)	1	DEAD LOAD	0.90
		3	WIND LOAD (+Z)	1.60
213	0.9*DL + 1.6*WL(-Z)	1	DEAD LOAD	0.90
		4	WIND LOAD (-Z)	1.60

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
214	0.9*DL + 1.6*WL(+X)	1	DEAD LOAD	0.90
		5	WIND LOAD (+X)	1.60
215	0.9*DL + 1.6*WL(-X)	1	DEAD LOAD	0.90
		6	WIND LOAD (-X)	1.60
216	0.9*DL + 1.0*EL(+X)	1	DEAD LOAD	0.90
		9	EL (+X)	1.00
217	0.9*DL - 1.0*EL(+X)	1	DEAD LOAD	0.90
		9	EL (+X)	-1.00
218	0.9*DL + 1.0*EL(+Z)	1	DEAD LOAD	0.90
		10	EL (+Z)	1.00
219	0.9*DL - 1.0*EL(+Z)	1	DEAD LOAD	0.90
		10	EL (+Z)	-1.00

Load Generators

There is no data of this type.

1 DEAD LOAD : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.210	0.296	0.034	-	-	-
2	-0.148	-0.948	0.071	-	-	-
3	0.289	-1.677	0.220	-	-	-
4	-0.251	-1.283	0.04	-	-	-
5	-0.001	-5.457	-0.174	-	-	-
6	0.011	-4.002	1.045	-	-	-
7	0.206	-5.412	0.989	-	-	-
8	0.003	-5.330	-0.403	-	-	-
9	0.486	-4.476	0.615	-	-	-
10	0.322	-7.159	-0.123	-	-	-
11	0.399	-7.463	0.695	-	-	-
12	0.248	-7.290	-0.116	-	-	-
13	1.279	-2.830	0.520	-	-	-
14	0.139	-5.183	0.133	-	-	-
15	0.274	-3.444	0.499	-	-	-
16	0.259	-4.701	-0.065	-	-	-
17	0.275	-2.151	0.046	-	-	-
18	-0.064	-1.290	-0.091	-	-	-

1 DEAD LOAD : Selfweight

Direction	Factor	Assigned Geometry
Y	-1.000	ALL

2 LIVE LOAD : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.033	-1.236	-0.227	-	-	-
2	0.155	-2.439	0.275	-	-	-
3	-0.042	-4.006	0.839	-	-	-
4	-0.291	-5.910	-0.963	-	-	-
5	-0.042	-4.856	-0.006	-	-	-
6	0.196	-4.240	0.012	-	-	-
7	-1.283	-10.718	2.299	-	-	-
8	-0.001	-8.372	-2.224	-	-	-
9	0.135	-9.716	0.797	-	-	-
10	0.064	-13.575	-0.801	-	-	-
11	-0.014	-7.761	0.782	-	-	-
12	-0.059	-11.806	-0.811	-	-	-
13	1.025	-6.432	0.843	-	-	-
14	0.009	-12.861	-0.941	-	-	-
15	0.031	-10.504	1.063	-	-	-
16	0.037	-10.917	-0.962	-	-	-
17	-0.006	-5.588	1.068	-	-	-
18	0.054	-5.687	-1.042	-	-	-

3 WIND LOAD (+Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.151	-0.077	0.402	-	-	-
2	-0.047	-0.467	0.402	-	-	-
3	-0.150	1.226	0.626	-	-	-
4	-0.05	-0.685	0.622	-	-	-
5	0.006	0.833	0.610	-	-	-
6	-0.003	-0.830	0.642	-	-	-
7	0.195	0.781	0.237	-	-	-
8	-0.003	-0.781	1.827	-	-	-
9	0.090	1.464	0.906	-	-	-
10	-0.053	-1.122	0.918	-	-	-
11	0.087	0.502	0.935	-	-	-
12	-0.049	-0.845	0.935	-	-	-
13	0.201	1.038	0.983	-	-	-
14	-0.022	-1.038	0.954	-	-	-
15	-0.044	0.700	0.604	-	-	-
16	0.058	-0.723	0.632	-	-	-
17	-0.044	0.563	0.481	-	-	-
18	-0.019	-0.540	0.471	-	-	-

4 WIND LOAD (-Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.151	0.077	-0.402	-	-	-
2	0.047	0.467	-0.402	-	-	-
3	0.150	-1.226	-0.626	-	-	-
4	0.05	0.685	-0.622	-	-	-
5	-0.006	-0.833	-0.610	-	-	-
6	0.003	0.830	-0.642	-	-	-
7	-0.195	-0.781	-0.237	-	-	-
8	0.003	0.781	-1.827	-	-	-
9	-0.090	-1.464	-0.906	-	-	-
10	0.053	1.122	-0.918	-	-	-
11	-0.087	-0.502	-0.935	-	-	-
12	0.049	0.845	-0.935	-	-	-
13	-0.201	-1.038	-0.983	-	-	-
14	0.022	1.038	-0.954	-	-	-
15	0.044	-0.700	-0.604	-	-	-
16	-0.058	0.723	-0.632	-	-	-
17	0.044	-0.563	-0.481	-	-	-
18	0.019	0.540	-0.471	-	-	-

5 WIND LOAD (+X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.910	1.782	-0.004	-	-	-
2	0.626	1.063	-0.004	-	-	-
3	1.130	-1.835	-0.051	-	-	-
4	0.666	-1.093	-0.014	-	-	-
5	-0.117	0.047	0.028	-	-	-
6	0.236	0.035	0.039	-	-	-
7	0.777	0.021	0.005	-	-	-
8	0.242	-0.021	0.029	-	-	-
9	0.487	1.274	0.012	-	-	-
10	0.595	0.277	0.016	-	-	-
11	0.487	-1.149	0.053	-	-	-
12	0.604	-0.402	0.045	-	-	-
13	0.780	0.100	0.036	-	-	-
14	-0.201	-0.100	0.022	-	-	-
15	0.535	0.519	0.009	-	-	-
16	0.715	-0.097	0.026	-	-	-
17	0.536	-0.741	-0.146	-	-	-
18	-0.151	0.319	-0.101	-	-	-

6 WIND LOAD (-X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-0.910	-1.782	0.004	-	-	-
2	-0.626	-1.063	0.004	-	-	-
3	-1.130	1.835	0.051	-	-	-
4	-0.666	1.093	0.014	-	-	-
5	0.117	-0.047	-0.028	-	-	-
6	-0.236	-0.035	-0.039	-	-	-
7	-0.777	-0.021	-0.005	-	-	-
8	-0.242	0.021	-0.029	-	-	-
9	-0.487	-1.274	-0.012	-	-	-
10	-0.595	-0.277	-0.016	-	-	-
11	-0.487	1.149	-0.053	-	-	-
12	-0.604	0.402	-0.045	-	-	-
13	-0.780	-0.100	-0.036	-	-	-
14	0.201	0.100	-0.022	-	-	-
15	-0.535	-0.519	-0.009	-	-	-
16	-0.715	0.097	-0.026	-	-	-
17	-0.536	0.741	0.146	-	-	-
18	0.151	-0.319	0.101	-	-	-

7 TL1 (TEMP LOAD +25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-3.310	-8.689	0.084	-	-	-
2	-0.012	0.007	0.097	-	-	-
3	-4.050	9.102	-3.109	-	-	-
4	-0.979	-0.510	3.105	-	-	-
5	0.548	0.194	-2.892	-	-	-
6	1.410	-0.105	3.275	-	-	-
7	-6.569	-0.069	-5.565	-	-	-
8	-0.065	0.069	5.387	-	-	-
9	-0.526	-0.609	-2.866	-	-	-
10	-1.127	2.867	2.777	-	-	-
11	0.729	0.225	-2.981	-	-	-
12	0.324	-2.484	2.682	-	-	-
13	5.354	-0.170	-2.950	-	-	-
14	0.835	0.170	2.958	-	-	-
15	2.520	2.867	-3.553	-	-	-
16	0.006	0.016	2.832	-	-	-
17	3.979	-2.868	-3.042	-	-	-
18	0.934	-0.015	3.761	-	-	-

8 TL2 (TEMP LOAD -25) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	3.310	8.689	-0.084	-	-	-
2	0.012	-0.007	-0.097	-	-	-
3	4.050	-9.102	3.109	-	-	-
4	0.979	0.510	-3.105	-	-	-
5	-0.548	-0.194	2.892	-	-	-
6	-1.410	0.105	-3.275	-	-	-
7	6.569	0.069	5.565	-	-	-
8	0.065	-0.069	-5.387	-	-	-
9	0.526	0.609	2.866	-	-	-
10	1.127	-2.867	-2.777	-	-	-
11	-0.729	-0.225	2.981	-	-	-
12	-0.324	2.484	-2.682	-	-	-
13	-5.354	0.170	2.950	-	-	-
14	-0.835	-0.170	-2.958	-	-	-
15	-2.520	-2.867	3.553	-	-	-
16	-0.006	-0.016	-2.832	-	-	-
17	-3.979	2.868	3.042	-	-	-
18	-0.934	0.015	-3.761	-	-	-

9 EL (+X) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	0.183	0.494	-0.002	-	-	-
2	0.06	0.167	-0.002	-	-	-
3	0.188	-0.500	-0.003	-	-	-
4	0.065	-0.158	-0.003	-	-	-
5	-0.005	-0.003	0.001	-	-	-
6	0.003	-	0.001	-	-	-
7	0.258	0.004	0.003	-	-	-
8	0.004	-0.004	0.006	-	-	-
9	0.121	0.603	0.002	-	-	-
10	0.161	0.299	0.004	-	-	-
11	0.120	-0.566	0.017	-	-	-
12	0.159	-0.336	0.014	-	-	-
13	0.261	0.031	0.009	-	-	-
14	0.073	-0.031	0.008	-	-	-
15	0.148	0.293	-0.004	-	-	-
16	0.067	-0.006	0.002	-	-	-
17	0.150	-0.350	-0.038	-	-	-
18	-	0.063	-0.014	-	-	-

10 EL (+Z) : Node Loads

Node	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
1	-	0.090	0.043	-	-	-
2	-0.125	-0.433	0.043	-	-	-
3	0.008	0.204	0.106	-	-	-
4	-0.133	0.137	0.099	-	-	-
5	0.006	0.242	0.105	-	-	-
6	-0.008	-0.240	0.110	-	-	-
7	0.072	0.243	0.013	-	-	-
8	-	-0.243	0.314	-	-	-
9	0.033	0.507	0.139	-	-	-
10	0.001	-0.335	0.140	-	-	-
11	0.032	0.154	0.145	-	-	-
12	0.002	-0.326	0.144	-	-	-
13	0.074	0.341	0.152	-	-	-
14	0.001	-0.341	0.148	-	-	-
15	0.002	0.242	0.114	-	-	-
16	0.043	-0.212	0.121	-	-	-
17	0.002	0.075	0.038	-	-	-
18	-0.01	-0.105	0.04	-	-	-

ATTACHMENT - 6 : STAAD INPUT

REAGENTS LIME STONE SILO FOUNDATION

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Job Information

	Engineer	Checked	Approved
Name:	SUE	BDA	AG
Date:	9/29/2020		

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	353	Highest Node	436
Number of Elements	8	Highest Beam	8
Number of Plates	308	Highest Plate	1307

Number of Basic Load Cases	86
Number of Combination Load Cases	0

Included in this printout are data for:

Property	R2: Rect 1.00x1.00

Included in this printout are results for load cases:

Type	L/C	Name
Primary	201	1.4*(SWDL + DL)
Primary	202	1.4*(SWDL + DL) + 1.7*LL
Primary	203	1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL
Primary	204	1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL
Primary	205	1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL
Primary	206	1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL
Primary	207	1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL
Primary	208	1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL
Primary	209	1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL
Primary	210	1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL
Primary	211	1.2*(SWDL + DL) + EL-0 + 0.5*LL
Primary	212	1.2*(SWDL + DL) + EL-45 + 0.5*LL
Primary	213	1.2*(SWDL + DL) + EL-90 + 0.5*LL
Primary	214	1.2*(SWDL + DL) + EL-135 + 0.5*LL
Primary	215	1.2*(SWDL + DL) + EL-180 + 0.5*LL
Primary	216	1.2*(SWDL + DL) + EL-225 + 0.5*LL
Primary	217	1.2*(SWDL + DL) + EL-270 + 0.5*LL
Primary	218	1.2*(SWDL + DL) + EL-315 + 0.5*LL
Primary	219	0.9*(SWDL + DL) + 1.6*WL-0
Primary	220	0.9*(SWDL + DL) + 1.6*WL-45
Primary	221	0.9*(SWDL + DL) + 1.6*WL-90
Primary	222	0.9*(SWDL + DL) + 1.6*WL-135
Primary	223	0.9*(SWDL + DL) + 1.6*WL-180
Primary	224	0.9*(SWDL + DL) + 1.6*WL-225
Primary	225	0.9*(SWDL + DL) + 1.6*WL-270
Primary	226	0.9*(SWDL + DL) + 1.6*WL-315
Primary	227	0.9*(SWDL + DL) + EL-0
Primary	228	0.9*(SWDL + DL) + EL-45

Job Information Cont...

Type	L/C	Name
Primary	232	0.9*(SWDL + DL) + EL-225
Primary	233	0.9*(SWDL + DL) + EL-270
Primary	234	0.9*(SWDL + DL) + EL-315

Nodes

Node	X (m)	Y (m)	Z (m)
1	0	909.850	0
2	3.107	909.850	0
3	5.000	909.850	0
4	8.107	909.850	0
5	0	909.850	3.107
6	3.107	909.850	3.107
7	5.000	909.850	3.107
8	8.107	909.850	3.107
9	0	914.107	0
10	3.107	914.107	0
11	5.000	914.107	0
12	8.107	914.107	0
13	0	914.107	3.107
14	3.107	914.107	3.107
15	5.000	914.107	3.107
16	8.107	914.107	3.107

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	1	9	4.257	2	0
2	2	10	4.257	2	0
3	3	11	4.257	2	0
4	4	12	4.257	2	0
5	5	13	4.257	2	0
6	6	14	4.257	2	0
7	7	15	4.257	2	0
8	8	16	4.257	2	0

Plates

There is no data of this type.

Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
2	Rect 1.00x1.00	10E+3	8.33E+6	8.33E+6	14.1E+6	CONCRETE

Plate Thickness

Prop	Node A (cm)	Node B (cm)	Node C (cm)	Node D (cm)	Material
1	65.000	65.000	65.000	65.000	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	205.000	0.300	7.83E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
1	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
2	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
3	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
4	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
5	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
6	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
7	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
8	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

There is no data of this type.

Primary Load Cases

Number	Name	Type
1	SELFWEIGHT DEAD LOAD (SWDL)	None
2	DEAD LOAD (DL)	None
3	LIVE LOAD (LL)	None
4	WIND LOAD - 0 (WL-0)	None
5	WIND LOAD - 45 (WL-45)	None
6	WIND LOAD - 90 (WL-90)	None
7	WIND LOAD - 135 (WL-135)	None
8	WIND LOAD - 180 (WL-180)	None
9	WIND LOAD - 225 (WL-225)	None
10	WIND LOAD - 270 (WL-270)	None
11	WIND LOAD - 315 (WL315)	None
12	EARTHQUAKE LOAD - 0 (EL-0)	None
13	EARTHQUAKE LOAD - 45 (EL-45)	None
14	EARTHQUAKE LOAD - 90 (EL-90)	None
15	EARTHQUAKE LOAD - 135 (EL-135)	None
16	EARTHQUAKE LOAD - 180 (EL-180)	None
17	EARTHQUAKE LOAD - 225 (EL-225)	None
18	EARTHQUAKE LOAD - 270 (EL-270)	None
19	EARTHQUAKE LOAD - 315 (EL315)	None
101	SWDL + DL + LL	None
102	SWDL + DL + 0.75*WL-0 + 0.75*LL	None
103	SWDL + DL + 0.75*WL-45 + 0.75*LL	None
104	SWDL + DL + 0.75*WL-90 + 0.75*LL	None
105	SWDL + DL + 0.75*WL-135 + 0.75*LL	None
106	SWDL + DL + 0.75*WL-180 + 0.75*LL	None
107	SWDL + DL + 0.75*WL-225 + 0.75*LL	None
108	SWDL + DL + 0.75*WL-270 + 0.75*LL	None
109	SWDL + DL + 0.75*WL-315 + 0.75*LL	None
110	SWDL + DL + 0.525*EL-0 + 0.75*LL	None
111	SWDL + DL + 0.525*EL-45 + 0.75*LL	None
112	SWDL + DL + 0.525*EL-90 + 0.75*LL	None
113	SWDL + DL + 0.525*EL-135 + 0.75*LL	None
114	SWDL + DL + 0.525*EL-180 + 0.75*LL	None
115	SWDL + DL + 0.525*EL-225 + 0.75*LL	None
116	SWDL + DL + 0.525*EL-270 + 0.75*LL	None
117	SWDL + DL + 0.525*EL-315 + 0.75*LL	None
118	0.6*(SWDL + DL) + WL-0	None
119	0.6*(SWDL + DL) + WL-45	None
120	0.6*(SWDL + DL) + WL-90	None
121	0.6*(SWDL + DL) + WL-135	None
122	0.6*(SWDL + DL) + WL-180	None
123	0.6*(SWDL + DL) + WL-225	None
124	0.6*(SWDL + DL) + WL-270	None
125	0.6*(SWDL + DL) + WL-315	None
126	0.6*(SWDL + DL) + 0.7*EL-0	None
127	0.6*(SWDL + DL) + 0.7*EL-45	None
128	0.6*(SWDL + DL) + 0.7*EL-90	None
129	0.6*(SWDL + DL) + 0.7*EL-135	None
130	0.6*(SWDL + DL) + 0.7*EL-180	None

Primary Load Cases Cont...

Number	Name	Type
131	0.6*(SWDL + DL) + 0.7*EL-225	None
132	0.6*(SWDL + DL) + 0.7*EL-270	None
133	0.6*(SWDL + DL) + 0.7*EL-315	None
201	1.4*(SWDL + DL)	None
202	1.4*(SWDL + DL) + 1.7*LL	None
203	1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL	None
204	1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL	None
205	1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL	None
206	1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL	None
207	1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL	None
208	1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL	None
209	1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL	None
210	1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL	None
211	1.2*(SWDL + DL) + EL-0 + 0.5*LL	None
212	1.2*(SWDL + DL) + EL-45 + 0.5*LL	None
213	1.2*(SWDL + DL) + EL-90 + 0.5*LL	None
214	1.2*(SWDL + DL) + EL-135 + 0.5*LL	None
215	1.2*(SWDL + DL) + EL-180 + 0.5*LL	None
216	1.2*(SWDL + DL) + EL-225 + 0.5*LL	None
217	1.2*(SWDL + DL) + EL-270 + 0.5*LL	None
218	1.2*(SWDL + DL) + EL-315 + 0.5*LL	None
219	0.9*(SWDL + DL) + 1.6*WL-0	None
220	0.9*(SWDL + DL) + 1.6*WL-45	None
221	0.9*(SWDL + DL) + 1.6*WL-90	None
222	0.9*(SWDL + DL) + 1.6*WL-135	None
223	0.9*(SWDL + DL) + 1.6*WL-180	None
224	0.9*(SWDL + DL) + 1.6*WL-225	None
225	0.9*(SWDL + DL) + 1.6*WL-270	None
226	0.9*(SWDL + DL) + 1.6*WL-315	None
227	0.9*(SWDL + DL) + EL-0	None
228	0.9*(SWDL + DL) + EL-45	None
229	0.9*(SWDL + DL) + EL-90	None
230	0.9*(SWDL + DL) + EL-135	None
231	0.9*(SWDL + DL) + EL-180	None
232	0.9*(SWDL + DL) + EL-225	None
233	0.9*(SWDL + DL) + EL-270	None
234	0.9*(SWDL + DL) + EL-315	None

Combination Load Cases

There is no data of this type.

Load Generators

There is no data of this type.

201 1.4*(SWDL + DL) : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.400
2	DEAD LOAD (DL)	1.400

202 1.4*(SWDL + DL) + 1.7*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.400
2	DEAD LOAD (DL)	1.400
3	LIVE LOAD (LL)	1.700

203 1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
4	WIND LOAD - 0 (WL-0)	1.600

204 1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
5	WIND LOAD - 45 (WL-45)	1.600

205 1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
6	WIND LOAD - 90 (WL-90)	1.600

206 1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
7	WIND LOAD - 135 (WL-135)	1.600

207 1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
8	WIND LOAD - 180 (WL-180)	1.600

208 1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
9	WIND LOAD - 225 (WL-225)	1.600

209 1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
10	WIND LOAD - 270 (WL-270)	1.600

210 1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
11	WIND LOAD - 315 (WL315)	1.600

211 1.2*(SWDL + DL) + EL-0 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
12	EARTHQUAKE LOAD - 0 (EL-0)	1.000

212 1.2*(SWDL + DL) + EL-45 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
13	EARTHQUAKE LOAD - 45 (EL-45)	1.000

213 1.2*(SWDL + DL) + EL-90 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
14	EARTHQUAKE LOAD - 90 (EL-90)	1.000

214 1.2*(SWDL + DL) + EL-135 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
15	EARTHQUAKE LOAD - 135 (EL-135)	1.000

215 1.2*(SWDL + DL) + EL-180 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
16	EARTHQUAKE LOAD - 180 (EL-180)	1.000

216 1.2*(SWDL + DL) + EL-225 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
17	EARTHQUAKE LOAD - 225 (EL-225)	1.000

217 1.2*(SWDL + DL) + EL-270 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
18	EARTHQUAKE LOAD - 270 (EL-270)	1.000

218 1.2*(SWDL + DL) + EL-315 + 0.5*LL : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	1.200
2	DEAD LOAD (DL)	1.200
3	LIVE LOAD (LL)	0.500
19	EARTHQUAKE LOAD - 315 (EL315)	1.000

219 0.9*(SWDL + DL) + 1.6*WL-0 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
4	WIND LOAD - 0 (WL-0)	1.600

220 0.9*(SWDL + DL) + 1.6*WL-45 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
5	WIND LOAD - 45 (WL-45)	1.600

221 0.9*(SWDL + DL) + 1.6*WL-90 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
6	WIND LOAD - 90 (WL-90)	1.600

222 0.9*(SWDL + DL) + 1.6*WL-135 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
7	WIND LOAD - 135 (WL-135)	1.600

223 0.9*(SWDL + DL) + 1.6*WL-180 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
8	WIND LOAD - 180 (WL-180)	1.600

224 0.9*(SWDL + DL) + 1.6*WL-225 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
9	WIND LOAD - 225 (WL-225)	1.600

225 0.9*(SWDL + DL) + 1.6*WL-270 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
10	WIND LOAD - 270 (WL-270)	1.600

226 0.9*(SWDL + DL) + 1.6*WL-315 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
11	WIND LOAD - 315 (WL315)	1.600

227 0.9*(SWDL + DL) + EL-0 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
12	EARTHQUAKE LOAD - 0 (EL-0)	1.000

228 0.9*(SWDL + DL) + EL-45 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
13	EARTHQUAKE LOAD - 45 (EL-45)	1.000

229 0.9*(SWDL + DL) + EL-90 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
14	EARTHQUAKE LOAD - 90 (EL-90)	1.000

230 0.9*(SWDL + DL) + EL-135 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
15	EARTHQUAKE LOAD - 135 (EL-135)	1.000

231 0.9*(SWDL + DL) + EL-180 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
16	EARTHQUAKE LOAD - 180 (EL-180)	1.000

232 0.9*(SWDL + DL) + EL-225 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
17	EARTHQUAKE LOAD - 225 (EL-225)	1.000

233 0.9*(SWDL + DL) + EL-270 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
18	EARTHQUAKE LOAD - 270 (EL-270)	1.000

234 0.9*(SWDL + DL) + EL-315 : Repeat Loads

Ref	Name	Factor
1	SELFWEIGHT DEAD LOAD (SWDL)	0.900
2	DEAD LOAD (DL)	0.900
19	EARTHQUAKE LOAD - 315 (EL315)	1.000

ATTACHMENT - 5 : STAAD OUTPUT

REAGENTS LIME STONE PLATFORM FOUNDATION

PROPRIETARY & CONFIDENTIAL

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*************************************************
*                                               *
*           STAAD.Pro CONNECT Edition           *
*           Version  22.00.00.15                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    OCT  1, 2020               *
*           Time=    12:48:39                   *
*                                               *
*  Licensed to: Larsen and Toubro Ltd.          *
*************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\MAADEN-GOLD-O19045\3860 REAGENTS LIME STONE\CIVIL\LATEST\3850_3860 - Lime Stone Platfor.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 9/25/2020
     4. JOB NAME MMGP
     5. JOB CLIENT MAADEN
     6. JOB NO O19045
     7. JOB PART 3850_3860 - PLATFORM SUPPORTING FOUNDATION
     8. ENGINEER NAME SUE
     9. CHECKER NAME BDA
    10. APPROVED NAME AG
    11. END JOB INFORMATION
    12. INPUT WIDTH 79
    13. UNIT METER MTON
    14. JOINT COORDINATES
    15. 1 -10.95 913.7 9.35; 2 -10.95 913.7 13.8; 3 -7.45 913.47 9.35
    16. 4 -7.45 913.47 13.8; 5 -2.25 913.47 9.35; 6 -2.25 913.47 13.8
    17. 7 1 913.47 3.025; 8 1 913.47 9.35; 9 10 913.47 3.025; 10 10 913.47 7.1
    18. 11 16 913.47 3.025; 12 16 913.47 7.1; 13 22 913.47 3.025; 14 22 913.47 7.1
    19. 15 26 913.47 7.1; 16 26 913.47 11.7; 17 32 913.47 7.1; 18 32 913.47 11.7
    20. 19 -10.95 910.85 9.35; 20 -10.95 911.45 13.8; 21 -7.45 910.85 9.35
    21. 22 -7.45 910.85 13.8; 23 -2.25 910.85 9.35; 24 -2.25 910.85 13.8
    22. 25 1 910.85 3.025; 26 1 910.85 9.35; 27 10 910.85 3.025; 28 10 910.85 7.1
    23. 29 16 910.85 3.025; 30 16 910.85 7.1; 31 22 910.85 3.025; 32 22 910.85 7.1
    24. 33 26 910.85 7.1; 34 26 910.85 11.7; 35 32 910.85 7.1; 36 32 910.85 11.7
    25. MEMBER INCIDENCES
    26. 1 19 1; 2 20 2; 3 21 3; 4 22 4; 5 23 5; 6 24 6; 7 25 7; 8 26 8; 9 27 9
    27. 10 28 10; 11 29 11; 12 30 12; 13 31 13; 14 32 14; 15 33 15; 16 34 16; 17 35 17
    28. 18 36 18
    29. START GROUP DEFINITION
    30. MEMBER
    31. _C1 7 8
    32. _C2 3 TO 6 9 TO 18
    33. _C3 1 2
    34. JOINT
    35. _F1 20
    36. _F2 19
    37. END GROUP DEFINITION
    38. DEFINE MATERIAL START
```

```
 39. ISOTROPIC CONCRETE
 40. E 2.21467E+06
 41. POISSON 0.17
 42. DENSITY 2.40261
 43. ALPHA 1E-05
 44. DAMP 0.05
 45. TYPE CONCRETE
 46. STRENGTH FCU 2812.28
 47. END DEFINE MATERIAL
 48. MEMBER PROPERTY AMERICAN
 49. 7 8 PRIS YD 0.7 ZD 0.7
 50. 3 TO 6 9 TO 18 PRIS YD 0.6 ZD 0.6
 51. MEMBER PROPERTY AMERICAN
 52. 1 2 PRIS YD 0.5 ZD 0.5
 53. CONSTANTS
 54. MATERIAL CONCRETE ALL
 55. SUPPORTS
 56. 19 TO 36 FIXED
 57. LOAD 1 DEAD LOAD
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
 58. SELFWEIGHT Y -1
 59. JOINT LOAD
 60. 1 FX 0.21 FY 0.296 FZ 0.034
 61. 2 FX -0.148 FY -0.948 FZ 0.071
 62. 3 FX 0.289 FY -1.677 FZ 0.22
 63. 4 FX -0.251 FY -1.283 FZ 0.04
 64. 5 FX -0.001 FY -5.457 FZ -0.174
 65. 6 FX 0.011 FY -4.002 FZ 1.045
 66. 7 FX 0.206 FY -5.412 FZ 0.989
 67. 8 FX 0.003 FY -5.33 FZ -0.403
 68. 9 FX 0.486 FY -4.476 FZ 0.615
 69. 10 FX 0.322 FY -7.159 FZ -0.123
 70. 11 FX 0.399 FY -7.463 FZ 0.695
 71. 12 FX 0.248 FY -7.29 FZ -0.116
 72. 13 FX 1.279 FY -2.83 FZ 0.52
 73. 14 FX 0.139 FY -5.183 FZ 0.133
 74. 15 FX 0.274 FY -3.444 FZ 0.499
 75. 16 FX 0.259 FY -4.701 FZ -0.065
 76. 17 FX 0.275 FY -2.151 FZ 0.046
 77. 18 FX -0.064 FY -1.29 FZ -0.091
 78. LOAD 2 LIVE LOAD
 79. JOINT LOAD
 80. 1 FX 0.033 FY -1.236 FZ -0.227
 81. 2 FX 0.155 FY -2.439 FZ 0.275
 82. 3 FX -0.042 FY -4.006 FZ 0.839
 83. 4 FX -0.291 FY -5.91 FZ -0.963
 84. 5 FX -0.042 FY -4.856 FZ -0.006
 85. 6 FX 0.196 FY -4.24 FZ 0.012
 86. 7 FX -1.283 FY -10.718 FZ 2.299
 87. 8 FX -0.001 FY -8.372 FZ -2.224
 88. 9 FX 0.135 FY -9.716 FZ 0.797
 89. 10 FX 0.064 FY -13.575 FZ -0.801
 90. 11 FX -0.014 FY -7.761 FZ 0.782
 91. 12 FX -0.059 FY -11.806 FZ -0.811
 92. 13 FX 1.025 FY -6.432 FZ 0.843
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93. 14 FX 0.009 FY -12.861 FZ -0.941
94. 15 FX 0.031 FY -10.504 FZ 1.063
95. 16 FX 0.037 FY -10.917 FZ -0.962
96. 17 FX -0.006 FY -5.588 FZ 1.068
97. 18 FX 0.054 FY -5.687 FZ -1.042
98. LOAD 3 WIND LOAD (+Z)
99. JOINT LOAD
100. 1 FX -0.151 FY -0.077 FZ 0.402
101. 2 FX -0.047 FY -0.467 FZ 0.402
102. 3 FX -0.15 FY 1.226 FZ 0.626
103. 4 FX -0.05 FY -0.685 FZ 0.622
104. 5 FX 0.006 FY 0.833 FZ 0.61
105. 6 FX -0.003 FY -0.83 FZ 0.642
106. 7 FX 0.195 FY 0.781 FZ 0.237
107. 8 FX -0.003 FY -0.781 FZ 1.827
108. 9 FX 0.09 FY 1.464 FZ 0.906
109. 10 FX -0.053 FY -1.122 FZ 0.918
110. 11 FX 0.087 FY 0.502 FZ 0.935
111. 12 FX -0.049 FY -0.845 FZ 0.935
112. 13 FX 0.201 FY 1.038 FZ 0.983
113. 14 FX -0.022 FY -1.038 FZ 0.954
114. 15 FX -0.044 FY 0.7 FZ 0.604
115. 16 FX 0.058 FY -0.723 FZ 0.632
116. 17 FX -0.044 FY 0.563 FZ 0.481
117. 18 FX -0.019 FY -0.54 FZ 0.471
118. LOAD 4 WIND LOAD (-Z)
119. JOINT LOAD
120. 1 FX 0.151 FY 0.077 FZ -0.402
121. 2 FX 0.047 FY 0.467 FZ -0.402
122. 3 FX 0.15 FY -1.226 FZ -0.626
123. 4 FX 0.05 FY 0.685 FZ -0.622
124. 5 FX -0.006 FY -0.833 FZ -0.61
125. 6 FX 0.003 FY 0.83 FZ -0.642
126. 7 FX -0.195 FY -0.781 FZ -0.237
127. 8 FX 0.003 FY 0.781 FZ -1.827
128. 9 FX -0.09 FY -1.464 FZ -0.906
129. 10 FX 0.053 FY 1.122 FZ -0.918
130. 11 FX -0.087 FY -0.502 FZ -0.935
131. 12 FX 0.049 FY 0.845 FZ -0.935
132. 13 FX -0.201 FY -1.038 FZ -0.983
133. 14 FX 0.022 FY 1.038 FZ -0.954
134. 15 FX 0.044 FY -0.7 FZ -0.604
135. 16 FX -0.058 FY 0.723 FZ -0.632
136. 17 FX 0.044 FY -0.563 FZ -0.481
137. 18 FX 0.019 FY 0.54 FZ -0.471
138. LOAD 5 WIND LOAD (+X)
139. JOINT LOAD
140. 1 FX 0.91 FY 1.782 FZ -0.004
141. 2 FX 0.626 FY 1.063 FZ -0.004
142. 3 FX 1.13 FY -1.835 FZ -0.051
143. 4 FX 0.666 FY -1.093 FZ -0.014
144. 5 FX -0.117 FY 0.047 FZ 0.028
145. 6 FX 0.236 FY 0.035 FZ 0.039
146. 7 FX 0.777 FY 0.021 FZ 0.005
147. 8 FX 0.242 FY -0.021 FZ 0.029
148. 9 FX 0.487 FY 1.274 FZ 0.012

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149. 10 FX 0.595 FY 0.277 FZ 0.016
150. 11 FX 0.487 FY -1.149 FZ 0.053
151. 12 FX 0.604 FY -0.402 FZ 0.045
152. 13 FX 0.78 FY 0.1 FZ 0.036
153. 14 FX -0.201 FY -0.1 FZ 0.022
154. 15 FX 0.535 FY 0.519 FZ 0.009
155. 16 FX 0.715 FY -0.097 FZ 0.026
156. 17 FX 0.536 FY -0.741 FZ -0.146
157. 18 FX -0.151 FY 0.319 FZ -0.101
158. LOAD 6 WIND LOAD (-X)
159. JOINT LOAD
160. 1 FX -0.91 FY -1.782 FZ 0.004
161. 2 FX -0.626 FY -1.063 FZ 0.004
162. 3 FX -1.13 FY 1.835 FZ 0.051
163. 4 FX -0.666 FY 1.093 FZ 0.014
164. 5 FX 0.117 FY -0.047 FZ -0.028
165. 6 FX -0.236 FY -0.035 FZ -0.039
166. 7 FX -0.777 FY -0.021 FZ -0.005
167. 8 FX -0.242 FY 0.021 FZ -0.029
168. 9 FX -0.487 FY -1.274 FZ -0.012
169. 10 FX -0.595 FY -0.277 FZ -0.016
170. 11 FX -0.487 FY 1.149 FZ -0.053
171. 12 FX -0.604 FY 0.402 FZ -0.045
172. 13 FX -0.78 FY -0.1 FZ -0.036
173. 14 FX 0.201 FY 0.1 FZ -0.022
174. 15 FX -0.535 FY -0.519 FZ -0.009
175. 16 FX -0.715 FY 0.097 FZ -0.026
176. 17 FX -0.536 FY 0.741 FZ 0.146
177. 18 FX 0.151 FY -0.319 FZ 0.101
178. LOAD 7 TL1 (TEMP LOAD +25)
179. JOINT LOAD
180. 1 FX -3.31 FY -8.689 FZ 0.084
181. 2 FX -0.012 FY 0.007 FZ 0.097
182. 3 FX -4.05 FY 9.102 FZ -3.109
183. 4 FX -0.979 FY -0.51 FZ 3.105
184. 5 FX 0.548 FY 0.194 FZ -2.892
185. 6 FX 1.41 FY -0.105 FZ 3.275
186. 7 FX -6.569 FY -0.069 FZ -5.565
187. 8 FX -0.065 FY 0.069 FZ 5.387
188. 9 FX -0.526 FY -0.609 FZ -2.866
189. 10 FX -1.127 FY 2.867 FZ 2.777
190. 11 FX 0.729 FY 0.225 FZ -2.981
191. 12 FX 0.324 FY -2.484 FZ 2.682
192. 13 FX 5.354 FY -0.17 FZ -2.95
193. 14 FX 0.835 FY 0.17 FZ 2.958
194. 15 FX 2.52 FY 2.867 FZ -3.553
195. 16 FX 0.006 FY 0.016 FZ 2.832
196. 17 FX 3.979 FY -2.868 FZ -3.042
197. 18 FX 0.934 FY -0.015 FZ 3.761
198. LOAD 8 TL2 (TEMP LOAD -25)
199. JOINT LOAD
200. 1 FX 3.31 FY 8.689 FZ -0.084
201. 2 FX 0.012 FY -0.007 FZ -0.097
202. 3 FX 4.05 FY -9.102 FZ 3.109
203. 4 FX 0.979 FY 0.51 FZ -3.105
204. 5 FX -0.548 FY -0.194 FZ 2.892
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205. 6 FX -1.41 FY 0.105 FZ -3.275
206. 7 FX 6.569 FY 0.069 FZ 5.565
207. 8 FX 0.065 FY -0.069 FZ -5.387
208. 9 FX 0.526 FY 0.609 FZ 2.866
209. 10 FX 1.127 FY -2.867 FZ -2.777
210. 11 FX -0.729 FY -0.225 FZ 2.981
211. 12 FX -0.324 FY 2.484 FZ -2.682
212. 13 FX -5.354 FY 0.17 FZ 2.95
213. 14 FX -0.835 FY -0.17 FZ -2.958
214. 15 FX -2.52 FY -2.867 FZ 3.553
215. 16 FX -0.006 FY -0.016 FZ -2.832
216. 17 FX -3.979 FY 2.868 FZ 3.042
217. 18 FX -0.934 FY 0.015 FZ -3.761
218. LOAD 9 EL (+X)
219. JOINT LOAD
220. 1 FX 0.183 FY 0.494 FZ -0.002
221. 2 FX 0.06 FY 0.167 FZ -0.002
222. 3 FX 0.188 FY -0.5 FZ -0.003
223. 4 FX 0.065 FY -0.158 FZ -0.003
224. 5 FX -0.005 FY -0.003 FZ 0.001
225. 6 FX 0.003 FY 0 FZ 0.001
226. 7 FX 0.258 FY 0.004 FZ 0.003
227. 8 FX 0.004 FY -0.004 FZ 0.006
228. 9 FX 0.121 FY 0.603 FZ 0.002
229. 10 FX 0.161 FY 0.299 FZ 0.004
230. 11 FX 0.12 FY -0.566 FZ 0.017
231. 12 FX 0.159 FY -0.336 FZ 0.014
232. 13 FX 0.261 FY 0.031 FZ 0.009
233. 14 FX 0.073 FY -0.031 FZ 0.008
234. 15 FX 0.148 FY 0.293 FZ -0.004
235. 16 FX 0.067 FY -0.006 FZ 0.002
236. 17 FX 0.15 FY -0.35 FZ -0.038
237. 18 FX 0 FY 0.063 FZ -0.014
238. LOAD 10 EL (+Z)
239. JOINT LOAD
240. 1 FX 0 FY 0.09 FZ 0.043
241. 2 FX -0.125 FY -0.433 FZ 0.043
242. 3 FX 0.008 FY 0.204 FZ 0.106
243. 4 FX -0.133 FY 0.137 FZ 0.099
244. 5 FX 0.006 FY 0.242 FZ 0.105
245. 6 FX -0.008 FY -0.24 FZ 0.11
246. 7 FX 0.072 FY 0.243 FZ 0.013
247. 8 FX 0 FY -0.243 FZ 0.314
248. 9 FX 0.033 FY 0.507 FZ 0.139
249. 10 FX 0.001 FY -0.335 FZ 0.14
250. 11 FX 0.032 FY 0.154 FZ 0.145
251. 12 FX 0.002 FY -0.326 FZ 0.144
252. 13 FX 0.074 FY 0.341 FZ 0.152
253. 14 FX 0.001 FY -0.341 FZ 0.148
254. 15 FX 0.002 FY 0.242 FZ 0.114
255. 16 FX 0.043 FY -0.212 FZ 0.121
256. 17 FX 0.002 FY 0.075 FZ 0.038
257. 18 FX -0.01 FY -0.105 FZ 0.04
258. ***LOAD COMBINATION FOR STRESS CALCULATION***
259. LOAD COMB 101 DL
260. 1 1.0
```

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261. ***
262. LOAD COMB 102 DL + LL + TL1
263. 1 1.0 2 1.0 7 1.0
264. LOAD COMB 103 DL + LL + TL2
265. 1 1.0 2 1.0 8 1.0
266. ***
267. LOAD COMB 104 DL + 0.75*WL(+Z) + 0.75*LL
268. 1 1.0 3 0.75 2 0.75
269. LOAD COMB 105 DL + 0.75*WL(-Z) + 0.75*LL
270. 1 1.0 4 0.75 2 0.75
271. LOAD COMB 106 DL + 0.75*WL(+X) + 0.75*LL
272. 1 1.0 5 0.75 2 0.75
273. LOAD COMB 107 DL + 0.75*WL(-X) + 0.75*LL
274. 1 1.0 6 0.75 2 0.75
275. ***
276. LOAD COMB 108 DL + 0.525*EL(+X) + 0.75*LL
277. 1 1.0 9 0.525 2 0.75
278. LOAD COMB 109 DL + 0.525*EL(-X) + 0.75*LL
279. 1 1.0 9 -0.525 2 0.75
280. LOAD COMB 110 DL + 0.525*EL(+Z) + 0.75*LL
281. 1 1.0 10 0.525 2 0.75
282. LOAD COMB 111 DL + 0.525*EL(-Z) + 0.75*LL
283. 1 1.0 10 -0.525 2 0.75
284. ***
285. LOAD COMB 112 0.6*DL + WL(+Z)
286. 1 0.6 3 1.0
287. LOAD COMB 113 0.6*DL +  WL(-Z)
288. 1 0.6 4 1.0
289. LOAD COMB 114 0.6*DL +  WL(+X)
290. 1 0.6 5 1.0
291. LOAD COMB 115 0.6*DL + WL(-X)
292. 1 0.6 6 1.0
293. ***
294. LOAD COMB 116 0.6*DL + 0.7*EL(+X)
295. 1 0.6 9 0.7
296. LOAD COMB 117 0.6*DL - 0.7*EL(+X)
297. 1 0.6 9 -0.7
298. LOAD COMB 118 0.6*DL + 0.7*EL(+Z)
299. 1 0.6 10 0.7
300. LOAD COMB 119 0.6*DL - 0.7*EL(+Z)
301. 1 0.6 10 -0.7
302. ***LOAD COMBINATION FOR STRENGTH CALCULATION***
303. LOAD COMB 201 1.4*DL
304. 1 1.4
305. LOAD COMB 202 1.4*(DL + TL1) + 1.7*LL
306. 1 1.4 7 1.4 2 1.7
307. LOAD COMB 203 1.4*(DL + TL2) + 1.7*LL
308. 1 1.4 8 1.4 2 1.7
309. ***
310. LOAD COMB 204 1.2*DL + 1.6*WL(+Z) + 0.5*LL
311. 1 1.2 3 1.6 2 0.5
312. LOAD COMB 205 1.2*DL + 1.6*WL(-Z) + 0.5*LL
313. 1 1.2 4 1.6 2 0.5
314. LOAD COMB 206 1.2*DL + 1.6*WL(+Z) + 0.5*LL
315. 1 1.2 5 1.6 2 0.5
316. LOAD COMB 207 1.2*DL + 1.6*WL(-Z) + 0.5*LL
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```
317. 1 1.2 6 1.6 2 0.5
318. ***
319. LOAD COMB 208 1.2*DL + 1.0*EL(+X) + 0.5*LL
320. 1 1.2 9 1.0 2 0.5
321. LOAD COMB 209 1.2*DL - 1.0*EL(+X) + 0.5*LL
322. 1 1.2 9 -1.0 2 0.5
323. LOAD COMB 210 1.2*DL + 1.0*EL(+Z) + 0.5*LL
324. 1 1.2 10 1.0 2 0.5
325. LOAD COMB 211 1.2*DL - 1.0*EL(+Z) + 0.5*LL
326. 1 1.2 10 -1.0 2 0.5
327. ***
328. LOAD COMB 212 0.9*DL + 1.6*WL(+Z)
329. 1 0.9 3 1.6
330. LOAD COMB 213 0.9*DL + 1.6*WL(-Z)
331. 1 0.9 4 1.6
332. LOAD COMB 214 0.9*DL + 1.6*WL(+X)
333. 1 0.9 5 1.6
334. LOAD COMB 215 0.9*DL + 1.6*WL(-X)
335. 1 0.9 6 1.6
336. ***
337. LOAD COMB 216 0.9*DL + 1.0*EL(+X)
338. 1 0.9 9 1.0
339. LOAD COMB 217 0.9*DL - 1.0*EL(+X)
340. 1 0.9 9 -1.0
341. LOAD COMB 218 0.9*DL + 1.0*EL(+Z)
342. 1 0.9 10 1.0
343. LOAD COMB 219 0.9*DL - 1.0*EL(+Z)
344. 1 0.9 10 -1.0
345. ***
346. PERFORM ANALYSIS PRINT STATICS CHECK


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS         36  NUMBER OF MEMBERS      18
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS     18

        Using 64-bit analysis engine.

        SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL      PRIMARY LOAD CASES =    10, TOTAL DEGREES OF FREEDOM =     108
TOTAL LOAD COMBINATION  CASES =    38  SO FAR.
```

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.176479929E+02
Y = 0.913473634E+03
Z = 0.450012700E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.110142466E+02
Y = 0.912993779E+03
Z = 0.760060480E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.862350712E+01
Y = 0.913476149E+03
Z = 0.578193747E+01

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 3.94
SUMMATION FORCE-Y = -110.76
SUMMATION FORCE-Z = 3.93

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4436.35 MY= -16.22 MZ= -4815.35

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -3.94
SUMMATION FORCE-Y = 110.76
SUMMATION FORCE-Z = -3.93

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4436.35 MY= 16.22 MZ= 4815.35

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	3.32128E-02	13
Y =	-2.82478E-03	11
Z =	2.71363E-02	6
RX=	1.49950E-04	6
RY=	0.00000E+00	0
RZ=	-1.83527E-04	13

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.129748986E+02
Y = 0.913476198E+03
Z = 0.754777746E+01

TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -136.62
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1032.13 MY= 0.01 MZ= -1773.64

TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 136.62
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1032.13 MY= -0.01 MZ= 1773.64

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	2.66170E-02	13
Y =	-4.46092E-03	10
Z =	3.26299E-02	7
RX=	1.78066E-04	7
RY=	0.00000E+00	0
RZ=	-1.47080E-04	13

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
WIND LOAD (+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.102009251E+02
Y = 0.913484034E+03
Z = 0.797652026E+01

TOTAL APPLIED LOAD 3

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 13.19

```
      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=       12077.93  MY=        -138.00  MZ=          -1.79

TOTAL REACTION LOAD    3

   ***TOTAL REACTION LOAD( MTON METE ) SUMMARY (LOADING     3 )
       SUMMATION FORCE-X =          -0.00
       SUMMATION FORCE-Y =           0.00
       SUMMATION FORCE-Z =         -13.19

      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=      -12077.93  MY=         138.00  MZ=           1.79


   MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING      3)
            MAXIMUMS    AT NODE
      X = -1.03150E-02       1
      Y =  4.81090E-04       9
      Z =  2.74610E-02       1
      RX=  1.41536E-04       1
      RY=  0.00000E+00       0
      RZ=  5.31639E-05       1


          STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.     4
          WIND LOAD (-Z)


           CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
         (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                        X =  0.102009251E+02
                        Y =  0.913484034E+03
                        Z =  0.797652026E+01
TOTAL APPLIED LOAD     4

   ***TOTAL APPLIED LOAD ( MTON METE ) SUMMARY (LOADING     4 )
       SUMMATION FORCE-X =          -0.00
       SUMMATION FORCE-Y =           0.00
       SUMMATION FORCE-Z =         -13.19

      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=      -12077.93  MY=         138.00  MZ=           1.79

TOTAL REACTION LOAD    4

   ***TOTAL REACTION LOAD( MTON METE ) SUMMARY (LOADING     4 )
       SUMMATION FORCE-X =           0.00
       SUMMATION FORCE-Y =          -0.00
       SUMMATION FORCE-Z =          13.19

      SUMMATION OF MOMENTS AROUND THE ORIGIN-
      MX=       12077.93  MY=        -138.00  MZ=          -1.79
```

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	1.03150E-02	1
Y =	-4.81090E-04	9
Z =	-2.74610E-02	1
RX=	-1.41536E-04	1
RY=	0.00000E+00	0
RZ=	-5.31639E-05	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
WIND LOAD (+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.636624698E+01
Y = 0.913509897E+03
Z = 0.793430315E+01

TOTAL APPLIED LOAD 5

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 8.86
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.01 MY= 73.69 MZ= -8112.32

TOTAL REACTION LOAD 5

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -8.86
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.01 MY= -73.69 MZ= 8112.32

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	6.21630E-02	1
Y =	9.17270E-04	1
Z =	-3.79129E-03	17
RX=	-2.09499E-05	17
RY=	0.00000E+00	0
RZ=	-3.20391E-04	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
WIND LOAD (-X)

STAAD SPACE -- PAGE NO. 12

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.636624698E+01
Y = 0.913509897E+03
Z = 0.793430315E+01

TOTAL APPLIED LOAD 6

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -8.86
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.01 MY= -73.69 MZ= 8112.32

TOTAL REACTION LOAD 6

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 8.86
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.01 MY= 73.69 MZ= -8112.32

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X	-6.21630E-02	1
Y	-9.17270E-04	1
Z	3.79129E-03	17
RX	2.09499E-05	17
RY	0.00000E+00	0
RZ	3.20391E-04	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
TL1 (TEMP LOAD +25)

TOTAL APPLIED LOAD 7

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 1.0002774E-03
SUMMATION FORCE-Y = -1.9997149E-03
SUMMATION FORCE-Z = 3.3795267E-07

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.07 MY= 0.02 MZ= -0.15

TOTAL REACTION LOAD 7

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -1.0002774E-03
SUMMATION FORCE-Y = 1.9997149E-03
SUMMATION FORCE-Z = -3.3795267E-07

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SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.07 MY= -0.02 MZ= 0.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	-2.26109E-01	1
Y =	-4.47259E-03	1
Z =	9.76648E-02	18
RX=	5.39676E-04	18
RY=	0.00000E+00	0
RZ=	1.16538E-03	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
TL2 (TEMP LOAD -25)

TOTAL APPLIED LOAD 8

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -1.0002774E-03
SUMMATION FORCE-Y = 1.9997149E-03
SUMMATION FORCE-Z = -3.3795267E-07

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.07 MY= -0.02 MZ= 0.15

TOTAL REACTION LOAD 8

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 1.0002774E-03
SUMMATION FORCE-Y = -1.9997149E-03
SUMMATION FORCE-Z = 3.3795267E-07

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.07 MY= 0.02 MZ= -0.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	2.26109E-01	1
Y =	4.47259E-03	1
Z =	-9.76648E-02	18
RX=	-5.39676E-04	18
RY=	0.00000E+00	0
RZ=	-1.16538E-03	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
EL (+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.102890876E+02
Y = 0.913497734E+03
Z = 0.655500976E+01

TOTAL APPLIED LOAD 9

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 2.02
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.91 MY= 13.91 MZ= -1851.07

TOTAL REACTION LOAD 9

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = -2.02
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.91 MY= -13.91 MZ= 1851.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X =	1.25009E-02	1
Y =	2.54283E-04	1
Z =	-9.86775E-04	17
RX=	-5.45272E-06	17
RY=	0.00000E+00	0
RZ=	-6.44304E-05	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
EL (+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.992765644E+01
Y = 0.913479832E+03
Z = 0.803213733E+01

TOTAL APPLIED LOAD 10

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 2.01

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1849.21 MY= -22.45 MZ= 0.03

TOTAL REACTION LOAD 10

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -2.01

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1849.21 MY= 22.45 MZ= -0.03

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)

	MAXIMUMS	AT NODE
X =	-4.25436E-03	2
Y =	-1.75962E-04	2
Z =	4.45663E-03	8
RX=	2.43205E-05	8
RY=	0.00000E+00	0
RZ=	2.74303E-05	2

********** END OF DATA FROM INTERNAL STORAGE **********

347. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 1,2020 TIME= 12:48:41 ****

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     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://www.bentley.com/en/support/                     *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2017 Bentley Systems, Inc.       *
     *                http://www.bentley.com                    *
     ************************************************************
```

ATTACHMENT - 6 : STAAD OUTPUT

REAGENTS LIME STONE SILO FOUNDATION

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PAGE NO. 1

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*           STAAD.Pro CONNECT Edition           *
*           Version  22.00.00.15                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    OCT  1, 2020               *
*           Time=    12: 3:57                   *
*                                               *
*  Licensed to: Larsen and Toubro Ltd.          *
*************************************************
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INPUT FILE: D:\MAADEN-GOLD-O19045\3860 REAGENTS LIME STONE\CIVIL\LATEST\3860_Lime Stone Silo_Plate Mat.. .STD
     2. START JOB INFORMATION
     3. JOB NAME MMGP
     4. JOB CLIENT MAADEN
     5. JOB NO O19074
     6. JOB PART 3860 - LIME STONE SILO FOUNDATION
     7. ENGINEER NAME SUE
     8. CHECKER NAME BDA
     9. APPROVED NAME AG
    10. ENGINEER DATE 9/29/2020
    11. END JOB INFORMATION
    12. INPUT WIDTH 79
    13. SET NL 90
    14. SET ITERLIM 40
    15. SET PRINT 1
    16. UNIT METER MTON
    17. JOINT COORDINATES
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    19. 5 0 909.85 3.107; 6 3.107 909.85 3.107; 7 5 909.85 3.107; 8 8.107 909.85 3.107
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    21. 13 0 914.107 3.107; 14 3.107 914.107 3.107; 15 5 914.107 3.107
    22. 16 8.107 914.107 3.107; 100 1.557 909.85 2.057; 101 4.05 909.85 2.057
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    25. 108 6.557 909.85 1.557; 109 3.607 909.85 2.057; 110 1.557 909.85 4.157
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    32. 129 5 909.85 -1.05; 130 6.05 909.85 -1.05; 131 7.607 909.85 -1.05
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172. 1108 259 254 243 2; 1109 142 256 244 139; 1110 143 373 246 140
173. 1111 373 257 247 246; 1112 374 260 250 249; 1113 261 144 367 251
174. 1114 262 145 368 410; 1115 263 264 411 252; 1116 264 146 369 411
175. 1117 265 147 370 253; 1118 144 266 371 367; 1119 267 262 410 258
176. 1120 145 412 141 368; 1121 412 268 372 141; 1122 269 375 254 259
177. 1123 375 413 142 254; 1124 413 377 256 142; 1125 377 263 252 256
178. 1126 146 414 143 369; 1127 414 270 373 143; 1128 378 265 253 257
179. 1129 147 271 374 370; 1130 266 267 258 371; 1131 268 376 255 372
180. 1132 376 269 259 255; 1133 270 378 257 373; 1134 271 272 260 374
181. 1135 149 415 266 144; 1136 415 150 267 266; 1137 150 273 262 267
182. 1138 273 151 145 262; 1139 151 106 412 145; 1140 106 416 268 412
183. 1141 416 152 376 268; 1142 152 153 269 376; 1143 153 274 375 269
184. 1144 274 107 413 375; 1145 417 154 263 377; 1146 154 275 264 263
185. 1147 275 155 146 264; 1148 155 108 414 146; 1149 108 418 270 414
186. 1150 418 156 378 270; 1151 156 157 265 378; 1152 158 419 271 147
187. 1153 419 159 272 271; 1154 276 277 149 148; 1155 277 160 415 149
188. 1156 160 278 150 415; 1157 278 161 273 150; 1158 161 279 151 273
189. 1159 162 280 152 416; 1160 280 281 153 152; 1161 282 283 154 417
190. 1162 283 163 275 154; 1163 163 284 155 275; 1164 164 285 156 418
191. 1165 285 286 157 156; 1166 286 287 158 157; 1167 287 165 419 158
192. 1168 165 288 159 419; 1169 379 289 277 276; 1170 289 290 160 277
193. 1171 290 380 278 160; 1172 432 291 279 161; 1173 291 433 100 279
194. 1174 433 292 162 100; 1175 292 293 280 162; 1176 293 381 281 280
195. 1177 381 382 109 281; 1178 382 434 101 109; 1179 434 166 282 101
196. 1180 166 383 283 282; 1181 435 294 284 163; 1182 294 436 102 284
197. 1183 436 295 164 102; 1184 295 296 285 164; 1185 296 384 286 285
198. 1186 384 297 287 286; 1187 297 298 165 287; 1188 298 385 288 165
199. 1189 299 300 289 379; 1190 300 301 290 289; 1191 301 5 380 290
200. 1192 302 167 433 291; 1193 167 303 292 433; 1194 303 304 293 292
201. 1195 304 6 381 293; 1196 6 305 382 381; 1197 168 306 166 434
202. 1198 306 7 383 166; 1199 307 169 436 294; 1200 169 308 295 436
203. 1201 308 309 296 295; 1202 309 8 384 296; 1203 8 310 297 384
204. 1204 310 311 298 297; 1205 311 312 385 298; 1206 313 420 300 299
205. 1207 420 170 301 300; 1208 170 314 5 301; 1209 314 171 386 5
206. 1210 171 421 302 386; 1211 422 172 303 167; 1212 172 315 304 303

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207. 1213 315 316 6 304; 1214 316 173 305 6; 1215 173 423 168 305
208. 1216 423 317 306 168; 1217 317 318 7 306; 1218 318 174 387 7
209. 1219 174 424 307 387; 1220 425 175 308 169; 1221 175 319 309 308
210. 1222 319 320 8 309; 1223 320 426 310 8; 1224 426 176 311 310
211. 1225 176 321 312 311; 1226 177 178 420 313; 1227 178 427 170 420
212. 1228 427 179 314 170; 1229 179 322 171 314; 1230 322 180 421 171
213. 1231 180 110 422 421; 1232 110 428 172 422; 1233 428 181 315 172
214. 1234 181 182 316 315; 1235 182 323 173 316; 1236 323 111 423 173
215. 1237 111 429 317 423; 1238 429 183 318 317; 1239 183 324 174 318
216. 1240 324 184 424 174; 1241 184 112 425 424; 1242 112 430 175 425
217. 1243 430 185 319 175; 1244 185 186 320 319; 1245 186 187 426 320
218. 1246 187 431 176 426; 1247 431 188 321 176; 1248 325 326 178 177
219. 1249 326 388 427 178; 1250 388 327 179 427; 1251 327 189 322 179
220. 1252 189 328 180 322; 1253 328 389 110 180; 1254 389 390 428 110
221. 1255 390 329 181 428; 1256 329 330 182 181; 1257 330 391 323 182
222. 1258 391 392 111 323; 1259 392 190 429 111; 1260 190 331 183 429
223. 1261 331 191 324 183; 1262 191 332 184 324; 1263 332 393 112 184
224. 1264 393 394 430 112; 1265 394 333 185 430; 1266 333 334 186 185
225. 1267 334 335 187 186; 1268 335 395 431 187; 1269 395 336 188 431
226. 1270 337 338 326 325; 1271 338 339 388 326; 1272 339 340 327 388
227. 1273 340 396 189 327; 1274 396 341 328 189; 1275 341 192 389 328
228. 1276 192 342 390 389; 1277 342 343 329 390; 1278 343 344 330 329
229. 1279 344 345 391 330; 1280 345 193 392 391; 1281 193 346 190 392
230. 1282 346 347 331 190; 1283 347 397 191 331; 1284 397 348 332 191
231. 1285 348 194 393 332; 1286 194 349 394 393; 1287 349 350 333 394
232. 1288 350 351 334 333; 1289 351 352 335 334; 1290 352 353 395 335
233. 1291 353 354 336 395; 1292 399 400 113 199; 1293 402 403 115 206
234. 1294 410 368 240 365; 1295 252 411 366 3; 1296 411 369 245 366
235. 1297 258 410 365 1; 1298 107 417 377 413; 1299 380 432 161 278
236. 1300 383 435 163 283; 1301 5 386 432 380; 1302 386 302 291 432
237. 1303 305 168 434 382; 1304 7 387 435 383; 1305 387 307 294 435
238. 1306 421 422 167 302; 1307 424 425 169 307
239. START GROUP DEFINITION
240. ELEMENT
241. _F5_BP 1000 TO 1307
242. _F5_BM 1000 TO 1074 1077 TO 1080 1083 1084 1087 TO 1090 1093 TO 1097 1099 -
243. 1102 1103 1105 1106 1109 TO 1190 1192 TO 1194 1197 1199 TO 1201 1204 TO 1207 -
244. 1210 TO 1212 1215 1216 1219 TO 1221 1224 TO 1294 1296 1298 TO 1300 1302 1303 -
245. 1305 TO 1307
246. _F5_SF 1000 TO 1051 1056 1057 1066 1067 1072 1073 1078 1079 1088 1089 1094 -
247. 1095 TO 1096 1099 1105 1110 1112 1113 1120 1121 1126 1127 1134 TO 1169 1173 -
248. 1174 1182 1183 1188 1189 1192 1193 1199 1200 1205 1206 1211 1220 1225 1226 -
249. 1231 1232 1241 1242 1247 TO 1293 1298 1306 1307
250. END GROUP DEFINITION
251. ELEMENT PROPERTY
252. 1000 TO 1307 THICKNESS 0.65
253. DEFINE MATERIAL START
254. ISOTROPIC CONCRETE
255. E 2.21467E+06
256. POISSON 0.17
257. DENSITY 2.40261
258. ALPHA 1E-05
259. DAMP 0.05
260. TYPE CONCRETE
261. STRENGTH FCU 2812.28
262. END DEFINE MATERIAL
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263. MEMBER PROPERTY INDIAN
264. 1 TO 8 PRIS YD 1 ZD 1
265. CONSTANTS
266. MATERIAL CONCRETE ALL
267. SUPPORTS
268. 1000 TO 1307 PLATE MAT DIRECT Y SUBGRADE 8040
269. LOAD 1 SELFWEIGHT DEAD LOAD (SWDL)
270. SELFWEIGHT Y -1
271. JOINT LOAD
272. 9 FX -0.3 FY -6.7 FZ -0.2
273. 10 FX 0.2 FY -6.7 FZ -0.3
274. 13 FX -0.3 FY -7.1 FZ 0.3
275. 14 FX 0.2 FY -7.1 FZ 0.2
276. 11 FX -0.2 FY -6.7 FZ -0.3
277. 12 FX 0.3 FY -6.7 FZ -0.2
278. 15 FX -0.2 FY -7.1 FZ 0.2
279. 16 FX 0.3 FY -7.1 FZ 0.3
280. ELEMENT LOAD
281. *** SOIL LOAD = (2.5-0.65)*1.9 = 3.515 MTON/M2 ***
282. 1000 TO 1307 PR GY -3.515
283. LOAD 2 DEAD LOAD (DL)
284. JOINT LOAD
285. 9 FX -0.1 FY -4 FZ -0.4
286. 10 FX 0.2 FY -3.6 FZ -0.3
287. 13 FX -0.3 FY -10.2 FZ 0.3
288. 14 FX 0.5 FY -9.6 FZ 0.3
289. 11 FX -0.2 FY -3.6 FZ -0.3
290. 12 FX 0.1 FY -4 FZ -0.3
291. 15 FX -0.5 FY -9.6 FZ 0.3
292. 16 FX 0.3 FY -10.2 FZ 0.3
293. LOAD 3 LIVE LOAD (LL)
294. JOINT LOAD
295. 9 FX -3.6 FY -96 FZ -3.2
296. 10 FX 2.5 FY -95 FZ -3.3
297. 13 FX -3.7 FY -100.7 FZ 3.3
298. 14 FX 2.7 FY -99.6 FZ 3.2
299. 11 FX -2.5 FY -95 FZ -3.3
300. 12 FX 3.6 FY -96 FZ -3.1
301. 15 FX -2.7 FY -99.5 FZ 3.2
302. 16 FX 3.7 FY -100.7 FZ 3.3
303. LOAD 4 WIND LOAD - 0 (WL-0)
304. JOINT LOAD
305. 9 FX 5.1 FY 56.7 FZ 1.9
306. 10 FX 5.1 FY -56.5 FZ -1.8
307. 13 FX 4.9 FY 57.8 FZ -1.7
308. 14 FX 4.9 FY -57.2 FZ 1.7
309. 11 FX 5.1 FY 56.6 FZ 1.8
310. 12 FX 5.1 FY -56.7 FZ -1.9
311. 15 FX 4.9 FY 57.2 FZ -1.7
312. 16 FX 4.9 FY -57.8 FZ 1.7
313. LOAD 5 WIND LOAD - 45 (WL-45)
314. JOINT LOAD
315. 9 FX 5.1 FY 80.8 FZ 4.9
316. 10 FX 2.6 FY 0.4 FZ 3.3
317. 13 FX 2 FY 0.3 FZ 2.3
318. 14 FX 4.5 FY -81 FZ 4.8
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319. 11 FX 4.6 FY 80.4 FZ 4.8
320. 12 FX 2.2 FY 0.6 FZ 2.2
321. 15 FX 2.5 FY 0 FZ 2.3
322. 16 FX 4.9 FY -81.5 FZ 4.7
323. LOAD 6 WIND LOAD - 90 (WL-90)
324. JOINT LOAD
325. 9 FX 2 FY 57.6 FZ 5
326. 10 FX -1.4 FY 57.1 FZ 5
327. 13 FX -2 FY -57.4 FZ 5
328. 14 FX 1.4 FY -57.3 FZ 5.1
329. 11 FX 1.4 FY 57.1 FZ 5
330. 12 FX -2.1 FY 57.6 FZ 5
331. 15 FX -1.4 FY -57.3 FZ 5
332. 16 FX 2 FY -57.4 FZ 5
333. LOAD 7 WIND LOAD - 135 (WL-135)
334. JOINT LOAD
335. 9 FX -2.2 FY 0.6 FZ 2.2
336. 10 FX -4.6 FY 80.4 FZ 4.8
337. 13 FX -4.9 FY -81.4 FZ 4.7
338. 14 FX -2.5 FY -0.1 FZ 2.3
339. 11 FX -2.6 FY 0.4 FZ 2.3
340. 12 FX -5.1 FY 80.8 FZ 4.9
341. 15 FX -4.5 FY -81 FZ 4.8
342. 16 FX -2 FY 0.3 FZ 2.3
343. LOAD 8 WIND LOAD - 180 (WL-180)
344. JOINT LOAD
345. 9 FX -5.1 FY -56.7 FZ -1.9
346. 10 FX -5.1 FY 56.5 FZ 1.8
347. 13 FX -4.9 FY -57.8 FZ 1.7
348. 14 FX -4.9 FY 57.2 FZ -1.7
349. 11 FX -5.1 FY -56.6 FZ -1.8
350. 12 FX -5.1 FY 56.7 FZ 1.9
351. 15 FX -4.9 FY -57.2 FZ 1.7
352. 16 FX -4.9 FY 57.8 FZ -1.7
353. LOAD 9 WIND LOAD - 225 (WL-225)
354. JOINT LOAD
355. 9 FX -5.1 FY -80.8 FZ -4.9
356. 10 FX -2.6 FY -0.4 FZ -2.3
357. 13 FX -2 FY -0.3 FZ -2.3
358. 14 FX -4.5 FY -81 FZ -4.8
359. 11 FX -4.6 FY -80.4 FZ -4.8
360. 12 FX -2.2 FY -0.6 FZ -2.2
361. 15 FX -2.5 FY 0 FZ -2.3
362. 16 FX -4.9 FY 81.5 FZ -4.7
363. LOAD 10 WIND LOAD - 270 (WL-270)
364. JOINT LOAD
365. 9 FX -2 FY -57.6 FZ -5
366. 10 FX 1.4 FY -57.1 FZ -5
367. 13 FX 2 FY 57.4 FZ -5
368. 14 FX -1.4 FY -57.3 FZ -5.1
369. 11 FX -1.4 FY 67.1 FZ -5
370. 12 FX 2.1 FY -57.6 FZ -5
371. 15 FX 1.4 FY 57.3 FZ -5
372. 16 FX -2 FY 57.4 FZ -5
373. LOAD 11 WIND LOAD - 315 (WL315)
374. JOINT LOAD
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375. 9 FX 2.2 FY -0.6 FZ -2.2
376. 10 FX 4.6 FY -80.4 FZ -4.8
377. 13 FX 4.9 FY 81.4 FZ -4.7
378. 14 FX 2.5 FY 0.1 FZ -2.3
379. 11 FX 2.6 FY -0.4 FZ -2.3
380. 12 FX 5.1 FY -80.8 FZ -4.9
381. 15 FX 4.5 FY -81 FZ -4.8
382. 16 FX 2 FY -0.3 FZ -2.3
383. LOAD 12 EARTHQUAKE LOAD - 0 (EL-0)
384. JOINT LOAD
385. 9 FX 3.8 FY 42.6 FZ 1.4
386. 10 FX 3.8 FY -42.5 FZ -1.3
387. 13 FX 3.7 FY 43.5 FZ -1.3
388. 14 FX 3.7 FY -43 FZ 1.3
389. 11 FX 3.8 FY 42.5 FZ 1.3
390. 12 FX 3.8 FY -42.6 FZ -1.4
391. 15 FX 3.7 FY 43 FZ -1.3
392. 16 FX 3.7 FY -43.5 FZ 1.3
393. LOAD 13 EARTHQUAKE LOAD - 45 (EL-45)
394. JOINT LOAD
395. 9 FX 3.8 FY 60.8 FZ 3.7
396. 10 FX 2 FY 0.3 FZ 1.7
397. 13 FX 1.5 FY 0.2 FZ 1.7
398. 14 FX 3.4 FY -60.9 FZ 3.6
399. 11 FX 3.5 FY 60.4 FZ 3.6
400. 12 FX 1.6 FY 0.5 FZ 1.7
401. 15 FX 1.8 FY 0 FZ 1.8
402. 16 FX 3.7 FY -61.3 FZ 3.6
403. LOAD 14 EARTHQUAKE LOAD - 90 (EL-90)
404. JOINT LOAD
405. 9 FX 1.5 FY 43.3 FZ 3.8
406. 10 FX -1.1 FY 43 FZ 3.8
407. 13 FX -1.5 FY -43.2 FZ 3.7
408. 14 FX 1.1 FY -43.1 FZ 3.8
409. 11 FX 1.1 FY 42.9 FZ 3.8
410. 12 FX -1.5 FY 43.3 FZ 3.8
411. 15 FX -1.1 FY -43.1 FZ 3.8
412. 16 FX 1.5 FY -43.2 FZ 3.8
413. LOAD 15 EARTHQUAKE LOAD - 135 (EL-135)
414. JOINT LOAD
415. 9 FX -1.6 FY 0.5 FZ 1.7
416. 10 FX -3.5 FY 60.4 FZ 3.6
417. 13 FX -3.7 FY -61.2 FZ 3.6
418. 14 FX -1.8 FY -0.1 FZ 1.8
419. 11 FX -2 FY 0.3 FZ 1.7
420. 12 FX -3.8 FY 60.8 FZ 3.7
421. 15 FX -3.4 FY -60.9 FZ 3.6
422. 16 FX -1.5 FY 0.2 FZ 1.7
423. LOAD 16 EARTHQUAKE LOAD - 180 (EL-180)
424. JOINT LOAD
425. 9 FX -3.8 FY -42.6 FZ -1.4
426. 10 FX -3.8 FY 42.5 FZ 1.3
427. 13 FX -3.7 FY -43.5 FZ 1.3
428. 14 FX -3.7 FY 43 FZ -1.3
429. 11 FX -3.8 FY -42.5 FZ -1.3
430. 12 FX -3.8 FY 42.6 FZ 1.4

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431. 15 FX -3.7 FY -43 FZ 1.3
432. 16 FX -3.7 FY 43.5 FZ -1.3
433. LOAD 17 EARTHQUAKE LOAD - 225 (EL-225)
434. JOINT LOAD
435. 9 FX -3.8 FY -60.8 FZ -3.7
436. 10 FX -2 FY -0.3 FZ -1.7
437. 13 FX -1.5 FY -0.2 FZ -1.7
438. 14 FX -3.4 FY 60.9 FZ -3.6
439. 11 FX -3.5 FY -60.4 FZ -3.6
440. 12 FX -1.6 FY -0.5 FZ -1.7
441. 15 FX -1.8 FY 0 FZ -1.8
442. 16 FX -3.7 FY 61.3 FZ -3.6
443. LOAD 18 EARTHQUAKE LOAD - 270 (EL-270)
444. JOINT LOAD
445. 9 FX -1.5 FY -43.3 FZ -3.8
446. 10 FX 1.1 FY -43 FZ -33.8
447. 13 FX 1.5 FY 43.2 FZ -3.7
448. 14 FX -1.1 FY 43.1 FZ -3.8
449. 11 FX -1.1 FY -42.9 FZ -3.8
450. 12 FX 1.5 FY -43.3 FZ -3.8
451. 15 FX 1.1 FY 43.1 FZ -3.8
452. 16 FX -1.5 FY 43.2 FZ -3.8
453. LOAD 19 EARTHQUAKE LOAD - 315 (EL315)
454. JOINT LOAD
455. 9 FX 1.6 FY -0.5 FZ -1.7
456. 10 FX 3.5 FY -60.4 FZ -3.6
457. 13 FX 3.7 FY 61.2 FZ -3.6
458. 14 FX 1.8 FY 0.1 FZ -1.8
459. 11 FX 2 FY -0.3 FZ -1.7
460. 12 FX 3.8 FY -60.8 FZ -3.7
461. 15 FX -3.4 FY -60.9 FZ -3.6
462. 16 FX 1.5 FY -0.2 FZ -1.7
463. ***
464. PERFORM ANALYSIS


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS        353  NUMBER OF MEMBERS       8
  NUMBER OF PLATES        308  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    345

        Using 64-bit analysis engine.

        SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL      PRIMARY LOAD CASES =    19, TOTAL DEGREES OF FREEDOM =    1083
TOTAL LOAD COMBINATION  CASES =     0  SO FAR.
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465. CHANGE
466. SUPPORTS
467. 1000 TO 1307 PLATE MAT DIRECT Y SUBGRADE 8040 COMPRESSION
468. ***LOAD COMBINATION FOR STRESS CALCULATION***
469. LOAD 101 SWDL + DL + LL
470. REPEAT LOAD
471. 1 1.0 2 1.0 3 1.0
472. ***
473. LOAD 102 SWDL + DL + 0.75*WL-0 + 0.75*LL
474. REPEAT LOAD
475. 1 1.0 2 1.0 4 0.75 3 0.75
476. LOAD 103 SWDL + DL + 0.75*WL-45 + 0.75*LL
477. REPEAT LOAD
478. 1 1.0 2 1.0 5 0.75 3 0.75
479. LOAD 104 SWDL + DL + 0.75*WL-90 + 0.75*LL
480. REPEAT LOAD
481. 1 1.0 2 1.0 6 0.75 3 0.75
482. LOAD 105 SWDL + DL + 0.75*WL-135 + 0.75*LL
483. REPEAT LOAD
484. 1 1.0 2 1.0 7 0.75 3 0.75
485. LOAD 106 SWDL + DL + 0.75*WL-180 + 0.75*LL
486. REPEAT LOAD
487. 1 1.0 2 1.0 8 0.75 3 0.75
488. LOAD 107 SWDL + DL + 0.75*WL-225 + 0.75*LL
489. REPEAT LOAD
490. 1 1.0 2 1.0 9 0.75 3 0.75
491. LOAD 108 SWDL + DL + 0.75*WL-270 + 0.75*LL
492. REPEAT LOAD
493. 1 1.0 2 1.0 10 0.75 3 0.75
494. LOAD 109 SWDL + DL + 0.75*WL-315 + 0.75*LL
495. REPEAT LOAD
496. 1 1.0 2 1.0 11 0.75 3 0.75
497. ***
498. LOAD 110 SWDL + DL + 0.525*EL-0 + 0.75*LL
499. REPEAT LOAD
500. 1 1.0 2 1.0 12 0.525 3 0.75
501. LOAD 111 SWDL + DL + 0.525*EL-45 + 0.75*LL
502. REPEAT LOAD
503. 1 1.0 2 1.0 13 0.525 3 0.75
504. LOAD 112 SWDL + DL + 0.525*EL-90 + 0.75*LL
505. REPEAT LOAD
506. 1 1.0 2 1.0 14 0.525 3 0.75
507. LOAD 113 SWDL + DL + 0.525*EL-135 + 0.75*LL
508. REPEAT LOAD
509. 1 1.0 2 1.0 15 0.525 3 0.75
510. LOAD 114 SWDL + DL + 0.525*EL-180 + 0.75*LL
511. REPEAT LOAD
512. 1 1.0 2 1.0 16 0.525 3 0.75
513. LOAD 115 SWDL + DL + 0.525*EL-225 + 0.75*LL
514. REPEAT LOAD
515. 1 1.0 2 1.0 17 0.525 3 0.75
516. LOAD 116 SWDL + DL + 0.525*EL-270 + 0.75*LL
517. REPEAT LOAD
518. 1 1.0 2 1.0 18 0.525 3 0.75
519. LOAD 117 SWDL + DL + 0.525*EL-315 + 0.75*LL
520. REPEAT LOAD
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521. 1 1.0 2 1.0 19 0.525 3 0.75
522. ***
523. LOAD 118 0.6*(SWDL + DL) + WL-0
524. REPEAT LOAD
525. 1 0.6 2 0.6 4 1.0
526. LOAD 119 0.6*(SWDL + DL) + WL-45
527. REPEAT LOAD
528. 1 0.6 2 0.6 5 1.0
529. LOAD 120 0.6*(SWDL + DL) + WL-90
530. REPEAT LOAD
531. 1 0.6 2 0.6 6 1.0
532. LOAD 121 0.6*(SWDL + DL) + WL-135
533. REPEAT LOAD
534. 1 0.6 2 0.6 7 1.0
535. LOAD 122 0.6*(SWDL + DL) + WL-180
536. REPEAT LOAD
537. 1 0.6 2 0.6 8 1.0
538. LOAD 123 0.6*(SWDL + DL) + WL-225
539. REPEAT LOAD
540. 1 0.6 2 0.6 9 1.0
541. LOAD 124 0.6*(SWDL + DL) + WL-270
542. REPEAT LOAD
543. 1 0.6 2 0.6 10 1.0
544. LOAD 125 0.6*(SWDL + DL) + WL-315
545. REPEAT LOAD
546. 1 0.6 2 0.6 11 1.0
547. ***
548. LOAD 126 0.6*(SWDL + DL) + 0.7*EL-0
549. REPEAT LOAD
550. 1 0.6 2 0.6 12 0.7
551. LOAD 127 0.6*(SWDL + DL) + 0.7*EL-45
552. REPEAT LOAD
553. 1 0.6 2 0.6 13 0.7
554. LOAD 128 0.6*(SWDL + DL) + 0.7*EL-90
555. REPEAT LOAD
556. 1 0.6 2 0.6 14 0.7
557. LOAD 129 0.6*(SWDL + DL) + 0.7*EL-135
558. REPEAT LOAD
559. 1 0.6 2 0.6 15 0.7
560. LOAD 130 0.6*(SWDL + DL) + 0.7*EL-180
561. REPEAT LOAD
562. 1 0.6 2 0.6 16 0.7
563. LOAD 131 0.6*(SWDL + DL) + 0.7*EL-225
564. REPEAT LOAD
565. 1 0.6 2 0.6 17 0.7
566. LOAD 132 0.6*(SWDL + DL) + 0.7*EL-270
567. REPEAT LOAD
568. 1 0.6 2 0.6 18 0.7
569. LOAD 133 0.6*(SWDL + DL) + 0.7*EL-315
570. REPEAT LOAD
571. 1 0.6 2 0.6 19 0.7
572. ***LOAD COMBINATION FOR STRENGTH CALCULATION***
573. LOAD 201 1.4*(SWDL + DL)
574. REPEAT LOAD
575. 1 1.4 2 1.4
576. ***
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577. LOAD 202 1.4*(SWDL + DL) + 1.7*LL
578. REPEAT LOAD
579. 1 1.4 2 1.4 3 1.7
580. ***
581. LOAD 203 1.2*(SWDL + DL) + 1.6*WL-0 + 0.5*LL
582. REPEAT LOAD
583. 1 1.2 2 1.2 4 1.6 3 0.5
584. LOAD 204 1.2*(SWDL + DL) + 1.6*WL-45 + 0.5*LL
585. REPEAT LOAD
586. 1 1.2 2 1.2 5 1.6 3 0.5
587. LOAD 205 1.2*(SWDL + DL) + 1.6*WL-90 + 0.5*LL
588. REPEAT LOAD
589. 1 1.2 2 1.2 6 1.6 3 0.5
590. LOAD 206 1.2*(SWDL + DL) + 1.6*WL-135 + 0.5*LL
591. REPEAT LOAD
592. 1 1.2 2 1.2 7 1.6 3 0.5
593. LOAD 207 1.2*(SWDL + DL) + 1.6*WL-180 + 0.5*LL
594. REPEAT LOAD
595. 1 1.2 2 1.2 8 1.6 3 0.5
596. LOAD 208 1.2*(SWDL + DL) + 1.6*WL-225 + 0.5*LL
597. REPEAT LOAD
598. 1 1.2 2 1.2 9 1.6 3 0.5
599. LOAD 209 1.2*(SWDL + DL) + 1.6*WL-270 + 0.5*LL
600. REPEAT LOAD
601. 1 1.2 2 1.2 10 1.6 3 0.5
602. LOAD 210 1.2*(SWDL + DL) + 1.6*WL-315 + 0.5*LL
603. REPEAT LOAD
604. 1 1.2 2 1.2 11 1.6 3 0.5
605. ***
606. LOAD 211 1.2*(SWDL + DL) + EL-0 + 0.5*LL
607. REPEAT LOAD
608. 1 1.2 2 1.2 12 1.0 3 0.5
609. LOAD 212 1.2*(SWDL + DL) + EL-45 + 0.5*LL
610. REPEAT LOAD
611. 1 1.2 2 1.2 13 1.0 3 0.5
612. LOAD 213 1.2*(SWDL + DL) + EL-90 + 0.5*LL
613. REPEAT LOAD
614. 1 1.2 2 1.2 14 1.0 3 0.5
615. LOAD 214 1.2*(SWDL + DL) + EL-135 + 0.5*LL
616. REPEAT LOAD
617. 1 1.2 2 1.2 15 1.0 3 0.5
618. LOAD 215 1.2*(SWDL + DL) + EL-180 + 0.5*LL
619. REPEAT LOAD
620. 1 1.2 2 1.2 16 1.0 3 0.5
621. LOAD 216 1.2*(SWDL + DL) + EL-225 + 0.5*LL
622. REPEAT LOAD
623. 1 1.2 2 1.2 17 1.0 3 0.5
624. LOAD 217 1.2*(SWDL + DL) + EL-270 + 0.5*LL
625. REPEAT LOAD
626. 1 1.2 2 1.2 18 1.0 3 0.5
627. LOAD 218 1.2*(SWDL + DL) + EL-315 + 0.5*LL
628. REPEAT LOAD
629. 1 1.2 2 1.2 19 1.0 3 0.5
630. ***
631. LOAD 219 0.9*(SWDL + DL) + 1.6*WL-0
632. REPEAT LOAD
```

```
633. 1 0.9 2 0.9 4 1.6
634. LOAD 220 0.9*(SWDL + DL) + 1.6*WL-45
635. REPEAT LOAD
636. 1 0.9 2 0.9 5 1.6
637. LOAD 221 0.9*(SWDL + DL) + 1.6*WL-90
638. REPEAT LOAD
639. 1 0.9 2 0.9 6 1.6
640. LOAD 222 0.9*(SWDL + DL) + 1.6*WL-135
641. REPEAT LOAD
642. 1 0.9 2 0.9 7 1.6
643. LOAD 223 0.9*(SWDL + DL) + 1.6*WL-180
644. REPEAT LOAD
645. 1 0.9 2 0.9 8 1.6
646. LOAD 224 0.9*(SWDL + DL) + 1.6*WL-225
647. REPEAT LOAD
648. 1 0.9 2 0.9 9 1.6
649. LOAD 225 0.9*(SWDL + DL) + 1.6*WL-270
650. REPEAT LOAD
651. 1 0.9 2 0.9 10 1.6
652. LOAD 226 0.9*(SWDL + DL) + 1.6*WL-315
653. REPEAT LOAD
654. 1 0.9 2 0.9 11 1.6
655. ***
656. LOAD 227 0.9*(SWDL + DL) + EL-0
657. REPEAT LOAD
658. 1 0.9 2 0.9 12 1.0
659. LOAD 228 0.9*(SWDL + DL) + EL-45
660. REPEAT LOAD
661. 1 0.9 2 0.9 13 1.0
662. LOAD 229 0.9*(SWDL + DL) + EL-90
663. REPEAT LOAD
664. 1 0.9 2 0.9 14 1.0
665. LOAD 230 0.9*(SWDL + DL) + EL-135
666. REPEAT LOAD
667. 1 0.9 2 0.9 15 1.0
668. LOAD 231 0.9*(SWDL + DL) + EL-180
669. REPEAT LOAD
670. 1 0.9 2 0.9 16 1.0
671. LOAD 232 0.9*(SWDL + DL) + EL-225
672. REPEAT LOAD
673. 1 0.9 2 0.9 17 1.0
674. LOAD 233 0.9*(SWDL + DL) + EL-270
675. REPEAT LOAD
676. 1 0.9 2 0.9 18 1.0
677. LOAD 234 0.9*(SWDL + DL) + EL-315
678. REPEAT LOAD
679. 1 0.9 2 0.9 19 1.0
680. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

STAAD SPACE -- PAGE NO. 15

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

*** LOAD CASE 118 -- START ITERATION NO. 2
*** LOAD CASE 118 -- START ITERATION NO. 3
*** LOAD CASE 118 -- START ITERATION NO. 4
*** LOAD CASE 118 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 118

*** LOAD CASE 119 -- START ITERATION NO. 2
*** LOAD CASE 119 -- START ITERATION NO. 3
*** LOAD CASE 119 -- START ITERATION NO. 4
*** LOAD CASE 119 -- START ITERATION NO. 5
*** LOAD CASE 119 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 119

*** LOAD CASE 120 -- START ITERATION NO. 2
*** LOAD CASE 120 -- START ITERATION NO. 3
*** LOAD CASE 120 -- START ITERATION NO. 4
*** LOAD CASE 120 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 120

*** LOAD CASE 121 -- START ITERATION NO. 2
*** LOAD CASE 121 -- START ITERATION NO. 3
*** LOAD CASE 121 -- START ITERATION NO. 4
*** LOAD CASE 121 -- START ITERATION NO. 5
*** LOAD CASE 121 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 121

*** LOAD CASE 122 -- START ITERATION NO. 2
*** LOAD CASE 122 -- START ITERATION NO. 3
*** LOAD CASE 122 -- START ITERATION NO. 4
*** LOAD CASE 122 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 122

*** LOAD CASE 123 -- START ITERATION NO. 2
*** LOAD CASE 123 -- START ITERATION NO. 3
*** LOAD CASE 123 -- START ITERATION NO. 4
*** LOAD CASE 123 -- START ITERATION NO. 5

STAAD SPACE -- PAGE NO. 16

**NOTE-Tension/Compression converged after 5 iterations, Case= 123

*** LOAD CASE 124 -- START ITERATION NO. 2
*** LOAD CASE 124 -- START ITERATION NO. 3
*** LOAD CASE 124 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 124

*** LOAD CASE 125 -- START ITERATION NO. 2
*** LOAD CASE 125 -- START ITERATION NO. 3
*** LOAD CASE 125 -- START ITERATION NO. 4
*** LOAD CASE 125 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 125

**NOTE-Tension/Compression converged after 1 iterations, Case= 126

*** LOAD CASE 127 -- START ITERATION NO. 2
*** LOAD CASE 127 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 127

*** LOAD CASE 128 -- START ITERATION NO. 2
*** LOAD CASE 128 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 128

*** LOAD CASE 129 -- START ITERATION NO. 2
*** LOAD CASE 129 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 129

**NOTE-Tension/Compression converged after 1 iterations, Case= 130

*** LOAD CASE 131 -- START ITERATION NO. 2
*** LOAD CASE 131 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 131

*** LOAD CASE 132 -- START ITERATION NO. 2
*** LOAD CASE 132 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 132

*** LOAD CASE 133 -- START ITERATION NO. 2
*** LOAD CASE 133 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 133

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

*** LOAD CASE 204 -- START ITERATION NO. 2
*** LOAD CASE 204 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 204

*** LOAD CASE 205 -- START ITERATION NO. 2
*** LOAD CASE 205 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 205

*** LOAD CASE 206 -- START ITERATION NO. 2
*** LOAD CASE 206 -- START ITERATION NO. 3
*** LOAD CASE 206 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

*** LOAD CASE 208 -- START ITERATION NO. 2
*** LOAD CASE 208 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 208

*** LOAD CASE 209 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 209

*** LOAD CASE 210 -- START ITERATION NO. 2
*** LOAD CASE 210 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

*** LOAD CASE 219 -- START ITERATION NO. 2
*** LOAD CASE 219 -- START ITERATION NO. 3
*** LOAD CASE 219 -- START ITERATION NO. 4
*** LOAD CASE 219 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 219

*** LOAD CASE 220 -- START ITERATION NO. 2
*** LOAD CASE 220 -- START ITERATION NO. 3
*** LOAD CASE 220 -- START ITERATION NO. 4
*** LOAD CASE 220 -- START ITERATION NO. 5
*** LOAD CASE 220 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 220

*** LOAD CASE 221 -- START ITERATION NO. 2
*** LOAD CASE 221 -- START ITERATION NO. 3
*** LOAD CASE 221 -- START ITERATION NO. 4

STAAD SPACE -- PAGE NO. 19

*** LOAD CASE 221 -- START ITERATION NO. 5
*** LOAD CASE 221 -- START ITERATION NO. 6
*** LOAD CASE 221 -- START ITERATION NO. 7

**NOTE-Tension/Compression converged after 7 iterations, Case= 221

*** LOAD CASE 222 -- START ITERATION NO. 2
*** LOAD CASE 222 -- START ITERATION NO. 3
*** LOAD CASE 222 -- START ITERATION NO. 4
*** LOAD CASE 222 -- START ITERATION NO. 5
*** LOAD CASE 222 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 222

*** LOAD CASE 223 -- START ITERATION NO. 2
*** LOAD CASE 223 -- START ITERATION NO. 3
*** LOAD CASE 223 -- START ITERATION NO. 4
*** LOAD CASE 223 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 223

*** LOAD CASE 224 -- START ITERATION NO. 2
*** LOAD CASE 224 -- START ITERATION NO. 3
*** LOAD CASE 224 -- START ITERATION NO. 4
*** LOAD CASE 224 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 224

*** LOAD CASE 225 -- START ITERATION NO. 2
*** LOAD CASE 225 -- START ITERATION NO. 3
*** LOAD CASE 225 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 225

*** LOAD CASE 226 -- START ITERATION NO. 2
*** LOAD CASE 226 -- START ITERATION NO. 3
*** LOAD CASE 226 -- START ITERATION NO. 4
*** LOAD CASE 226 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 226

**NOTE-Tension/Compression converged after 1 iterations, Case= 227

*** LOAD CASE 228 -- START ITERATION NO. 2
*** LOAD CASE 228 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 228

STAAD SPACE -- PAGE NO. 20

*** LOAD CASE 229 -- START ITERATION NO. 2
*** LOAD CASE 229 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 229

*** LOAD CASE 230 -- START ITERATION NO. 2
*** LOAD CASE 230 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 230

**NOTE-Tension/Compression converged after 1 iterations, Case= 231

*** LOAD CASE 232 -- START ITERATION NO. 2
*** LOAD CASE 232 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 232

*** LOAD CASE 233 -- START ITERATION NO. 2
*** LOAD CASE 233 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 233

*** LOAD CASE 234 -- START ITERATION NO. 2
*** LOAD CASE 234 -- START ITERATION NO. 3
*** LOAD CASE 234 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 234

681. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 1,2020 TIME= 12: 5: 1 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://www.bentley.com/en/support/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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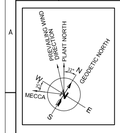
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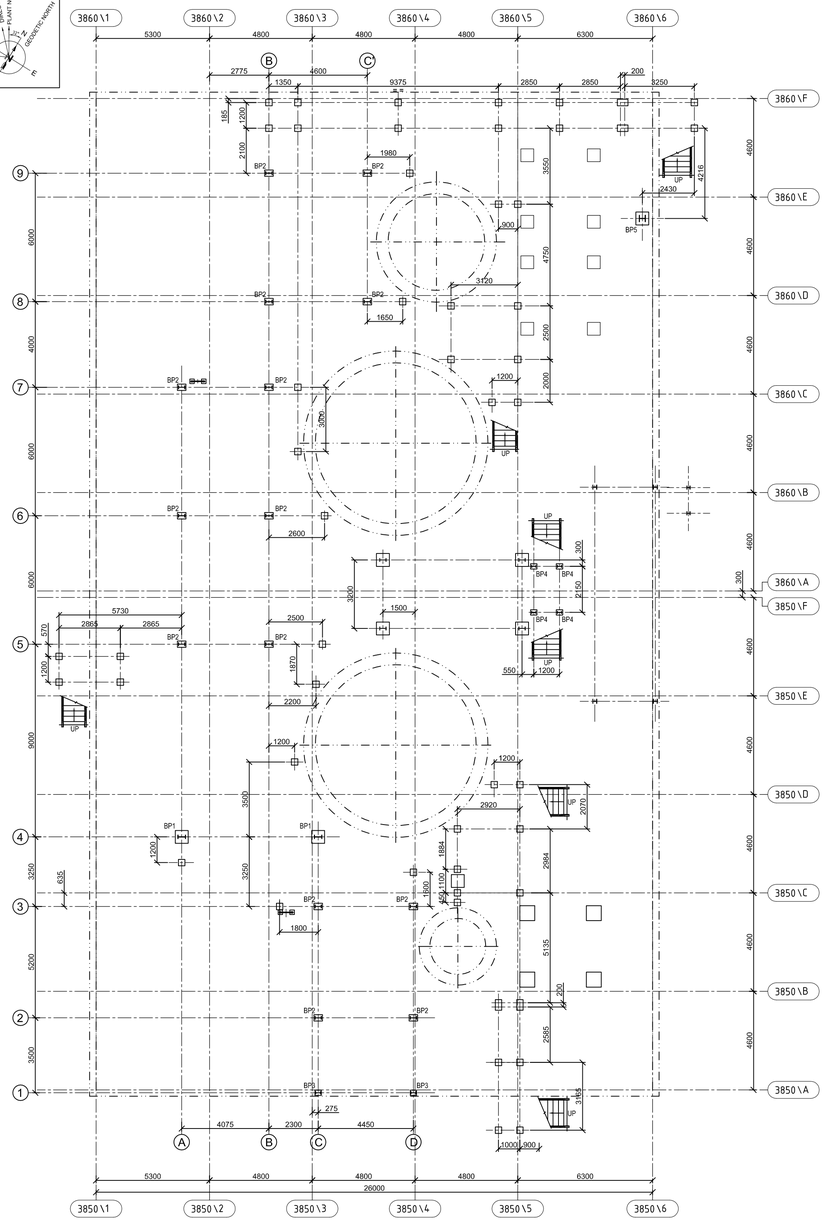
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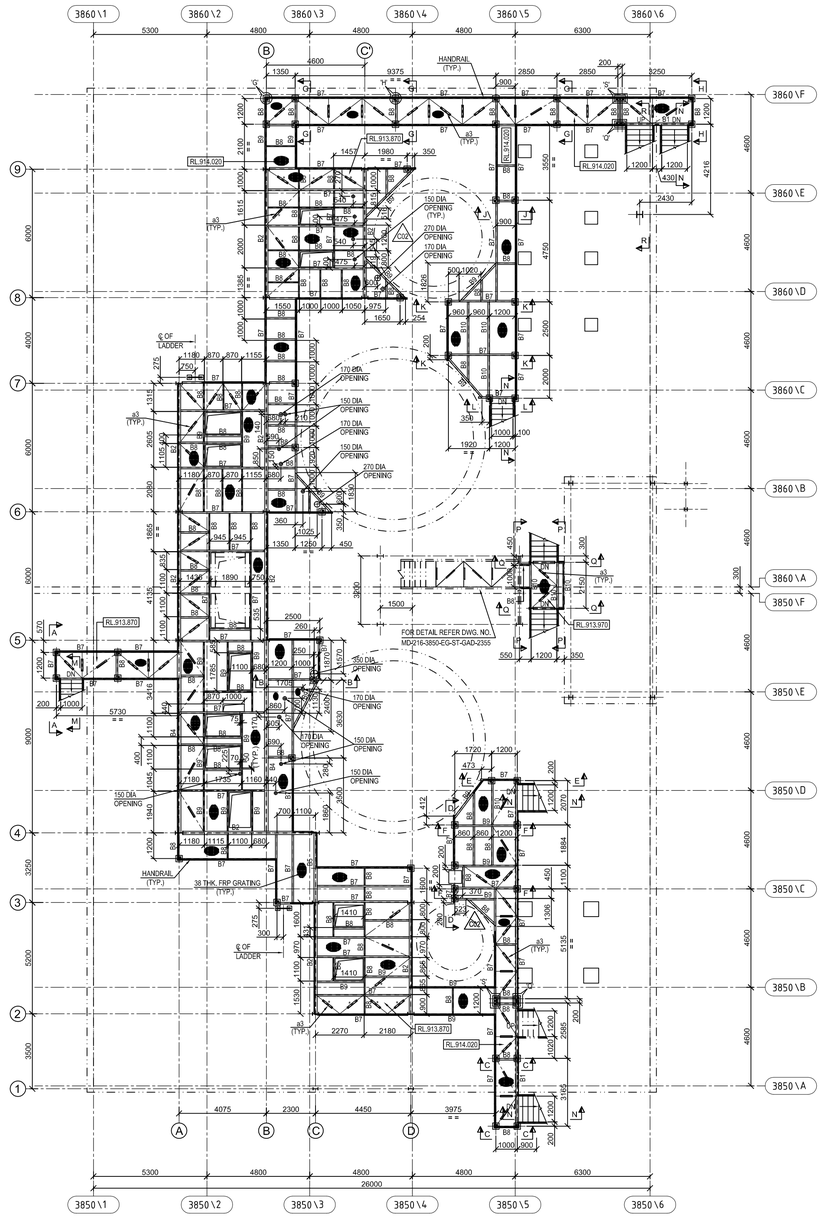
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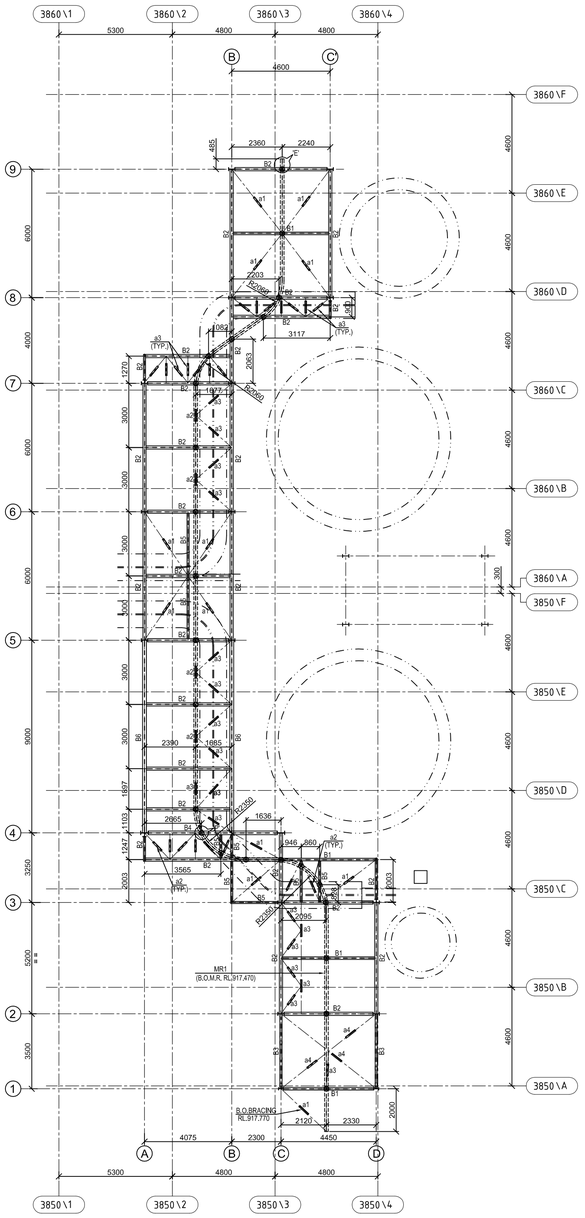




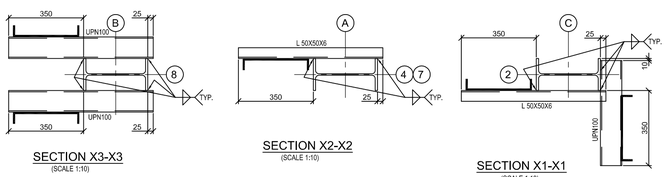
PLAN AT BASE LEVEL
(SCALE 1:100)



PLAN AT RL 913.870, RL 914.020 & RL 913.970 LEVEL
(TOP OF FLOOR BRACING SHOULD BE KEPT 50mm BELOW T.O.S. U.N.O.)
(TOP OF '*' MARKED FLOOR BRACING SHOULD BE KEPT 150mm BELOW T.O.S.)
(SCALE 1:100)



PLAN AT RL 918.210(T.O.S.) LEVEL
(TOP OF BRACING SHOULD BE KEPT 50mm BELOW T.O.S. U.N.O.)
(SCALE 1:100)



SECTION X3-X3
(SCALE 1:10)

SECTION X2-X2
(SCALE 1:10)

SECTION X1-X1
(SCALE 1:10)

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO. MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR), DRAWING NO. MD - 216 - 3850 -EG -ST - GAD - 2004 & 2005
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL,
F.G.L. - FINISHED GROUND LEVEL,
F.F.L. - FINISHED FLOOR LEVEL,
TYP. - TYPICAL,
B.O.M.R. - BOTTOM OF MONORAIL,
MKD. - MARKED,
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
REF. - REFER
DRG. NO. - DRAWING NUMBER
FRP - FIBERGLASS REINFORCED PLASTIC

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3850-EG-ST-GAD-2355	PLAN, ELEVATIONS AND DETAILS
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850/3860/3-6/A-F,A-C, 3D AND SECTIONS
MD-216-3850-EG-ST-GAD-1001	PLAN VIEW 3850/1-3, 3D VIEW AND SECTIONS
MD-216-3850-EG-ME-GAD-1009	SECTION C-C
MD-216-3850-EG-ME-GAD-1008	SECTION B-B
MD-216-3850-EG-ME-GAD-1007	SECTION A-A
MD-216-3850-EG-ME-GAD-1014	PLAN AT EL. +912.350
MD-216-3850-EG-ME-GAD-1004	PLAN AT EL. +912.350

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	30/11/2023	AS BUILT	SUS	SYC	APL	PRI	
C02	26/09/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED △3	SUS	SYC	APL	PRI	
C01	01/09/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED △2	SUS	SYC	APL	PRI	
A01	01/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

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MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

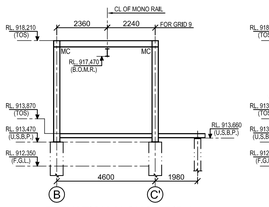
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OUTOTEC DRAWING NUMBER: 900707__GAU01__AA03_MTL01__38514 — DOCUMENT ID: OU100689592 — REVISION: 3

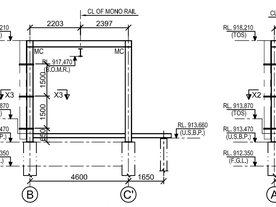
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CHECKED	SYC	31/07/2020
ENGINEER	APL	30/07/2020
APPROVED	PRI	01/08/2020
EMPLOYER		
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MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
LIME SLAKING AREA
MONORAIL BEAM SUPRTS & WLKWAY DET

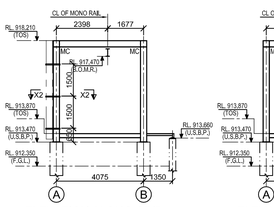
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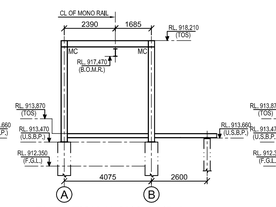
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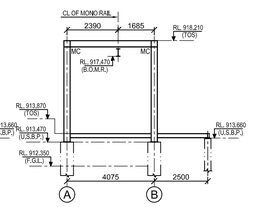
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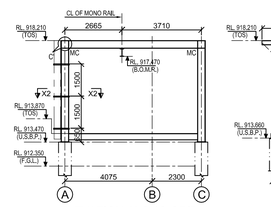
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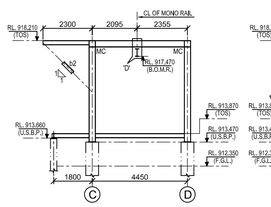
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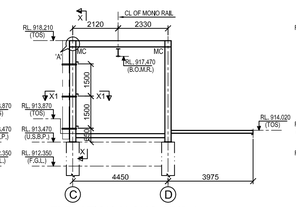
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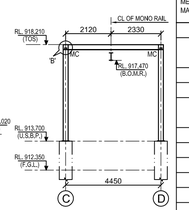
ELEVATION ALONG GRID 4
SCALE 1:100



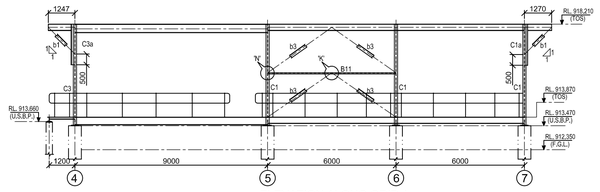
ELEVATION ALONG GRID 3
SCALE 1:100



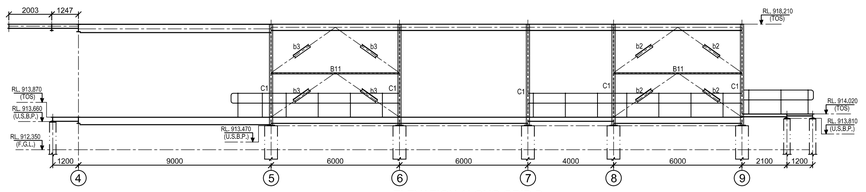
ELEVATION ALONG GRID 2
SCALE 1:100



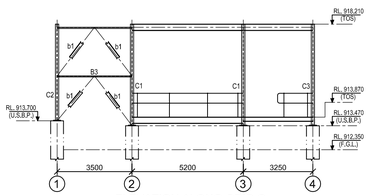
ELEVATION ALONG GRI
SCALE 1:100



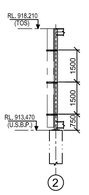
ELEVATION ALONG GRID A
SCALE 1:100



ELEVATION ALONG GRID B
SCALE 1:100



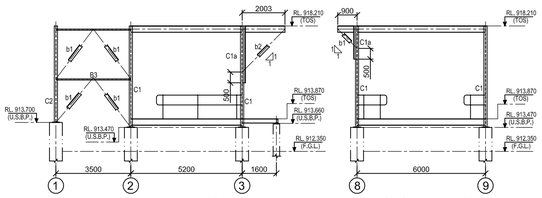
ELEVATION ALONG GRID C
SCALE 1:100



VIEW X-X
SCALE 1:100

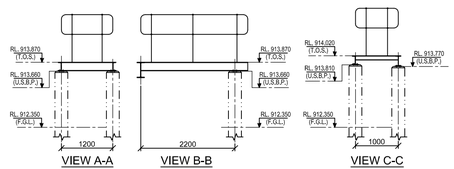
SCHEDULE OF MEMBERS					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(kN)	FACTORED SHEAR FORCE(kN)	REMARKS
C1	IPE 300	I	-	-	GUSSET PLATE 8 THK.
C1a	1. IPE 300 2. 1/2 CUT IPE 300	I, I_2	-	-	
C2	IPE 240	I	-	-	
C3	IPE 360	I	-	-	
C3a	1. IPE 360 2. 1/2 CUT IPE 360	I, I_2	-	-	
C4	IPE 200	I	-	-	
B1	IPE 240	I	60	70	
B2	IPE 300	I	150	110	
B3	UPN 100	[	90	-	
B4	IPE 360	I	140	90	
B5	IPE 200	I	40	30	
B6	IPE 400	I	90	100	
B7	UPN 200	[	70	40	
B8	UPN 100	[	30	20	
B9	UPN 140	[	30	20	
B10	UPN 120	[	-	-	
B11	UPN 140	[	120	-	
a1	L 90X90X8	⌜	60	-	
a2	L 75X75X8	⌜	90	-	
a3	L 65X65X6	L	45	-	
a4	L 100X100X10	L	30	-	
b1	L 90X90X8	L	90	-	
b2	L 100X100X10	L	90	-	
b3	L 130X130X10	L	150	-	
b4	L 65X65X6	L	30	-	
ST1	UPN 200	[	-	-	
ST2	UPN 200	[	-	-	
MR1	IPE 300	I	-	-	

GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED
FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos.
M20 BOLTS SHALL BE PROVIDED.



ELEVATION ALONG GRID D
SCALE 1:100

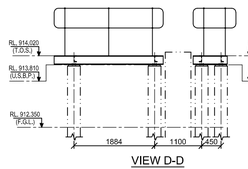
ELEVATION ALONG GRID C'
SCALE 1:100



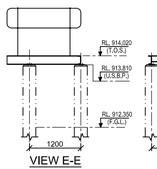
VIEW A-A
SCALE 1:50

VIEW B-B
SCALE 1:50

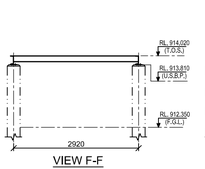
VIEW C-C
SCALE 1:50



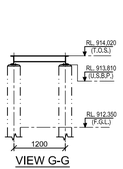
VIEW D-D
SCALE 1:50



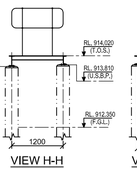
VIEW E-E
SCALE 1:50



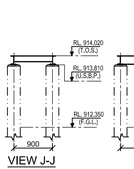
VIEW F-F
SCALE 1:50



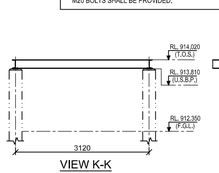
VIEW G-G
SCALE 1:50



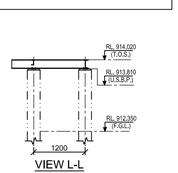
VIEW H-H
SCALE 1:50



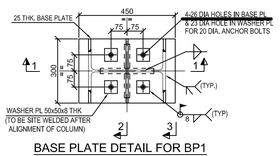
VIEW J-J
SCALE 1:50



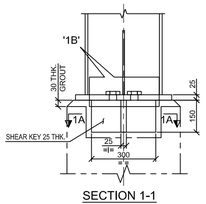
VIEW K-K
SCALE 1:50



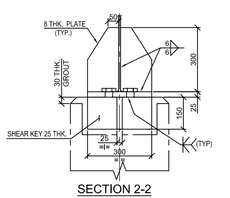
VIEW L-L
SCALE 1:50



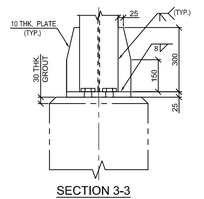
BASE PLATE DETAIL FOR BP1
(SCALE - 1:10)



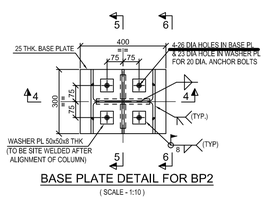
SECTION 1-1
(SCALE - 1:10)



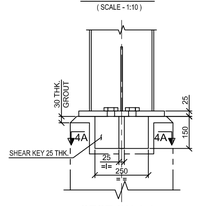
SECTION 2-2
(SCALE - 1:10)



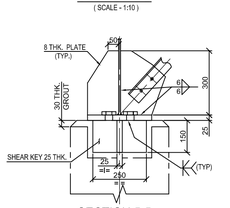
SECTION 3-3
(SCALE - 1:10)



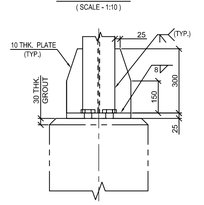
BASE PLATE DETAIL FOR BP2
(SCALE - 1:10)



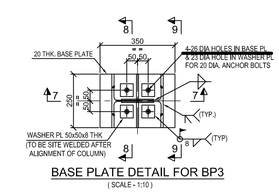
SECTION 4-4
(SCALE - 1:10)



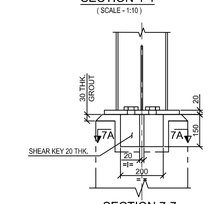
SECTION 5-5
(SCALE - 1:10)



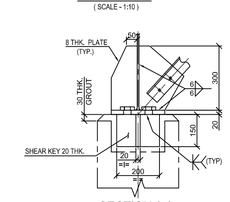
SECTION 6-6
(SCALE - 1:10)



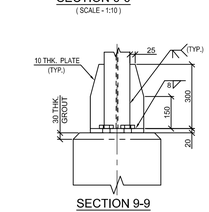
BASE PLATE DETAIL FOR BP3
(SCALE - 1:10)



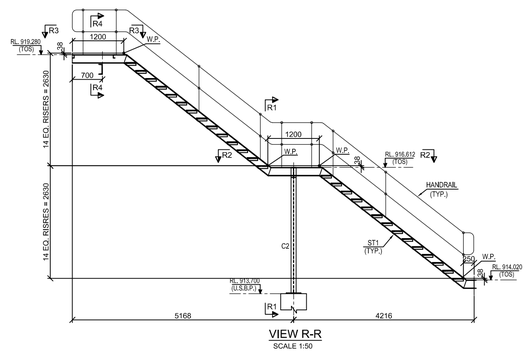
SECTION 7-7
(SCALE - 1:10)



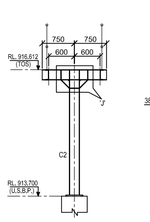
SECTION 8-8
(SCALE - 1:10)



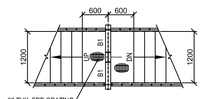
SECTION 9-9
(SCALE - 1:10)



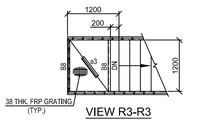
VIEW R-R
SCALE 1:50



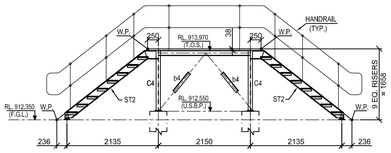
VIEW R1-R1
SCALE 1:50



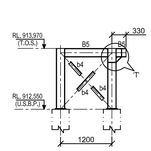
VIEW R2-R2
SCALE 1:50



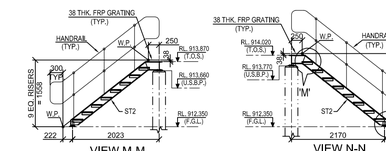
VIEW R3-R3
SCALE 1:50



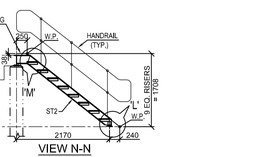
VIEW P-P
SCALE 1:50



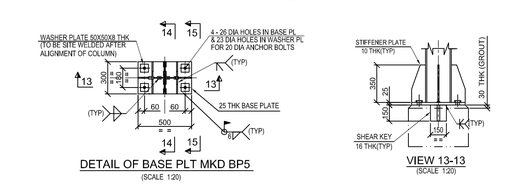
VIEW Q-Q
SCALE 1:50



VIEW M-M
SCALE 1:50

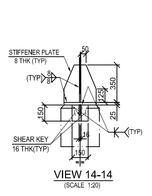


VIEW N-N
SCALE 1:50

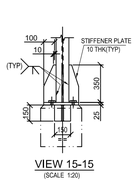


DETAIL OF BASE PLT MKD BP5
(SCALE 1:20)

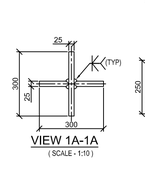
VIEW 13-13
(SCALE 1:20)



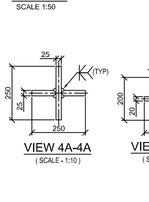
VIEW 14-14
(SCALE 1:20)



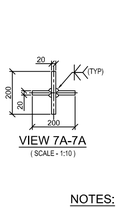
VIEW 15-15
(SCALE 1:20)



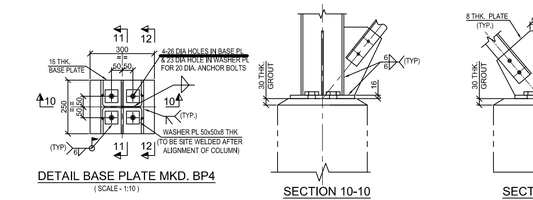
VIEW 1A-1A
(SCALE - 1:10)



VIEW 4A-4A
(SCALE - 1:10)

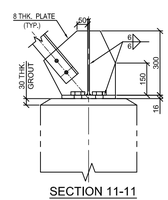


VIEW 7A-7A
(SCALE - 1:10)

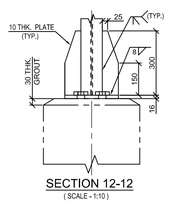


DETAIL BASE PLATE MKD. BP4
(SCALE - 1:10)

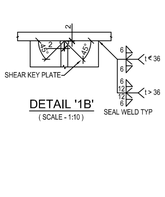
SECTION 10-10
(SCALE - 1:10)



SECTION 11-11
(SCALE - 1:10)



SECTION 12-12
(SCALE - 1:10)



DETAIL '1B'
(SCALE - 1:10)

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO. MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3850 -EG -ST - GAD - 2003 & 2005

LEGENDS :-

R.L - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S - TOP OF STEEL.
F.G.L - FINISHED GROUND LEVEL.
F.F.L - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
REF. - REFER
DRG. NO. - DRAWING NUMBER
FRP - FIBERGLASS REINFORCED PLASTIC

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3850-EG-ST-GAD-2355	PLAN, ELEVATIONS AND DETAILS
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850/3860/3-6/A-F,A-C, 3D AND SECTIONS
MD-216-3850-EG-ST-GAD-1001	PLAN VIEW 3850/1-3, 3D VIEW AND SECTIONS
MD-216-3850-EG-ME-GAD-1009	SECTION C-C
MD-216-3850-EG-ME-GAD-1008	SECTION B-B
MD-216-3850-EG-ME-GAD-1007	SECTION A-A
MD-216-3850-EG-ME-GAD-1014	PLAN AT EL +912.350
MD-216-3850-EG-ME-GAD-1004	PLAN AT EL +912.350

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	30/11/2023	AS BUILT	SUS	SYC	APL	PRI	
C02	26/09/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 2	SUS	SYC	APL	PRI	
C01	01/09/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 1	SUS	SYC	APL	PRI	
A01	01/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة معادن العربية السعودية

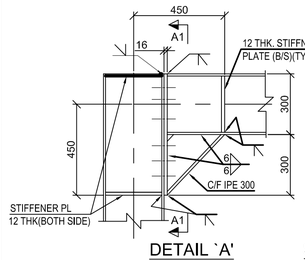
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LARSEN & TOUBRO DRAWING NUMBER	019045_450_S_G_26143_002
REVISION	3
OUTOTEC DRAWING NUMBER	900707__GAU01__AA03_MTL01__38515
DOCUMENT ID	OU100689593
REVISION	3
DRAWN	SUS 29/07/2020
CHECKED	SYC 31/07/2020
ENGINEER	APL 30/07/2020
APPROVED	PRI 01/08/2020
EMPLOYER	
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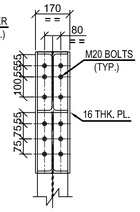
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
LIME SLAKING AREA
MONORAIL BEAM SUPRTS & WLKWAY DET

DRAWING NUMBER: MD - 216 - 3850 - EG - ST - GAD - 2004
REV: D01

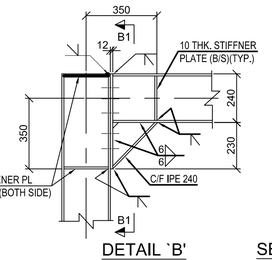
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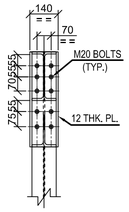
DETAIL `A'
(SCALE 1:15)



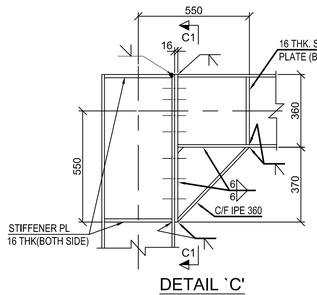
SECTION A1-A1
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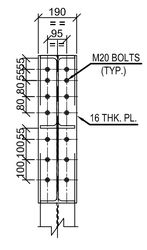
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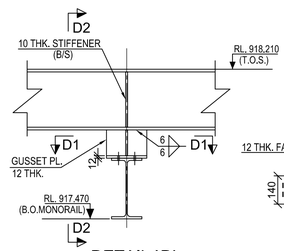
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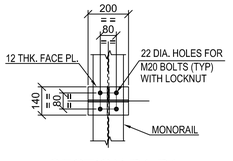
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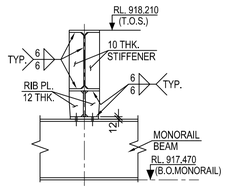
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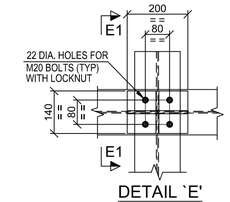
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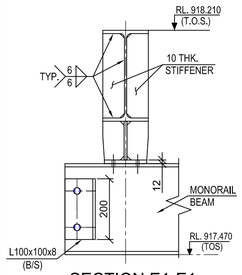
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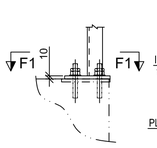
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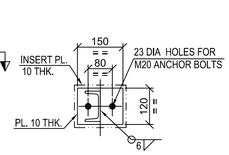
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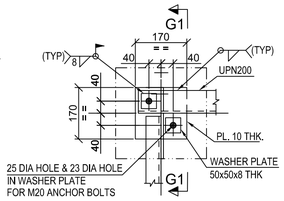
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(SCALE 1:10)



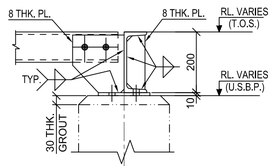
DETAIL `F'
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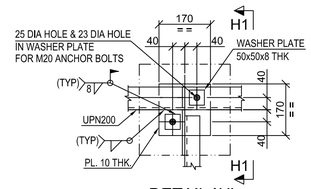
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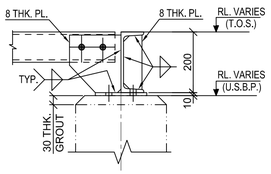
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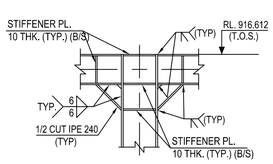
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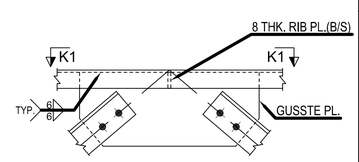
DETAIL `H'
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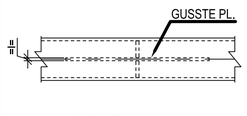
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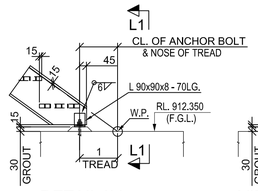
DETAIL `J'
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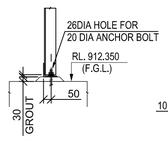
DETAIL 'K'
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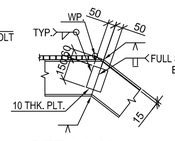
SECTION K1-K1
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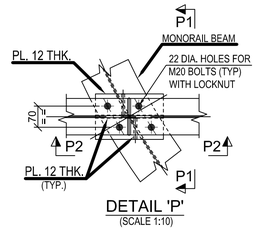
DETAIL `L'
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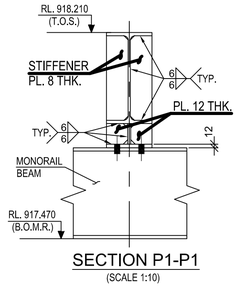
VIEW L1-L1
(SCALE 1:20)



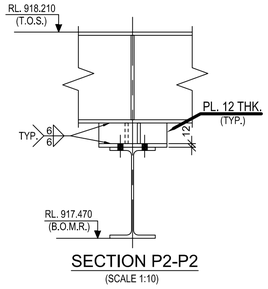
DETAIL `M'
(SCALE 1:20)



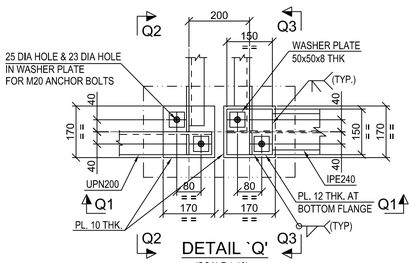
DETAIL 'P'
(SCALE 1:10)



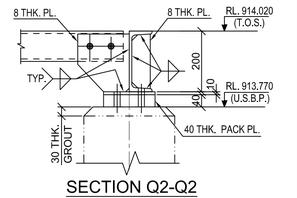
SECTION P1-P1
(SCALE 1:10)



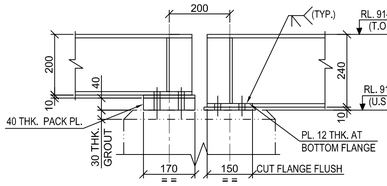
SECTION P2-P2
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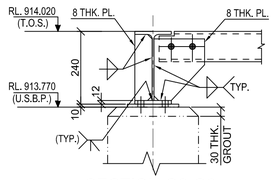
DETAIL `Q'
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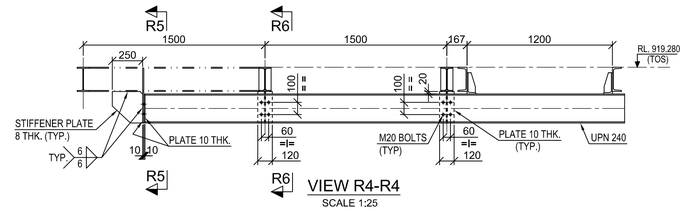
SECTION Q2-Q2
(SCALE 1:10)



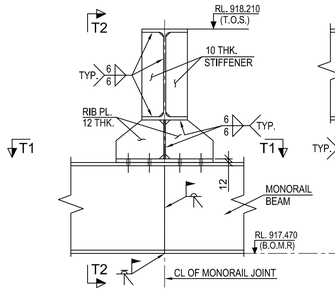
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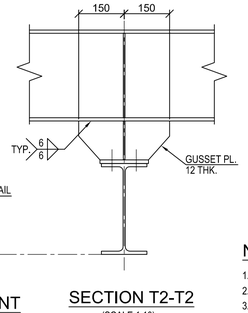
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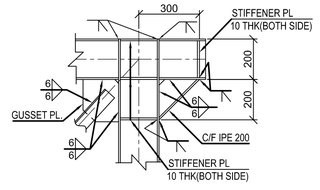
VIEW R4-R4
SCALE 1:25



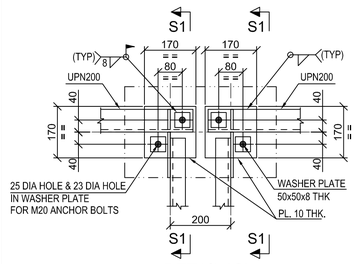
DETAIL OF MONORAIL SPLICE JOINT
(MONORAIL JOINT TO BE DONE AT SUPPORT LOCATIONS ONLY)
(SCALE 1:10)



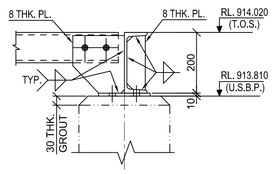
SECTION T2-T2
(SCALE 1:10)



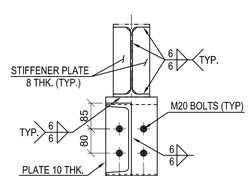
DETAIL `T'
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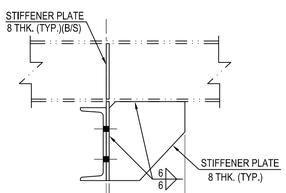
DETAIL `S'
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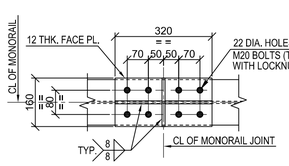
SECTION S1-S1
(SCALE 1:10)



SECTION R5-R5
(SCALE 1:10)



SECTION R6-R6
(SCALE 1:10)



SECTION T1-T1
(SCALE 1:10)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
REF. - REFER
DRG. NO. - DRAWING NUMBER
FRP - FIBERGLASS REINFORCED PLASTIC

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001.
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3850 -EG -ST - GAD - 2003 & 2004.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3850-EG-ST-GAD-2355	PLAN, ELEVATIONS AND DETAILS								
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850/3860/3-6/A-F,A-C, 3D AND SECTIONS								
MD-216-3850-EG-ST-GAD-1001	PLAN VIEW 3850/1-3, 3D VIEW AND SECTIONS								
MD-216-3850-EG-ME-GAD-1009	SECTION C-C								
MD-216-3850-EG-ME-GAD-1008	SECTION B-B	D01	30/11/2023	AS BUILT	SUS	SYC	APL	PRI	
MD-216-3850-EG-ME-GAD-1007	SECTION A-A	C02	26/09/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED △C02	SUS	SYC	APL	PRI	
MD-216-3850-EG-ME-GAD-1014	PLAN AT EL. +912.350	C01	01/09/2020	ISSUED FOR CONSTRUCTION	SUS	SYC	APL	PRI	
MD-216-3850-EG-ME-GAD-1004	PLAN AT EL. +912.350	A01	01/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26143_003 — REVISION: 3

OUTOTEC DRAWING NUMBER: 900707__GAU01__AA03_MTL01__38516 — DOCUMENT ID: OU100689594 — REVISION: 3

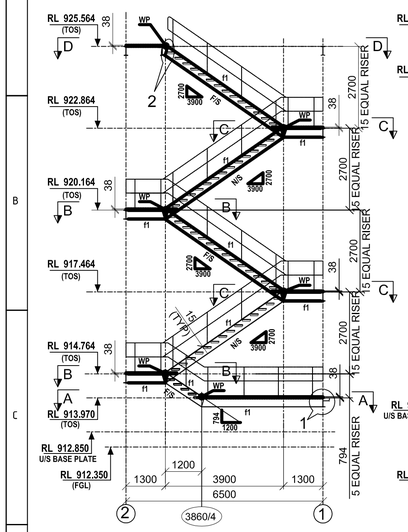
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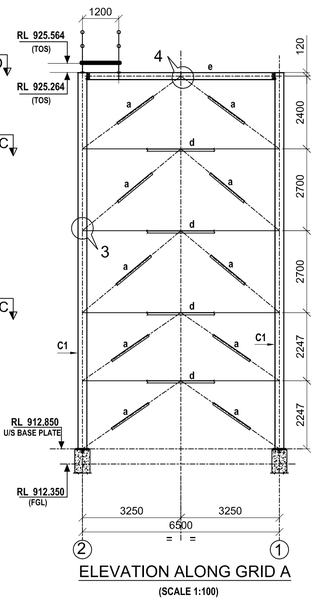
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Saudi Arabian Mining Company شركة التعدين العربية السعودية
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DRAWN	SUS	29/07/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	31/07/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APL	30/07/2020	LIME SLAKING AREA
APPROVED	PRI	01/08/2020	MONORAIL BEAM SUPRTS & WLKWAY DET
EMPLOYER			
SCALE 1:25,20, 15,10	A1		DRAWING NUMBER: MD - 216 - 3850 - EG - ST - GAD - 2005 — REV: D01

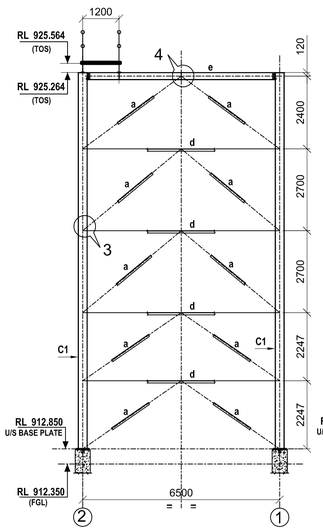
DRAWING NUMBER: MD-216-3850-EG-ST-GAD-2355 | REV: D01



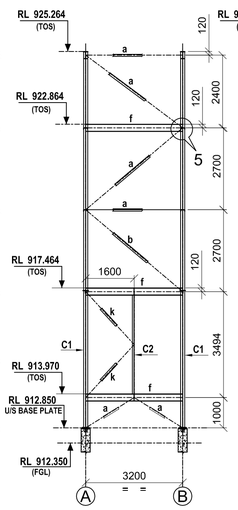
ELEVATION OF STAIR
(SCALE 1:100)



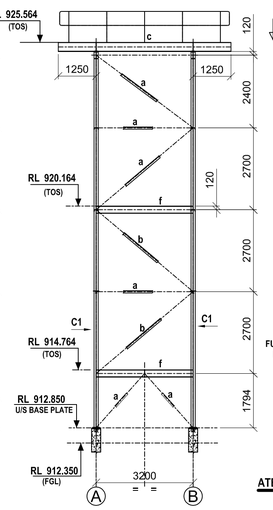
ELEVATION ALONG GRID A
(SCALE 1:100)



ELEVATION ALONG GRID B
(SCALE 1:100)

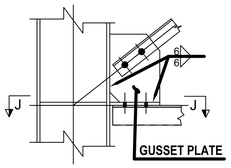


ELEVATION ALONG GRID 1
(SCALE 1:100)

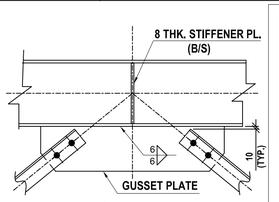


ELEVATION ALONG GRID 2
(SCALE 1:100)

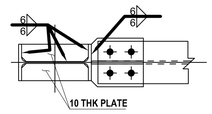
ATE



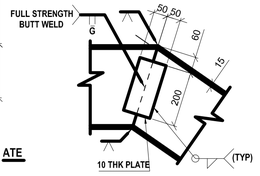
DETAIL - 3
(SCALE 1:10)



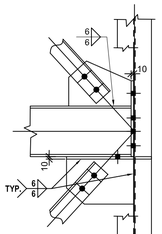
DETAIL - 4
(SCALE 1:10)



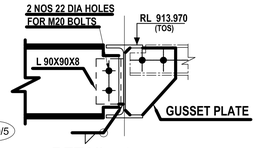
PLAN J-J
(SCALE 1:10)



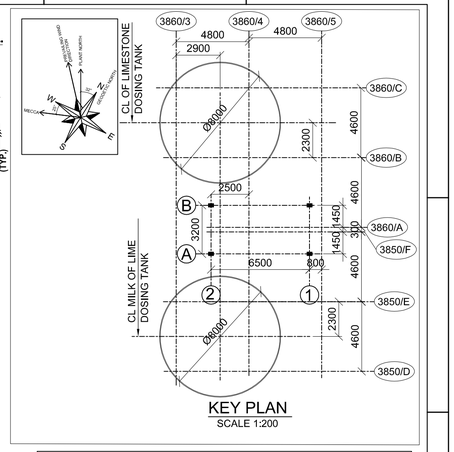
DETAIL - 2
(SCALE 1:10)



DETAIL - 5
(SCALE 1:10)



DETAIL - 1
(SCALE 1:10)

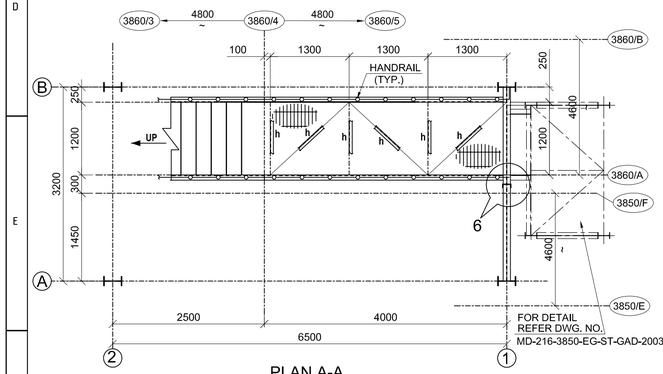


KEY PLAN
SCALE 1:200

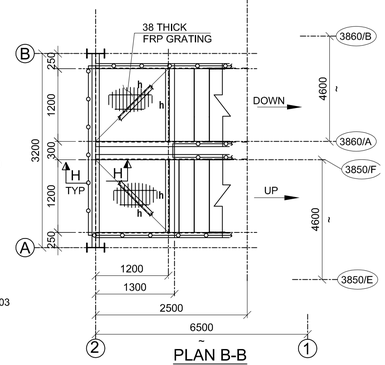
SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE240	I	150	-	MATERIAL GRADE:S275 JR
C2	UPN140	]	90	-	
a	2Ls 65x65x6	T	110	-	
b	2Ls 75x75x8	T	70	-	
c	IPE300	I	50	50	
d	UPN140	⊓	30	-	
d1	UPN100	]	-	20	
e	IPE240	I	20	35	
f	IPE240	I	60	50	
f1	UPN240	[	-	40	
g	L 65x65x6	⌝	20	-	
h	L 65x65x6	⌝	20	-	
k	L 75x75x8	⌝	60	-	

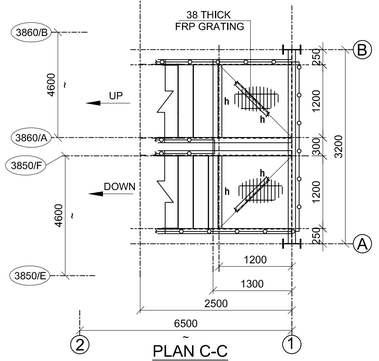
ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



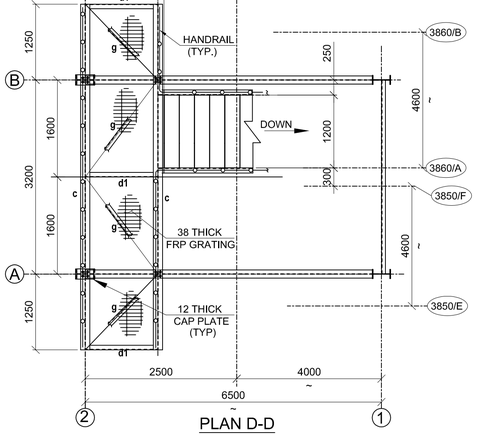
PLAN A-A
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



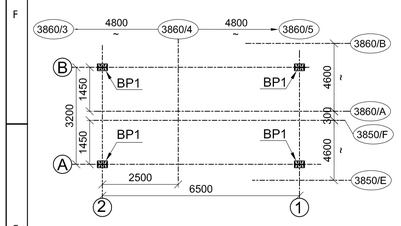
PLAN B-B
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



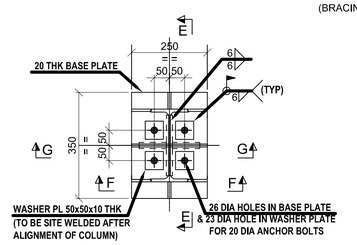
PLAN C-C
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



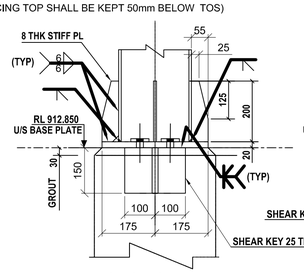
PLAN D-D
SCALE 1:50
(BRACING TOP SHALL BE KEPT 50mm BELOW TOS)



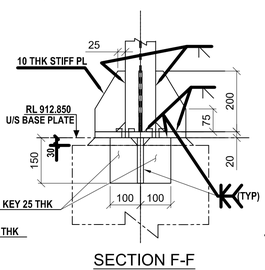
PLAN AT BASE LEVEL AT RL 912.850 (USBP)
SCALE 1:100



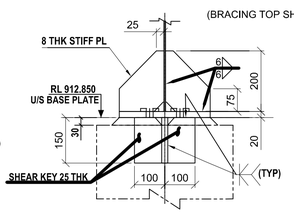
DETAIL OF BASE PLATE MKD BP1
(SCALE 1:10)



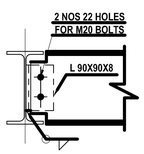
SECTION E-E
(SCALE 1:10)



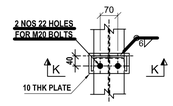
SECTION F-F
(SCALE 1:10)



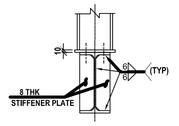
SECTION G-G
(SCALE 1:10)



VIEW H-H
(SCALE 1:10)



DETAIL - 6
(SCALE 1:10)



SECTION K-K
(SCALE 1:10)

LEGEND:

TOS - TOP OF STEEL
CL - CENTER LINE
FGL - FINISHED GROUND LEVEL.
PL - PLATE
STIFF - STIFENER
THK - THICKNESS
UNO - UNLESS NOTED OTHERWISE
MM - MILLIMETER
TYP - TYPICAL
GA - GENERAL ARRANGEMENT
DRG - DRAWING
WP - WORKING POINT
N/S - NEAR SIDE
F/S - FAR SIDE
TOG - TOP OF GRATING
MKD - MARKED
USBP - UNDER SIDE BASE PLATE
U/S - UNDER SIDE
FRP - FIBERGLASS REINFORCED PLASTIC

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36,
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ME-GAD-1002	STRUCTURAL STANDARD STAIR DETAILS								
MD-216-3850-EG-ME-GAD-1015	LIME SLAKING AND LIMESTONE PLAN AT EL. +924.100								
MD-216-3850-EG-ME-GAD-1014	LIME SLAKING AND LIMESTONE PLAN AT EL. +912.350	D01	30/11/2023	AS BUILT	SUS	SYC	APL	PRI	
MD-216-3850-EG-ST-GAD-1002	PLAN VIEW 3850 - 3860, 3-6, A-F,A-C, 3D AND SECTIONS	C01	25/08/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER	SUS	SYC	APL	PRI	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS			COMMENTS DATED 13/08/2020 & REVISED AS MARKED					
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	08/05/2020	ISSUED FOR REFERENCE	GST	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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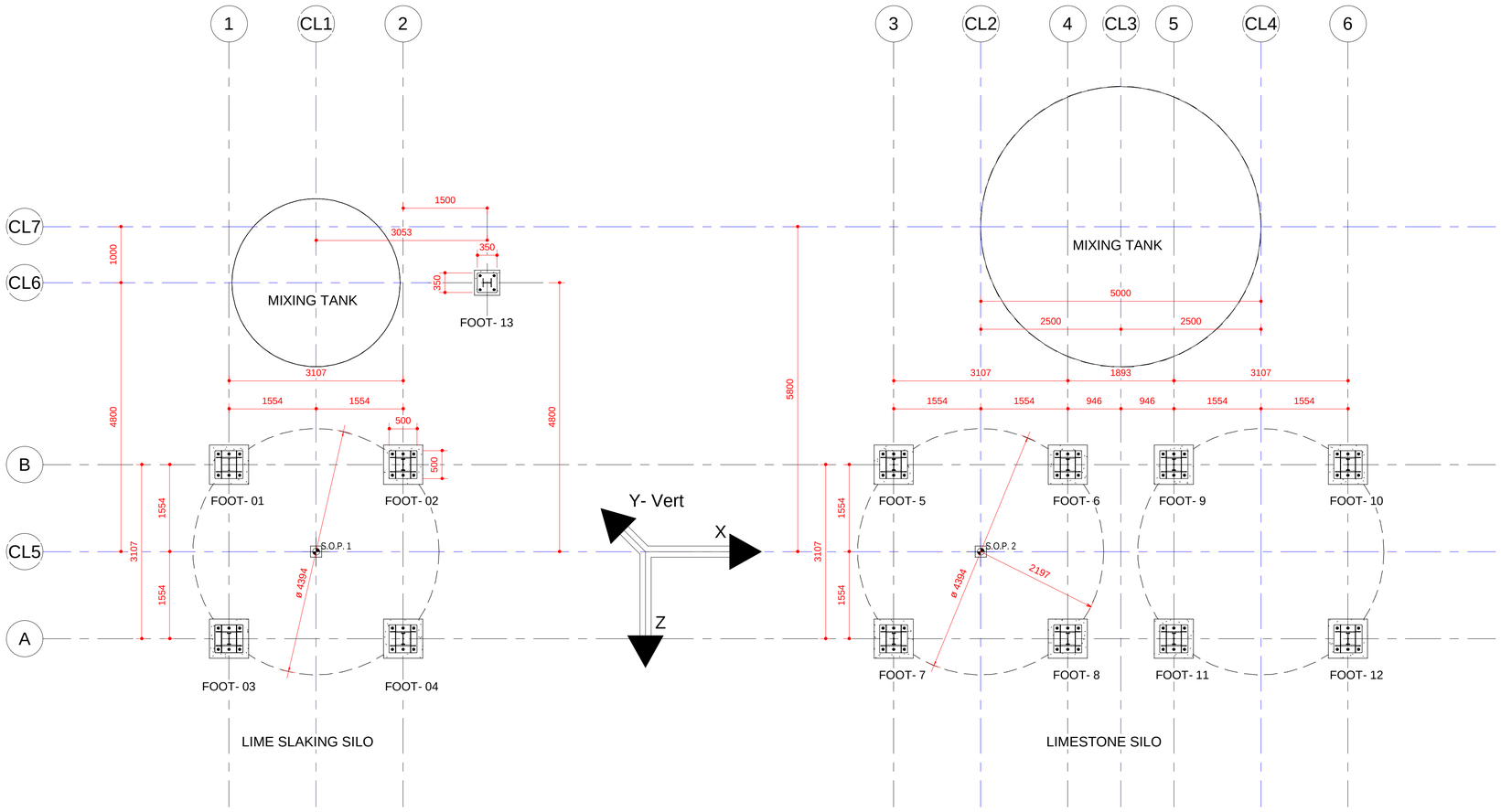
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DRAWN	GST	06/05/2020	
CHECKED	SYC	07/05/2020	
ENGINEER	APL	04/05/2020	
APPROVED	PRI	08/05/2020	
EMPLOYER			
SCALE	1:100,50,10		A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26075_001 | REVISION: 2
OUTOTEC DRAWING NUMBER: 900707_GAU01_AA03_MTL01_38508 | DOCUMENT ID: OU100682983 | REVISION: 2

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
REAGENTS - LIME SLAKING STAIR TOWER
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3850 - EG - ST - GAD - 2355 | REV: D01



1 CIVIL LOADING PLAN

1 : 50

	UNFACTORED FOOT REACTIONS																																								
	FOOT 1			FOOT 2			FOOT 3			FOOT 4			FOOT 5			FOOT 6			FOOT 7			FOOT 8			FOOT 9			FOOT 10			FOOT 11			FOOT 12			FOOT 13		
	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)	Rx (kN)	Rz (kN)	Ry (kN)
SWDL	2	2	66	-2	2	62	2	-2	63	-2	-2	69	3	2	67	-2	3	67	3	-3	71	-2	-2	71	2	3	67	-3	2	67	2	-2	71	-3	-3	71	0	0	2
DL	5	-2	83	8	2	99	-5	5	181	6	8	181	1	4	40	-2	3	36	3	-3	102	-5	-3	96	2	3	36	-1	3	40	5	-3	96	-3	-3	102	0	1	14
LL	33	30	1011	-26	34	933	27	-29	959	-26	-31	1054	36	32	960	-25	33	950	37	-33	1007	-27	-32	996	25	33	950	-36	31	960	27	-32	995	-37	-33	1007	0	0	10
WL0	-33	-19	-437	-39	20	435	-46	10	-484	-46	-13	486	-51	-19	-567	-51	18	565	-49	17	-578	-49	-17	572	-51	-18	-566	-51	19	567	-49	17	-572	-49	-17	578			
WL45	-32	-47	-652	-16	-16	-6	-24	-26	0	-44	-36	658	-51	-49	-808	-26	-23	-4	-20	-23	-3	-45	-48	810	-46	-48	-804	-22	-22	-6	-25	-23	0	-49	-47	815			
WL90	-13	-47	-486	16	-42	-444	12	-47	485	-16	-38	445	-20	-50	-576	14	-50	-571	20	-50	574	-14	-51	573	-14	-50	-571	21	-50	-576	14	-50	573	-20	-50	574			
WL135	14	-19	-35	39	-44	-621	41	-40	686	21	-18	-29	22	-22	-6	46	-48	-804	49	-47	814	25	-23	1	26	-23	-4	51	-49	-808	45	-48	810	20	-23	-3			
WL180	33	19	437	39	-20	-435	46	-10	484	46	13	-486	51	19	567	51	-18	-565	49	-17	578	49	17	-572	51	18	566	51	-19	-567	49	-17	572	49	17	-578			
WL225	32	47	652	16	16	6	24	26	0	44	36	-658	51	49	808	26	23	4	20	23	3	45	48	-810	46	48	804	22	22	6	25	23	0	49	47	-815			
WL270	13	47	486	-16	42	444	-12	47	-485	16	38	-445	20	50	576	-14	50	571	-20	50	-574	14	51	-573	14	50	571	-21	50	576	-14	50	-573	20	50	-574			
WL315	-14	19	35	-39	44	621	-41	40	-686	-21	18	29	-22	22	6	-46	48	804	-49	47	-814	-25	23	-1	-26	23	4	-51	49	808	-45	48	-810	-20	23	3			
E0	-25	-14	-328	-29	15	327	-34	8	-364	-35	-10	365	-38	-14	-426	-38	13	425	-37	13	-435	-37	-13	430	-38	-13	-425	-38	14	426	-37	13	-430	-37	-13	435			
E45	-24	-35	-490	-12	-12	-5	-18	-19	0	-33	-27	495	-38	-37	-608	-20	-17	-3	-15	-17	-2	-34	-36	609	-35	-36	-604	-16	-17	-5	-18	-18	0	-37	-36	613			
E90	-9	-35	-365	12	-32	-334	9	-35	365	-12	-28	334	-15	-38	-433	11	-38	-430	15	-37	432	-11	-38	431	-11	-38	-429	15	-38	-433	11	-38	431	-15	-38	432			
E135	11	-15	-26	29	-33	-467	30	-30	515	16	-13	-22	16	-17	-5	35	-36	-604	37	-36	612	18	-18	1	20	-17	-3	38	-37	-608	34	-36	609	15	-17	-2			
E180	25	14	328	29	-15	-327	34	-8	364	35	10	-365	38	14	426	38	-13	-425	37	-13	435	37	13	-430	38	13	425	38	-14	-426	37	-13	430	37	13	-435			
E225	24	35	490	12	12	5	18	19	0	33	27	-495	38	37	608	20	17	3	15	17	2	34	36	-609	35	36	604	16	17	5	18	18	0	37	36	-613			
E270	9	35	365	-12	32	334	-9	35	-365	12	28	-334	15	38	433	-11	38	430	-15	37	-432	11	38	-431	11	38	429	-15	38	433	-11	38	-431	15	38	-432			
E315	-11	15	26	-29	33	467	-30	30	-515	-16	13	22	-16	17	5	-35	36	604	-37	36	-612	-18	18	-1	-20	17	3	-38	37	608	-34	36	-609	-15	17	2			

NOTES
Rx and Ry - horizontal loads - Loads can be applied in any direction
Ry - Vertical loads positive down and negative up
All loads are unfactored. Seismic and Wind loads are factored ACC to SBC301 - EQ 2.4.1-5
All loads are applied to the concrete at structural steel base plate level.

GENERAL

All structural steel to comply with Grade: EN10025-2-S355JR +AR

PROJECT SPECIFICATION DOCUMENTS

Structural:
- MD-101 -SMEM-EG-ST-SPC-0001 REV. C-Steel Fabrication Specification
- MD-101 -SMEM-EG-ST-SPC-0002 REV. C -Steel Erection
- MD-101-SMEM-EG-ST-SPC-0003 REV. C -Grout
- MD-101-SMEM-EG-ST-SPC-0009 REV. C - Protective Coating of Structural Steelwork
- MD-101-SMEM-EG-ST-SPC-0010 REV. C - Fire Retardant Coatings for Structures
- MD-101-SMEM-EG-GE-SPC-0002 REV. C – Welding of plate and sections

STRUCTURAL STEEL CONTRACTORS TO ENSURE THAT:

Two prints of workshop Detail drawings to be submitted to the Engineer for approval prior to proceeding with the fabrication of the steelwork
Steel Sections to manufactured to EN10025-2 or EN 10155.

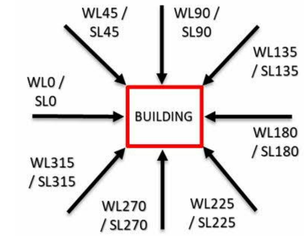
UNLESS OTHERWISE NOTED:

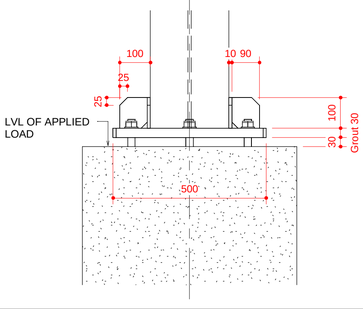
Welding:

All Weld To Comply With AWS D1.1 Latest Edition,
Electrode Classification E70xx
100% Visual Inspection Required On All Welds
10% Liquid Penetration Test (Dyepen)
Fillet Welds 6mm Continuous.

Bolts M16 Grade 8.8 Hot-Dipped Galvanised with Oversized Nuts.

WIND AND EARTHQUAKE LOAD DIRECTIONS





INFORMATION ONLY

REV	DATE	BY	DESCRIPTION	CHK	APD
A	2020-05-12		ISSUED FOR INFORMATION		

ARCHITECT:

CLIENT: TECHMATIC INDUSTRIALS

DRAWING STATUS: INFORMATION

If drawing status = construction, a signed copy of this drawing (either in hardcopy or electronic format) is available at the office of origin and at the office of issue.

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PROJECT: QUTOTEC - Mansourah Massarah Gold Project

TITLE: CIVIL LOADING PLAN

SCALE @ A1 :	CHECKED:	APPROVED:
As indicated	I NAIDOO	
DESIGN:	DRAWN:	DATE:
J KEYSER	J KEYSER	2020-05-12
PROJECT No:	DRAWING No:	REV:
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CESA Drawing Control Unit ISO

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REVIT FILE NAME: C:\Users\ZAJN02777\Documents\WSP Revit Local\Techmatic Qutotec Sitio\Techmatic Qutotec S&A.rvt

PLOT DATE: 20200524 15:53:34 - DO NOT SCALE - DIMENSIONS IN MILLIMETERS (mm) - IF IN DOUBT, ASK